

# INVARIANT MEASURES ON THE SPACE OF MEASURED LAMINATIONS FOR SUBGROUPS OF MAPPING CLASS GROUP

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**ABSTRACT.** Let  $\Gamma$  be a non-elementary subgroup of the mapping class group of a surface. In this paper, we study the dynamics of the  $\Gamma$ -action on the space of measured laminations. We prove that on the space  $\mathcal{R}_\Gamma$  of  $\Gamma$ -recurrent measured laminations, there is at most one  $\Gamma$ -invariant Radon measure (up to scaling). In particular, when  $\Gamma$  is of divergence type, we show that the  $\Gamma$ -action on  $\mathcal{R}_\Gamma$  is uniquely ergodic, by explicitly constructing the ergodic measure. This generalizes Lindenstrauss–Mirzakhani’s result and Hamenstädt’s result for the full mapping class group, in which case the ergodic measure is the Thurston measure. When  $\Gamma$  is convex cocompact, we also prove that every invariant ergodic Radon measure on the space of all measured laminations is either the unique measure on  $\mathcal{R}_\Gamma$ , or a counting measure on the orbit of a non-recurrent measured lamination.

Our method is geometric and does not rely on continuous or homogeneous flows on the ambient space or a dynamical system associated with a finite measure space. This leads to a unifying approach for various metric spaces, including Teichmüller spaces and partially  $\text{CAT}(-1)$  spaces.

In the appendix, we prove non-arithmeticity of the Teichmüller length spectrum of a non-elementary subgroup of the mapping class group. We prove this by introducing the notion of cross-ratios on  $\mathcal{MF}$  and  $\mathcal{PMF}$ , and studying their geometric and dynamical properties, despite the lack of negatively curved features of the Teichmüller space and of conformal geometry on  $\mathcal{PMF}$ . As an application, we show topological mixing of the Teichmüller geodesic flow for every non-elementary subgroup.

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## 1. INTRODUCTION

Let  $S$  be a connected orientable surface of genus  $g$  with  $p$  punctures, where  $3g - 3 + p \geq 1$ , i.e.,  $S$  is a finite-type surface which is not a sphere with at most 3 punctures or a torus. The Teichmüller space  $\mathcal{T} = \mathcal{T}(S)$  is the space of all marked Riemann surface structures on  $S$ , or equivalently, the space of all marked hyperbolic structures on  $S$ . The mapping class group  $\text{Mod}(S)$  is the group of isotopy classes of orientation-preserving self-diffeomorphisms of  $S$ . The Teichmüller space  $\mathcal{T}$  is equipped with a natural metric called the Teichmüller metric, under which the natural action of  $\text{Mod}(S)$  is isometric. The  $\text{Mod}(S)$ -action is properly discontinuous, and its quotient  $\mathcal{M} := \text{Mod}(S) \backslash \mathcal{T}$  is the moduli space of Riemann surfaces.

In Thurston's theory on surfaces ([FLP79], [Thu22]), the mapping class group and the Teichmüller space of  $S$  are closely related to *measured laminations* on  $S$ . Fix a complete hyperbolic structure on  $S$  of finite area. A geodesic lamination on  $S$  is a compact subset of  $S$  foliated by simple geodesics. A *measured lamination* is a geodesic lamination equipped with a transverse measure. This notion generalizes simple closed curves.

The space of measured laminations on  $S$  is denoted by  $\mathcal{ML} = \mathcal{ML}(S)$ . Endowed with the weak\*-topology,  $\mathcal{ML}$  is homeomorphic to  $\mathbb{S}^{6g-7+2p} \times (0, +\infty) \simeq \mathbb{R}^{6g-6+2p} \setminus \{0\}$ , where the  $(0, +\infty)$ -component corresponds to the scaling of transverse measures on each geodesic lamination. Here, the unit sphere  $\mathbb{S}^{6g-7+2p}$  is identified with the space  $\mathcal{PML} = \mathcal{PML}(S)$  of *projective measured laminations* on  $S$ . The space  $\mathcal{PML}$  can be regarded as the boundary of  $\mathcal{T}$  as shown by Thurston ([Thu88], [Thu97]), and hence  $\mathcal{PML}$  is also referred to as the *Thurston boundary*.

Another interpretation of  $\mathcal{ML}$  is given by the celebrated theorem of Hubbard and Masur [HM79]. Fixing a Riemann surface structure  $x \in \mathcal{T}$ , the Hubbard–Masur theorem asserts that the space  $\mathcal{ML}$  is homeomorphic to the space of holomorphic quadratic differentials  $\mathcal{Q}(S, x)$  on the Riemann surface  $(S, x)$ . Each quadratic differential  $q \in \mathcal{Q}(S, x)$  determines a Teichmüller geodesic ray emanating from  $x \in \mathcal{T}$ . The projective class  $[\xi] \in \mathcal{PML}$  of the measured lamination  $\xi \in \mathcal{ML}$  corresponding to  $q$ , given by the Hubbard–Masur theorem, is an accumulation point of this Teichmüller geodesic ray.

The space  $\mathcal{ML}$  admits a natural  $\text{Mod}(S)$ -action, which encodes the global geometry and dynamics of the Teichmüller space. Indeed, the induced  $\text{Mod}(S)$ -action on  $\mathcal{PML}$  is a continuous extension of the  $\text{Mod}(S)$ -action on  $\mathcal{T}$ . Using the train track coordinates on  $\mathcal{ML}$ , or a natural symplectic structure on  $\mathcal{ML}$ , Thurston defined the  $\text{Mod}(S)$ -invariant measure

$$\mu_{\text{Th}} \quad \text{on} \quad \mathcal{ML}$$

which is in the Lebesgue measure class [Thu22]. The measure  $\mu_{\text{Th}}$  is now called the *Thurston measure*, and the  $\text{Mod}(S)$ -action on  $(\mathcal{ML}, \mu_{\text{Th}})$  is ergodic as shown by Masur [Mas85, Theorem 2]. The Thurston measure has been a central object in the study of the geometry and dynamics of  $\text{Mod}(S)$

and  $\mathcal{T}$ . For example, Mirzakhani [Mir08b], Athreya–Bufetov–Eskin–Mirzakhani [ABEM12], and Erlandsson–Souto [ES16] studied counting and equidistribution of  $\text{Mod}(S)$ -orbits using  $\mu_{\text{Th}}$ . It also played a major role in Mirzakhani’s ergodic theory of Thurston’s earthquake flow [Mir08a].

**1.1. Measure classification for full  $\text{Mod}(S)$ : finite covolume.** It is natural to ask whether the Thurston measure is the only measure on  $\mathcal{ML}$  that is suited to the study of the dynamics of  $\text{Mod}(S)$ . Lindenstrauss and Mirzakhani [LM08], and Hamenstädt [Ham09], independently classified all  $\text{Mod}(S)$ -invariant ergodic Radon measures on  $\mathcal{ML}$ . While the  $\text{Mod}(S)$ -action on  $\mathcal{ML}$  is not uniquely ergodic due to the cuspidal feature of  $\text{Mod}(S)$  or of  $\mathcal{M} = \text{Mod}(S) \setminus \mathcal{T}$ , the Thurston measure turns out to be the unique  $\text{Mod}(S)$ -invariant Radon measure on the region where the Teichmüller geodesic flow shows interesting dynamics.

More precisely, using the identification  $\mathcal{ML} = \mathcal{Q}(S, x)$  provided by the Hubbard–Masur theorem, we define the *recurrence locus* for  $\text{Mod}(S)$  as the set  $\mathcal{R}_{\text{Mod}(S)} \subset \mathcal{ML}$  of all measured laminations  $\xi \in \mathcal{ML}$  such that the Teichmüller geodesic ray determined by the corresponding quadratic differential  $q_\xi \in \mathcal{Q}(S, x)$  recurs to a compact subset in the quotient  $\mathcal{M} = \text{Mod}(S) \setminus \mathcal{T}$ . One of the main steps in the measure classification by both Lindenstrauss–Mirzakhani and Hamenstädt is as follows.

**Theorem 1.1** ([LM08], [Ham09]). *Let  $\mu$  be a  $\text{Mod}(S)$ -invariant Radon measure on  $\mathcal{R}_{\text{Mod}(S)}$ . Then  $\mu$  is a constant multiple of  $\mu_{\text{Th}}$ .*

In the exceptional case when  $S$  is a once-punctured torus or a 4-punctured sphere (i.e.,  $3g - 3 + p = 1$ ), homogeneous dynamics comes into play. In this case, the Teichmüller space is equal to the hyperbolic plane  $\mathbb{H}^2$  and  $\text{Mod}(S)$  is equal to  $\text{SL}(2, \mathbb{Z})$ . Then the  $\text{Mod}(S)$ -action on  $\mathcal{ML}$  corresponds to the  $\text{SL}(2, \mathbb{Z})$ -action on the *horospherical foliation* of the unit tangent bundle  $T^1 \mathbb{H}^2$ , or the *unipotent flow (or horocyclic flow)* on  $\text{PSL}(2, \mathbb{Z}) \backslash \text{PSL}(2, \mathbb{R})$ .

Furstenberg first proved that the unipotent flow on  $\Gamma \backslash \text{PSL}(2, \mathbb{R})$  is uniquely ergodic with respect to the Haar measure when  $\Gamma < \text{PSL}(2, \mathbb{R})$  is a uniform lattice [Fur73]. This was extended by Veech to uniform lattices in semi-simple Lie groups [Vee77]. Generalizing these results, Dani proved for lattices in reductive groups that the Haar measure is the unique invariant ergodic Radon measure on the recurrence locus ([Dan78], [Dan81]).

**1.2. Measure classification for subgroups: infinite covolume.** Given Theorem 1.1, one can seek an analogous measure classification on  $\mathcal{ML}$  for general subgroups of  $\text{Mod}(S)$ , as asked by Lindenstrauss and Mirzakhani [LM08, Remark 1.4(2)]. We aim to study this question in this paper.

Before we proceed, let us mention two important ingredients for both Lindenstrauss–Mirzakhani’s and Hamenstädt’s complete measure classification for the full  $\text{Mod}(S)$ -action. First, the  $\text{Mod}(S)$ -action has finite covolume with respect to the Masur–Veech volume form on the fiber bundle  $\mathcal{Q}^1 \mathcal{T}$  of unit-area holomorphic quadratic differentials over  $\mathcal{T}$  ([Mas82a], [Vee82]).

The bundle  $\mathcal{Q}^1\mathcal{T}$  can be regarded as the unit cotangent bundle over  $\mathcal{T}$ , and the Masur–Veech volume is also induced from the Thurston measure. Second, the Teichmüller horocyclic flow on the quotient bundle  $\text{Mod}(S)\backslash\mathcal{Q}^1\mathcal{T}$  exhibits some non-divergence, as proved by Minsky and Weiss [MW02].

On the other hand, the quotient  $\Gamma\backslash\mathcal{Q}^1\mathcal{T}$  has infinite volume for a general subgroup  $\Gamma < \text{Mod}(S)$ , and it is hard to expect the non-divergence of horocyclic flows on  $\Gamma\backslash\mathcal{Q}^1\mathcal{T}$ . Even in the case where  $S$  is a once-punctured torus or a 4-punctured sphere, when  $\mathcal{T}$  is the hyperbolic plane, neither holds true. These serve as obstructions to applying the arguments of [LM08] or [Ham09] to subgroups of  $\text{Mod}(S)$ .

On the hyperbolic plane  $\mathbb{H}^2$ , the first infinite-covolume examples studied for this problem are due to Burger [Bur90]. Burger considered a *convex cocompact* subgroup  $\Gamma < \text{PSL}(2, \mathbb{R})$  whose critical exponent is strictly greater than  $1/2$  and showed that there exists a unique  $\Gamma$ -invariant ergodic Radon measure on the recurrent horospherical foliation of  $\mathbb{H}^2$ . This was later generalized by Roblin [Rob03] for discrete groups of isometries of  $\text{CAT}(-1)$  spaces under assumptions on finite Bowen–Margulis–Sullivan measures and the existence of certain coverings.

Let us get back to the Teichmüller space  $\mathcal{T}$ , where tools from homogeneous dynamics or negatively curved geometry do not apply immediately. Before presenting our precise statements, which require the construction of a specific Radon measure, we summarize our measure classification results for non-elementary  $\Gamma < \text{Mod}(S)$ :

- (1) There exists at most one  $\Gamma$ -invariant Radon measure on  $\mathcal{ML}$  supported on the recurrence locus for  $\Gamma$ .
- (2) When  $\Gamma$  is of divergence type, the  $\Gamma$ -action on the recurrence locus is uniquely ergodic.<sup>1</sup>
- (3) When  $\Gamma$  is a convex cocompact subgroup, we classify all  $\Gamma$ -invariant Radon measures on  $\mathcal{ML}$ .

In this paper, we develop machinery for a general metric space with partial hyperbolicity, without any assumption on its global geometry. We use them to deduce versions of (1) and (2) in that setting.

**1.3. Main statements.** We mainly consider a *non-elementary* subgroup  $\Gamma < \text{Mod}(S)$ , i.e.,  $\Gamma$  is not virtually cyclic and contains a pseudo-Anosov mapping class. There exists  $0 < \delta_\Gamma < +\infty$ , called the critical exponent of  $\Gamma$ , which is the abscissa of convergence of the Poincaré series  $s \mapsto \sum_{g \in \Gamma} e^{-sd(x, gx)}$ , for (any)  $x \in \mathcal{T}$ . The finiteness of the critical exponent is due to Kaimanovich and Masur [KM96], and its positivity is due to McCarthy [McC85]. We say that  $\Gamma$  is of *divergence type* if the Poincaré series diverges at  $s = \delta_\Gamma$ . Otherwise,  $\Gamma$  is said to be of convergence type.

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<sup>1</sup>By unique ergodicity, we mean that there exists a unique ergodic invariant Radon measure up to a constant multiple.

Generalizing the ergodicity of the  $\text{Mod}(S)$ -action on  $(\mathcal{ML}, \mu_{\text{Th}})$  due to Masur [Mas85], we construct a Radon measure for non-elementary subgroups of  $\text{Mod}(S)$  and show its ergodicity.

**Theorem 1.2** (Ergodicity). *Let  $\Gamma < \text{Mod}(S)$  be a non-elementary subgroup of divergence type. Then there exists a nonzero,  $\Gamma$ -invariant Radon measure  $\mu_\Gamma$  on  $\mathcal{ML}$  such that*

*the  $\Gamma$ -action on  $(\mathcal{ML}, \mu_\Gamma)$  is ergodic.*

Theorem 1.2 also applies to certain normal subgroups of a divergence-type group, which are not necessarily of divergence type. See Theorem 8.2 and Section 9.2.

The measure  $\mu_\Gamma$  is very explicit. Delaying its construction, we first discuss unique ergodicity, which is the main result of this paper. Similarly to  $\mathcal{R}_{\text{Mod}(S)}$ , we define the *recurrence locus*  $\mathcal{R}_\Gamma \subset \mathcal{ML}$  for a subgroup  $\Gamma < \text{Mod}(S)$  as follows: fixing  $x \in \mathcal{T}$ ,

$$\mathcal{R}_\Gamma := \left\{ \xi \in \mathcal{ML} : \begin{array}{l} \text{Teichmüller geodesic ray given by } q_\xi \in \mathcal{Q}(S, x) \\ \text{recurs to a compact subset in } \Gamma \backslash \mathcal{T} \end{array} \right\}$$

where  $q_\xi \in \mathcal{Q}(S, x)$  is the quadratic differential on the Riemann surface  $(S, x)$  corresponding to  $\xi \in \mathcal{ML}$ , given by the Hubbard–Masur theorem. This set is  $\Gamma$ -invariant and does not depend on the choice of  $x \in \mathcal{T}$ . We construct  $\mu_\Gamma$  so that it is  $\Gamma$ -invariant and is supported on the recurrence locus  $\mathcal{R}_\Gamma$ . Moreover, the  $\Gamma$ -action on  $(\mathcal{R}_\Gamma, \mu_\Gamma)$  is uniquely ergodic.

**Theorem 1.3** (Unique ergodicity). *Let  $\Gamma < \text{Mod}(S)$  be a non-elementary subgroup. Suppose that there exists a nonzero,  $\Gamma$ -invariant Radon measure  $\mu$  on  $\mathcal{ML}$  that is supported on  $\mathcal{R}_\Gamma$ . Then  $\Gamma$  is of divergence type and*

*$\mu$  is a constant multiple of  $\mu_\Gamma$ .*

*In other words,*

- (1) *if  $\Gamma$  is of convergence type, then there does not exist a nonzero,  $\Gamma$ -invariant Radon measure on  $\mathcal{R}_\Gamma$ .*
- (2) *if  $\Gamma$  is of divergence type, then*

*the  $\Gamma$ -action on  $(\mathcal{R}_\Gamma, \mu_\Gamma)$  is uniquely ergodic.*

The full mapping class group  $\text{Mod}(S)$  is of divergence type thanks to [Mas82a] and [Vee82]. It contains many independent pseudo-Anosov elements (cf. [Thu88]), so it is non-elementary. Thus, Theorem 1.3 extends Theorem 1.1.

A subgroup  $\Gamma < \text{Mod}(S)$  is said to have a *non-arithmetic* length spectrum if the Teichmüller (stable) translation lengths of pseudo-Anosov elements of  $\Gamma$  generate a dense additive subgroup of  $\mathbb{R}$ . As in many other measure classification results (e.g. [Rob03]), non-arithmeticity of the length spectrum is an essential ingredient in the classification. Indeed, such a measure classification does not hold without non-arithmeticity in general, as explained

in Remark 1.14. For this reason, we prove the non-arithmeticity of the Teichmüller length spectrum of any non-elementary subgroup of  $\text{Mod}(S)$ .

**Theorem 1.4** (Non-arithmeticity). *Let  $\Gamma < \text{Mod}(S)$  be a non-elementary subgroup. Then  $\Gamma$  has non-arithmetic length spectrum.*

The length spectrum can be similarly defined for any isometric action on a metric space. Its non-arithmeticity is a fundamentally important property from many geometric and dynamical viewpoints. Indeed, non-arithmeticity is a necessary ingredient for mixing, equidistribution, counting results for geodesic flows, dynamics of horospherical foliations, and measure classification of horospherical-invariant measures. In general negatively curved settings, these geometry and dynamics were studied in ([Dal00], [Rob03], etc.).

Apart from the full mapping class group  $\text{Mod}(S)$  which contains a Veech group acting on a Teichmüller disk as a non-elementary Fuchsian group, Theorem 1.4 is the first non-arithmeticity result for Teichmüller length spectra of subgroups of  $\text{Mod}(S)$ .

*Remark 1.5.* We also note that, to the best of the authors' knowledge, it is an open problem whether the length spectrum of a non-elementary discrete subgroup  $\Gamma < \text{Isom}(X)$  is always non-arithmetic when  $X$  is a simply connected Riemannian manifold with curvature at most  $-1$ . See Section 1.5 for some known cases.

As an application of our non-arithmeticity theorem (Theorem 1.4), we prove that the Teichmüller geodesic flow is topologically mixing for a non-elementary subgroup  $\Gamma < \text{Mod}(S)$ . Recall that  $\mathcal{Q}^1\mathcal{T}$  is the bundle of unit-area (holomorphic) quadratic differentials over  $\mathcal{T}$ . Then the  $\text{Mod}(S)$ -action on  $\mathcal{T}$  naturally extends to  $\mathcal{Q}^1\mathcal{T}$ , and moreover  $\mathcal{Q}^1\mathcal{T}$  is identified with the unit cotangent bundle over  $\mathcal{T}$ , via a  $\text{Mod}(S)$ -equivariant map. In this regard, the Teichmüller geodesic flow is naturally defined on  $\mathcal{Q}^1\mathcal{T}$  and descends to the quotient  $\Gamma \backslash \mathcal{Q}^1\mathcal{T}$  for any subgroup  $\Gamma < \text{Mod}(S)$ .

In particular, the quotient  $\text{Mod}(S) \backslash \mathcal{Q}^1\mathcal{T}$  is identified with the cotangent bundle over the moduli space  $\text{Mod}(S) \backslash \mathcal{T}$ . Dynamical properties of the Teichmüller geodesic flow on  $\text{Mod}(S) \backslash \mathcal{Q}^1\mathcal{T}$ , such as mixing results, turned out to play a significant role in the study of counting problems for the mapping class group  $\text{Mod}(S)$  ([EM11], [ABEM12], [EMR19], [EMM22], [AH23], etc.).

Unless  $\Gamma$  is the full mapping class group, interesting dynamics only occurs in a certain subset of  $\Gamma \backslash \mathcal{Q}^1\mathcal{T}$  in general. To describe this subset, we note that there exists a unique  $\Gamma$ -minimal subset  $\Lambda(\Gamma) \subset \mathcal{PMF}$ , called the limit set of  $\Gamma$  [MP89]. We then consider the subset

$$\Omega_\Gamma \subset \Gamma \backslash \mathcal{Q}^1\mathcal{T}$$

of quadratic differentials whose two associated projective measured foliations belong to  $\Lambda(\Gamma)$ . Then  $\Omega_\Gamma$  is a non-wandering domain for the Teichmüller geodesic flow, and we indeed prove its topological mixing.

**Theorem 1.6** (Topological mixing). *Let  $\Gamma < \text{Mod}(S)$  be a non-elementary subgroup. Then the Teichmüller geodesic flow on  $\Omega_\Gamma \subset \Gamma \backslash \mathcal{Q}^1\mathcal{T}$  is topologically mixing.*

We note that topological mixing does not require  $\Gamma$  to be of divergence type.

**Example 1.7** (Divergence-type subgroups). Examples of divergence-type subgroups of  $\text{Mod}(S)$  include (1) the full group  $\text{Mod}(S)$  and (2) convex cocompact subgroups as discussed in more detail below.

(3) Stabilizers of many Teichmüller disks are another source of subgroups of divergence type, which are also referred to as Veech groups. To give explicit examples of pseudo-Anosovs, Thurston considered in [Thu88] the subgroup  $\Gamma < \text{Mod}(S)$  that is generated by multitwists about a filling pair of multicurves on  $S$ . He showed that  $\Gamma$  stabilizes a Teichmüller disk, which is an isometrically embedded copy of  $\mathbb{H}^2$  in  $\mathcal{T}$ . In particular,  $\Gamma$  is a geometrically finite Fuchsian group and is of divergence type with respect to the Teichmüller metric [Pat76]. Later, Veech constructed many other families of lattice stabilizers of Teichmüller disks [Vee89]. Note that these subgroups contain multitwists and are not convex cocompact, and also that any finitely generated Veech group is of divergence type. As Teichmüller disks are much smaller than the ambient Teichmüller space, they are “infinite-covolume subgroups.”

(4) Given a collection  $\mathcal{C} = \{f_1, f_2, \dots\} \subset \text{Mod}(S)$  of independent pseudo-Anosovs, a ping-pong argument using the contracting properties of  $f_i$ ’s guarantees the following: for suitably large  $n_i \in \mathbb{N}$  for each  $i$ ,  $\langle f_i^{n_i} : f_i \in \mathcal{C} \rangle$  becomes a free subgroup of  $\text{Mod}(S)$  (cf. [FM02], [Mos02]). When  $\mathcal{C}$  is finite, the resulting subgroup is convex cocompact. When  $\mathcal{C}$  is infinite, one can choose  $n_i$ ’s carefully so that the resulting subgroup is of divergence type, which is not convex cocompact since it is not finitely generated.

(5) Moreover, many divergence-type subgroups can be constructed using coverings between surfaces [BS24].

(6) Being of divergence type is also related to the growth of orbits. Concretely, when a non-elementary subgroup  $\Gamma < \text{Mod}(S)$  has *purely exponential growth*, i.e.,  $\#\{g \in \Gamma : d(x, gx) \leq R\} \asymp e^{\delta_\Gamma R}$  as  $R \rightarrow +\infty$ , for (any)  $x \in \mathcal{T}$ , it is easy to see that  $\Gamma$  is of *divergence type*. This is indeed the case for  $\text{Mod}(S)$  [ABEM12, Theorem 1.2] and for convex cocompact subgroups [Coo93, Théorème 7.2] (cf. [Gek13, Theorem 1.1]).

Purely exponential growth also motivated the work of Schapira and Tapie [ST21], and of Yang [Yan19], on strongly positively recurrent (SPR) groups, also known as statistically convex cocompact (SCC) groups. These groups are known to have purely exponential growth [Yan19, Theorem B] (cf. [ST21, Theorem 7.26]), and therefore they are of *divergence type*. Yang also constructed many examples of SCC subgroups in [Yan19]. We will not define the notion of SPR(=SCC) groups, but let us mention that there are

non-convex-cocompact examples which are SCC [Yan19, Proposition 6.6]. Our theory applies to these examples as well.

We now come back to our discussion of measure classifications. Farb and Mosher [FM02] introduced another important family of subgroups of divergence type. A finitely generated  $\Gamma < \text{Mod}(S)$  is called *convex cocompact* if it has a quasi-convex orbit in  $\mathcal{T}$ . We establish a measure classification for such  $\Gamma$  on the *entire* space of measured laminations.

**Theorem 1.8** (Convex cocompact subgroups). *Let  $\Gamma < \text{Mod}(S)$  be a non-elementary convex cocompact subgroup. Then every  $\Gamma$ -invariant ergodic Radon measure on  $\mathcal{ML}$  is either*

$$\mu_\Gamma \quad \text{or} \quad \sum_{g \in \Gamma} D_{g \cdot \xi} \text{ for some } \xi \in \mathcal{ML} \setminus \mathcal{R}_\Gamma$$

*up to scaling, where  $D_{g \cdot \xi}$  is the Dirac measure at  $g \cdot \xi \in \mathcal{ML} \setminus \mathcal{R}_\Gamma$ .*

In Theorem 10.1, we also obtain the classification of  $\Gamma$ -orbit closures in  $\mathcal{ML}$ . This is equivalent to considering the quotient  $\Gamma \backslash \mathcal{Q}^1 \mathcal{T}$  and classifying stable manifolds under the Teichmüller geodesic flow.

Our complete measure classification for convex cocompact  $\Gamma$  is based on an explicit understanding of the  $\Gamma$ -action on  $\mathcal{ML} \setminus \mathcal{R}_\Gamma$ . It would be of independent interest to understand such an action for a general subgroup, which is also relevant to the structure of limit sets in the Thurston boundary  $\mathcal{PM}\mathcal{L}$ .

*Remark 1.9.* We note that Theorem 1.8 can also be studied using the finiteness of the Bowen–Margulis–Sullivan (BMS) measures for convex cocompact subgroups due to Gekhtman [Gek13], and their strong geometric properties deduced from quasi-convexity of orbits in  $\mathcal{T}$ . Indeed, Roblin provided in [Rob03] a strategy based on finite BMS measures and certain geometric conditions. The authors thank Gekhtman and Oh for sharing this viewpoint with them. It would be interesting to pursue this alternative approach and see what byproducts it yields.

**1.4. Explicit construction of invariant Radon measures.** We now explain the construction of the measure  $\mu_\Gamma$  for non-elementary  $\Gamma < \text{Mod}(S)$ .

Fixing a basepoint  $x \in \mathcal{T}$ , a Borel probability measure  $\nu$  on  $\mathcal{PM}\mathcal{L}$  is called a  $\delta_\Gamma$ -dimensional conformal measure of  $\Gamma$  if for every  $g \in \Gamma$ ,

$$\frac{dg_* \nu}{d\nu}([\xi]) = \left( \sqrt{\frac{\text{Ext}_x(\xi)}{\text{Ext}_{gx}(\xi)}} \right)^{\delta_\Gamma} \quad \text{for a.e. } [\xi] \in \mathcal{PM}\mathcal{L}$$

where  $\text{Ext}_x(\xi)$  denotes the extremal length of  $\xi \in \mathcal{ML}$  on the Riemann surface  $(S, x)$  (see Equation (3.1) for the precise definition). Such a measure was constructed by Athreya–Bufetov–Eskin–Mirzakhani [ABEM12] for  $\Gamma = \text{Mod}(S)$ , by Gekhtman [Gek13] for  $\Gamma$  convex cocompact, and by Coulon [Cou24] and by Yang [Yan24] for general  $\Gamma$ .

Once we fix a basepoint  $x \in \mathcal{T}$ , we can explicitly write the homeomorphism  $\mathcal{ML} \rightarrow \mathcal{PML} \times (0, +\infty)$  using the Hubbard–Masur theorem as follows:

$$\xi \mapsto ([\xi], \sqrt{\text{Ext}_x(\xi)}).$$

For a non-elementary subgroup  $\Gamma < \text{Mod}(S)$  and a  $\delta_\Gamma$ -dimensional conformal measure  $\nu$  of  $\Gamma$  on  $\mathcal{PML}$ , we define the Radon measure  $\mu_\nu$  on  $\mathcal{ML}$  by

$$(1.1) \quad d\mu_\nu(\xi) := \left(\sqrt{\text{Ext}_x(\xi)}\right)^{\delta_\Gamma-1} \cdot d\nu([\xi]) d\text{Leb}_{\mathbb{R}}\left(\sqrt{\text{Ext}_x(\xi)}\right).$$

It follows from the conformality of  $\nu$  that  $\mu_\nu$  is  $\Gamma$ -invariant. Moreover, while the conformal measure  $\nu$  depends on the choice of  $x \in \mathcal{T}$ , the measure  $\mu_\nu$  is independent of  $x$ .

When  $\Gamma < \text{Mod}(S)$  is a non-elementary subgroup of divergence type, Coulon [Cou24] and Yang [Yan24] showed that the  $\delta_\Gamma$ -dimensional conformal measure  $\nu$  of  $\Gamma$  is unique. Hence, we write

$$\mu_\Gamma := \mu_\nu \quad \text{on} \quad \mathcal{ML}$$

in this case. Furthermore,  $\mu_\Gamma$  is supported on the recurrence locus  $\mathcal{R}_\Gamma$ .

**1.5. Horospherical foliations of CAT(−1) spaces.** Before discussing the proof, we present an application to CAT(−1) spaces.

Let  $X$  be a proper geodesic CAT(−1) space. Following [Rob03], the space  $SX$  of isometric embeddings  $\mathbb{R} \rightarrow X$  serves as the unit tangent bundle over  $X$ . It admits a geodesic flow  $SX \curvearrowright \mathbb{R}$ , which we denote by  $a_t : SX \rightarrow SX$  for  $t \in \mathbb{R}$ . For  $u \in SX$ , its stable horosphere (or stable manifold) is the set  $H^-(u)$  of  $v \in SX$  such that the distance between  $ua_t$  and  $va_t$  tends to 0 as  $t \rightarrow +\infty$ . Stable horospheres form a foliation on  $SX$ , called the *horospherical foliation* of  $X$ . The space of its leaves is denoted by  $\mathcal{H}$ , which admits a natural action of the isometry group  $\text{Isom}(X)$ .

Roblin studied measure classification for discrete subgroups of  $\text{Isom}(X)$ , acting on  $\mathcal{H}$ . The notions of non-elementary and divergence-type subgroups are defined analogously. For a discrete subgroup  $\Gamma < \text{Isom}(X)$ , its recurrence locus  $\mathcal{R}_\Gamma \subset \mathcal{H}$  is also defined similarly:

$$\mathcal{R}_\Gamma := \{H^-(u) \in \mathcal{H} : ua_t \text{ recurs to a compact subset in } \Gamma \backslash X \text{ as } t \rightarrow +\infty\}.$$

We say that  $\Gamma$  has *non-arithmetic* length spectrum if the (stable) translation lengths of elements in  $\Gamma$  generate a dense additive subgroup of  $\mathbb{R}$ . Examples include the following cases: let  $\Gamma < \text{Isom}(X)$  be non-elementary and denote by  $\Lambda(\Gamma) \subset \partial X$  its limit set, the set of accumulation points of a  $\Gamma$ -orbit in  $X$ .

- (1)  $\Gamma \backslash X$  is a negatively curved surface [Dal99],
- (2)  $X$  is a rank-one symmetric space [Kim06].
- (3)  $\Lambda(\Gamma)$  possesses a non-singleton connected component [Bou95].
- (4)  $\Gamma$  contains a parabolic isometry [DP98].

**Theorem 1.10** (CAT(−1) spaces). *Let  $X$  be a proper geodesic CAT(−1) space and  $\Gamma < \text{Isom}(X)$  a non-elementary discrete subgroup. Suppose that*

- (1)  $\Gamma$  is of divergence type, and
- (2)  $\Gamma$  has non-arithmetic length spectrum.

*Then*

*the  $\Gamma$ -action on  $\mathcal{R}_\Gamma$  is uniquely ergodic.*

An analogous problem was extensively studied for rank-one symmetric spaces; see Burger [Bur90], Roblin [Rob03], Sarig [Sar04], Ledrappier–Sarig [LS07], Landesberg–Lindenstrauss [LL22], Landesberg–Lee–Lindenstrauss–Oh [LLLO23] etc. For a general CAT(−1) space, it was proved by Roblin [Rob03] under the stronger assumptions that  $\Gamma$  has a finite Bowen–Margulis–Sullivan measure and that a certain covering exists. These unique invariant measures are called the Burger–Roblin measures in these settings.

Our novelty in this CAT(−1) setting is the removal of these finiteness and covering assumptions, which was possible because we do not rely on mixing of continuous flows or certain dynamical systems with finite measures.

In [CK26], we extend our techniques further to classify horospherical invariant measures on higher-rank homogeneous spaces of infinite volume. This resolves open problems proposed by Landesberg–Lee–Lindenstrauss–Oh [LLLO23] and by Oh [Oh]. Such a higher-rank homogeneous space is CAT(0) and the plethora of higher-dimensional flats rules out negatively curved geometry, even partial. Still, our techniques developed in this paper can be extended further in that setting by employing more geometric aspects of certain infinite covolume discrete subgroups of Lie groups.

**1.6. More examples.** Let us mention one interesting family of examples, in addition to the mapping class groups and CAT(−1) groups. In [GS19], Genevois and Stocker considered a geodesically complete proper CAT(0) space  $X$  containing a point that has a CAT(−1) neighborhood, and a non-elementary subgroup  $\Gamma < \text{Isom}(X)$  acting geometrically on  $X$ . They proved that  $\Gamma$  contains a contracting isometry whose quasi-axis passes through the aforementioned CAT(−1) neighborhood several times. In fact, this contracting isometry is squeezing (see below for the definition). Meanwhile, since  $\Gamma$  acts non-elementarily and geometrically on  $X$ , it is necessarily of divergence type [Yan19, Theorem B].

**1.7. Metric spaces with squeezing isometries.** Our proofs of Theorem 1.2, Theorem 1.3, and Theorem 1.10 are based on the special geometric feature of the axis of a pseudo-Anosov mapping class and the axis of a loxodromic isometry on a CAT(−1) space, which we call the *squeezing property* in this paper.

In the rest of the introduction, let  $(X, d)$  be a proper geodesic metric space. For a geodesic  $\gamma \subset X$ , we denote by  $\pi_\gamma(\cdot)$  the *nearest-point projection* map onto  $\gamma$ . We consider two notions of hyperbolicity for  $\gamma$ .

- We say that a geodesic  $\gamma \subset X$  is *contracting* if there exists  $K > 0$  such that, for every geodesic  $\eta \subset X$ , points that are  $K$ -deep in the convex hull of  $\pi_\gamma(\eta)$  are  $K$ -close to  $\eta$ .
- We say that a geodesic  $\gamma \subset X$  is *squeezing* if for each  $\epsilon > 0$  there exists  $K = K(\epsilon) > 0$  such that, for every geodesic  $\eta \subset X$ , points that are  $K$ -deep in the convex hull of  $\pi_\gamma(\eta)$  are  $\epsilon$ -close to  $\eta$ .

We say that an isometry  $g \in \text{Isom}(X)$  is contracting (squeezing, resp.) if it admits a contracting (squeezing, resp.) axis. The squeezing property can be regarded as a quantitative version of the contracting property. See Figure 1 for a rough sketch, and Section 5 for precise definitions.

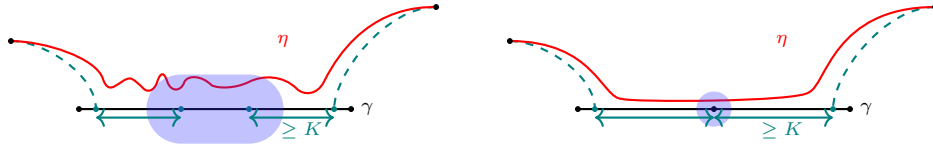


FIGURE 1. A contracting geodesic (left) and a squeezing geodesic (right)

It is well known that every geodesic in a  $\text{CAT}(-1)$  space is squeezing. Hence, every loxodromic isometry is squeezing there. Furthermore, in the mapping class group viewed as the isometry group of Teichmüller space, every pseudo-Anosov mapping class is contracting due to Minsky [Min96], and moreover, it is squeezing. See Section 4.1 for further context.

We prove the aforementioned results for Teichmüller spaces and  $\text{CAT}(-1)$  spaces by studying ergodic theory under the presence of squeezing isometries. We mainly consider a *non-elementary* subgroup  $\Gamma < \text{Isom}(X)$ , i.e., a non-virtually-cyclic subgroup acting properly on  $X$  that contains a contracting isometry. The notion of divergence type is defined analogously to the one for subgroups of  $\text{Mod}(S)$ . We say that the *squeezing spectrum* of  $\Gamma$

$$\text{Spec}_{sq}(\Gamma) := \{\text{translation length of } g : g \in \Gamma \text{ is squeezing}\} \subset \mathbb{R}$$

is *non-arithmetic* if  $\text{Spec}_{sq}(\Gamma)$  generates a dense additive subgroup of  $\mathbb{R}$ .

It remains to define an object corresponding to the space of measured laminations on a surface and to the horospherical foliation of a  $\text{CAT}(-1)$  space. We consider the product space <sup>2</sup>

$$\mathcal{H} := \partial^h X \times \mathbb{R}$$

where  $\partial^h X$  is the horofunction boundary of  $X$ . Each point in  $\partial^h X$  is identified with a 1-Lipschitz cocycle on  $X$ , and this induces a natural  $\text{Isom}(X)$ -action on the space  $\mathcal{H}$ : the action on the  $\mathbb{R}$ -component is given by the cocycle determined by the  $\partial^h X$ -component. See Section 5.6 for details.

Theorem 1.2 will follow from the following more general statement.

<sup>2</sup>We use the same notation as in the  $\text{CAT}(-1)$  case, because this is precisely the horospherical foliation when  $X$  is  $\text{CAT}(-1)$ .

**Theorem 1.11** (Ergodicity). *Let  $(X, d)$  be a proper geodesic metric space and let  $\Gamma < \text{Isom}(X)$  be a non-elementary subgroup. Suppose that*

- (1)  $\Gamma$  is of divergence type, and
- (2) the squeezing spectrum  $\text{Spec}_{sq}(\Gamma)$  is non-arithmetic.

*Then there exists a nonzero,  $\Gamma$ -invariant Radon measure  $\mu_\Gamma$  on  $\mathcal{H}$  such that*

*the  $\Gamma$ -action on  $(\mathcal{H}, \mu_\Gamma)$  is ergodic.*

*Remark 1.12.* As with Theorem 1.2, our ergodicity result applies to normal subgroups of a divergence-type group, which are not necessarily of divergence type (Theorem 8.2).

As in the setting of Teichmüller spaces, the measure  $\mu_\Gamma$  is explicitly constructed, and it turns out that  $\mu_\Gamma$  is supported on the region corresponding to the recurrence locus. This region is given by

$$\Lambda_c(\Gamma) \times \mathbb{R} \subset \mathcal{H}$$

where  $\Lambda_c(\Gamma) \subset \partial^h X$  is the *conical limit set* of  $\Gamma$  (Definition 5.17). Delaying the construction of  $\mu_\Gamma$ , we state the unique ergodicity theorem, from which Theorem 1.3 and Theorem 1.10 follow.

**Theorem 1.13** (Unique ergodicity). *Let  $(X, d)$  be a proper geodesic metric space and let  $\Gamma < \text{Isom}(X)$  be a non-elementary subgroup such that  $\text{Spec}_{sq}(\Gamma)$  is non-arithmetic. Suppose that there exists a nonzero,  $\Gamma$ -invariant Radon measure  $\mu$  on  $\mathcal{H}$  supported on  $\Lambda_c(\Gamma) \times \mathbb{R}$ . Then  $\Gamma$  is of divergence type and*

*$\mu$  is a constant multiple of  $\mu_\Gamma$ .*

*In other words,*

- (1) *if  $\Gamma$  is of convergence type, then there does not exist a nonzero,  $\Gamma$ -invariant Radon measure on  $\Lambda_c(\Gamma) \times \mathbb{R}$ .*
- (2) *if  $\Gamma$  is of divergence type, then*

*the  $\Gamma$ -action on  $(\Lambda_c(\Gamma) \times \mathbb{R}, \mu_\Gamma)$  is uniquely ergodic.*

*Remark 1.14.* We note that non-arithmeticity of the squeezing spectrum is essential. In general, the horospherical foliation is not uniquely ergodic without non-arithmeticity: there are uncountably many mutually singular invariant measures on the horospherical foliation of a standard Cayley graph of a free group.

For non-elementary subgroups of mapping class groups, non-arithmeticity of their Teichmüller length spectra (Theorem 1.4) is proved in Appendix A, and we also observe that pseudo-Anosov mapping classes are squeezing (Proposition 4.1). In this sense, we use not only the “partial hyperbolicity” of  $\text{Mod}(S)$  but also certain features of  $\text{Mod}(S)$  shared with Zariski dense subgroups of Lie groups.

**1.8. On the proof of the measure classification.** A key point of our argument is to show the relationship between  $\Gamma$ -invariant measures on the horospherical foliation and  $\Gamma$ -conformal measures on the horoboundary. The latter measures always give the former; our main contribution is proving that the former measures are always given by the latter, and that the former ones are ergodic. The proof proceeds in four steps; the two main contributions are Steps (1) and (4), while Steps (2) and (3) are based on previous results. We describe them for the action of  $\Gamma < \text{Mod}(S)$  on  $\mathcal{ML}$ .

- (1) There is a natural one-parameter flow  $\{T_a\}_{a \in \mathbb{R}}$  on  $\mathcal{ML}$  such that  $T_a$  scales each measured lamination by  $e^a$ . Suppose now that there exists an ergodic  $\Gamma$ -invariant Radon measure  $\mu$  on the space  $\mathcal{R}_\Gamma$  of  $\Gamma$ -recurrent measured laminations. We prove that  $\mu$  is then *quasi-invariant* under the flow  $\{T_a\}_{a \in \mathbb{R}}$ , namely,  $T_a^* \mu$  and  $\mu$  are mutually absolutely continuous for any  $a \in \mathbb{R}$ . We prove the quasi-invariance in Section 7, and this is the main argument of this paper.
- (2) Once we have the quasi-invariance as in Step (1), it follows that  $\mu$  is the product of a conformal measure  $\nu$  of  $\Gamma$  on  $\mathcal{PM}\mathcal{L}$  and the Lebesgue measure on  $(0, +\infty)$ , under the identification  $\mathcal{ML} = \mathcal{PM}\mathcal{L} \times (0, +\infty)$ , as in Equation (1.1). This is based on the work [ANSS02] for skew-product dynamical systems, which was developed further in the negatively-curved setting in [Sar04]. For the Teichmüller space, this strategy was first implemented by Hamenstädt [Ham09] using train tracks.
- (3) After Step (2), we deduce the essential uniqueness of  $\mu$  from the essential uniqueness of the conformal measure  $\nu$ . We also deduce that  $\Gamma$  is of divergence type. This is based on a recent version of Patterson–Sullivan theory due to Coulon [Cou24] and Yang [Yan24].
- (4) Conversely, we begin with  $\Gamma$  of divergence type and the measure  $\mu$  of the form in Equation (1.1). Then we prove that  $\mu$  is  $\Gamma$ -ergodic. As shown by Schmidt [Sch77], the two main ingredients for the ergodicity of  $\mu$  are the ergodicity of  $\nu$  and the fullness of the essential subgroup  $E_\nu(\Gamma)$  for  $\Gamma$  and  $\nu$  (see Definition 8.3). The former ingredient is given by Coulon [Cou24] and Yang [Yan24]. Our contribution towards the ergodicity is about the latter ingredient, showing that the essential subgroup is full. To do this, we construct an inductive partition of a given compact subset of  $\mathcal{ML}$  satisfying certain measure conditions. This technical construction is the content of Lemma 8.5.

Our approach is (coarse) geometric, and this strategy is completely different from works of Lindenstrauss–Mirzakhani [LM08] and of Hamenstädt [Ham09] using dynamical properties of geodesic and unipotent flows. This also applies to  $\text{CAT}(-1)$  spaces and general metric spaces with squeezing isometries, as discussed above. While special cases of  $\text{CAT}(-1)$  spaces were studied via a more dynamical approach in many literature, for instance

[Rob03], our strategy in this paper provides a different argument which is applicable in general, not only for those special cases.

Let us describe some more details about Step (1). In Theorem 7.5, we first show that  $\mu$  is in fact supported on a smaller subspace  $\mathcal{S}_\Gamma$  of  $\mathcal{R}_\Gamma$  which fibers over the squeezed limit set we introduce in Definition 7.6. Intuitively, along the Teichmüller geodesics towards measured laminations in  $\mathcal{S}_\Gamma$ , geodesics are more and more “squeezed.” This is a genuinely new step required for Teichmüller space. Indeed, this step does not appear in previous work on CAT(−1) or homogeneous settings using dynamics of flows.

We then prove the quasi-invariance of  $\mu$  under  $T_a$  for various  $a \in \mathbb{R}$ . For this, we use the squeezing property of pseudo-Anosov mapping classes and a fine control of translations along pseudo-Anosov axes. In addition, we need to construct a nice covering of  $\mathcal{S}_\Gamma$  into “almost disjoint” basic open sets. This construction is another of our main contributions, and is outlined in Theorem 7.10. This covering is also crucially used in Step (4), when we show that  $E_\nu(\Gamma) = \mathbb{R}$ .

In contrast to previous literature which heavily uses dynamics of continuous flows, we do not make use of dynamical properties (ergodic theorems, mixing, equidistribution, etc.) of the geodesic/unipotent flows on the Teichmüller space, or the theory of train tracks and the curve complex. Instead, we employ a geometric approach based on the contracting property of axes of pseudo-Anosov mapping classes due to Minsky [Min96], and our observation that the contracting property can be enhanced to the *squeezing* property (Proposition 4.1). Because of this, we can deal with *general non-elementary subgroups* despite the lack of homogeneity or negative curvature of the Teichmüller space.

Finally, we emphasize that the current techniques can be extended further to yield horospherical measure classification results for certain infinite-covolume higher-rank discrete subgroups. See [CK26] for details.

**1.9. On the proof of non-arithmeticity.** To prove Theorem 1.4, we introduce the notion of cross-ratios on the space  $\mathcal{MF}$  of measured foliations and the space  $\mathcal{PMF}$  of projective measured foliations on  $S$ , and study their geometric and dynamical properties. See Definition A.2 for precise definitions of cross-ratios.

Showing the non-arithmeticity using cross-ratios was already done by Kim<sup>3</sup> [Kim06] for rank-one symmetric spaces. For instance in real hyperbolic case, Kim proved that if a discrete subgroup of isometries does not have non-arithmetic length spectrum, then its limit set is contained in a discrete union of proper spheres in the boundary. This can also be seen as a consequence of negatively curved geometry and the conformality of the boundary action. Kim then deduced that the discrete subgroup cannot be non-elementary, based on a structure theory of rank-one Lie groups.

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<sup>3</sup>Inkang Kim, not the second-named author.

In contrast, in the setting of Teichmüller geometry, we do not have a negatively curved geometry, homogeneity of the space, or conformality of the boundary action. To overcome this difficulty, after we introduce the notion of cross-ratios on  $\mathcal{MF}$  and  $\mathcal{PMF}$ , we also observe useful relationships between cross-ratios and geometric and dynamical properties of limit sets. Moreover, we further investigate the piecewise-linear structure of  $\mathcal{MF}$  and localize dynamics by using the theory of train tracks, and then complete the proof of the non-arithmeticity by proceeding with certain inductive arguments.

*Remark 1.15.* We remark that an approach to proving the non-arithmeticity of Teichmüller length spectra (Theorem 1.4) via train tracks and reduction to linear algebra was attempted in an earlier version [GM23a] of the work of Gekhtman–Ma [GM23b] as well as in Gekhtman’s UChicago Ph.D. thesis [Gek14], but contains a mistake; the argument was purported to work even for non-elementary semigroups of mapping class groups. In this current paper, we take a different approach, as described above.

**1.10. Non-geodesic spaces.** There is an important class of metric spaces that we will not discuss in detail. Let  $\Gamma$  be a non-elementary relatively hyperbolic group. Let  $m$  be a finitely supported, admissible, symmetric probability measure on  $\Gamma$ . There is a left-invariant metric  $d_m$  on  $\Gamma$  associated with  $m$ , called the *Green metric*. This metric is quasi-isometric to the word metric and is Gromov hyperbolic, but is not geodesic in general. Nonetheless, it is *roughly geodesic*. In a sense, every loxodromic element of  $\Gamma$  is a *squeezing* isometry with respect to  $d_m$ , even if it does not possess an axis. Hence, under the non-arithmeticity assumption, we expect that there exists a unique nonzero,  $\Gamma$ -invariant ergodic Radon measure on the  $d_m$ -horospherical foliation. This unique measure is in the measure class of  $(m\text{-stationary measure}) \times \text{Leb}_{\mathbb{R}}$ .

In order to apply our theory to Green metrics, one needs to replace geodesics with rough geodesics. We leave this for future work.

**1.11. Organization.** In Section 2, we present a brief overview of the Teichmüller space, especially its boundary. We define the Hubbard–Masur coordinates for the space of measured laminations in Section 3. In Section 4, we discuss some aspects of pseudo-Anosov mapping classes, including contracting and squeezing properties of their axes. We discuss the geometry of a general metric space with contracting and squeezing isometries in Section 5. Section 6 is devoted to the Patterson–Sullivan theory with squeezing isometries. In Section 7, we prove the unique ergodicity (Theorem 1.13). The ergodicity (Theorem 1.11) is proved in Section 8. In Section 9, we deduce our measure classifications on the space of measured laminations (Theorem 1.2, Theorem 1.3, and Theorem 1.8). Our orbit closure classification is presented in Section 10. In Appendix A, we establish non-arithmeticity of the length spectra of non-elementary subgroups of  $\text{Mod}(S)$  (Theorem 1.4). We then deduce in Appendix B topological mixing of the Teichmüller geodesic flow for non-elementary subgroups (Theorem 1.6).

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## 2. TEICHMÜLLER THEORY

In this section, we review basic Teichmüller theory. Our explanation is minimal and we refer interested readers to [Mas09], [FM12] and [GM23b].

In the rest of this paper, let  $S$  be a connected orientable surface of genus  $g$  with  $p$  punctures, where  $3g - 3 + p \geq 1$ . Recall from the introduction that the Teichmüller space  $\mathcal{T} = \mathcal{T}(S)$  is the space of all marked Riemann surface structures on  $S$ . More precisely,

$$\mathcal{T} := \left\{ (X, f) : \begin{array}{l} X \text{ is a Riemann surface and} \\ f : S \rightarrow X \text{ is a diffeomorphism (marking)} \end{array} \right\} / \sim$$

where  $(X, f) \sim (Y, h)$  if  $h \circ f^{-1} : X \rightarrow Y$  is isotopic to a biholomorphic map. We equip  $\mathcal{T}$  with the *Teichmüller metric*  $d_{\mathcal{T}}$ , defined as

$$d_{\mathcal{T}}((X, f), (Y, h)) := \frac{1}{2} \log \inf \left\{ K(\phi) : \begin{array}{l} \text{quasiconformal } \phi : X \rightarrow Y \\ \text{isotopic to } h \circ f^{-1} \end{array} \right\}$$

where  $K(\phi) \geq 1$  denotes the quasiconformal dilatation of  $\phi$ . The metric  $d_{\mathcal{T}}$  is Finsler, proper, uniquely geodesic but not Riemannian. The Teichmüller space  $\mathcal{T}$  is homeomorphic to  $\mathbb{R}^{6g-6+2p}$ .

The mapping class group  $\text{Mod}(S)$  of  $S$  is the group of isotopy classes of orientation-preserving diffeomorphisms of  $S$ . It acts on  $(\mathcal{T}, d_{\mathcal{T}})$  by

$$g \cdot (X, f) := (X, f \circ g^{-1}) \quad \text{for } g \in \text{Mod}(S) \text{ and } (X, f) \in \mathcal{T}.$$

We note that  $\text{Mod}(S)$  is more or less the full isometry group of  $(\mathcal{T}, d_{\mathcal{T}})$ , as shown by Royden [Roy71, Theorem 2] and by Earle and Kra ([EK74a], [EK74b]). See also Ivanov's work [Iva01]. The quotient

$$\mathcal{M} = \mathcal{M}(S) := \text{Mod}(S) \backslash \mathcal{T}$$

is the moduli space of Riemann surface structures on  $S$ .

Note that in the exceptional case when  $3g - 3 + p = 1$  (i.e.,  $S$  is once-punctured torus or 4-punctured sphere),  $\mathcal{T} = \mathbb{H}^2$ . Hence, the main case of interest is when  $3g - 3 + p \geq 2$ .

**2.1. Thurston boundary.** Recall the space  $\mathcal{ML} = \mathcal{ML}(S)$  of measured laminations on  $S$ , which is homeomorphic to  $\mathbb{R}^{6g-6+2p} \setminus \{0\}$ . We denote by  $\mathcal{PML} = \mathcal{PML}(S)$  the space of projective measured laminations on  $S$ , where the projectivization is given by the scaling of transverse measures. The space  $\mathcal{PML}$  can also be identified with the unit sphere in  $\mathcal{ML} \simeq \mathbb{R}^{6g-7+2p} \setminus \{0\}$ , and hence  $\mathcal{PML} \simeq \mathbb{S}^{6g-7+2p}$ . Thurston compactified the Teichmüller space using  $\mathcal{PML}$  and showed that the  $\text{Mod}(S)$ -action on  $\mathcal{T}$  continuously extends to the compactification  $\mathcal{T} \cup \mathcal{PML}$ . In this regard,  $\mathcal{PML}$  is also referred to as the *Thurston boundary* of  $\mathcal{T}$  ([Thu88], [Thu97]).

There is another notion called *measured foliation* on surfaces. The space of measured foliations (up to equivalence) on  $S$  is denoted by  $\mathcal{MF} = \mathcal{MF}(S)$ , and we also denote its projectivization by  $\mathcal{PMF} = \mathcal{PMF}(S)$ . In fact,  $\mathcal{PML}$  and  $\mathcal{PMF}$  give rise to the same compactification of  $\mathcal{T}$ . By this we mean that not only are the two spaces homeomorphic via a map  $\mathcal{PML} \rightarrow \mathcal{PMF}$ , but also the convergence in  $\mathcal{T} \cup \mathcal{PML}$  is equivalent to the convergence in  $\mathcal{T} \cup \mathcal{PMF}$ .

While we stick to measured laminations throughout, they are interchangeable with measured foliations depending on the readers' preference.

**2.2. Gardiner–Masur boundary.** Gardiner and Masur proposed another compactification of the Teichmüller space using the so-called *Gardiner–Masur boundary*  $\partial^{GM}\mathcal{T}$  in [GM91]. This compactification  $\mathcal{T} \cup \partial^{GM}\mathcal{T}$  is now called the Gardiner–Masur compactification. The  $\text{Mod}(S)$ -action on  $\mathcal{T}$  continuously extends to  $\mathcal{T} \cup \partial^{GM}\mathcal{T}$  as well.

Both the Thurston compactification and the Gardiner–Masur compactification are obtained by embedding  $\mathcal{T}$  into the space  $\mathbb{R}_{\geq 0}^{\mathcal{S}}$  of functions on the set  $\mathcal{S}$  of isotopy classes of essential simple closed curves on  $S$ , and then taking the closure in the projective space  $\mathbb{P}(\mathbb{R}_{\geq 0}^{\mathcal{S}})$ . Recalling that  $\mathcal{T}$  can also be viewed as the space of all marked hyperbolic structures on  $S$ , Thurston embedded  $\mathcal{T}$  using the hyperbolic length, and Gardiner–Masur embedded  $\mathcal{T}$  using the extremal length (see Equation (3.1)). In this regard, Gardiner and Masur proved that the Thurston boundary  $\mathcal{PML}$  sits in the Gardiner–Masur boundary  $\partial^{GM}\mathcal{T}$  as a proper subset [GM91, Theorem 7.1].

**2.3. Uniquely ergodic laminations and boundary comparison.** A measured lamination is called *uniquely ergodic* if the underlying geodesic lamination admits a unique transverse measure up to scaling. We denote by  $\mathcal{UE} = \mathcal{UE}(S) \subset \mathcal{PML}$  the subset of projective classes of uniquely ergodic measured laminations on  $S$ .

On the subset  $\mathcal{UE} \subset \mathcal{PML}$ , the accumulation of points in  $\mathcal{T}$  is well-behaved: if  $[\xi] \in \mathcal{UE}$ , then for each  $x \in \mathcal{T}$ , there exists a *unique* Teichmüller (geodesic) ray  $\gamma_x : [0, +\infty) \rightarrow \mathcal{T}$  based at  $x$  such that  $\lim_{t \rightarrow +\infty} \gamma_x(t) = [\xi]$  in

the Thurston compactification ([HM79], [Mas80], [Mas82b]). Furthermore, for every  $x, y \in \mathcal{T}$ , the two rays  $\gamma_x$  and  $\gamma_y$  are asymptotic, i.e., there exists  $T \in \mathbb{R}$  such that  $\lim_{t \rightarrow +\infty} d_{\mathcal{T}}(\gamma_x(t), \gamma_y(t + T)) = 0$  [Mas80].

Moreover, while the identity map  $\mathcal{T} \rightarrow \mathcal{T}$  does not continuously extend to an embedding of the Thurston compactification into the Gardiner–Masur compactification, it extends to a homeomorphism

$$(2.1) \quad \begin{array}{ccc} \mathcal{T} \cup \mathcal{UE} & \longrightarrow & \mathcal{T} \cup \mathcal{UE} \\ \cap & & \cap \\ \mathcal{T} \cup \mathcal{PML} & & \mathcal{T} \cup \partial^{GM} \mathcal{T} \end{array}$$

as proved by Masur [Mas82b] and recovered by Miyachi [Miy13].

Thanks to these facts, we can regard  $\mathcal{UE}$  as a topological subspace of  $\mathcal{PML}$  and as a topological subspace of  $\partial^{GM} \mathcal{T}$  at the same time. As  $\partial^{GM} \mathcal{T}$  is identified with the horofunction boundary of  $(\mathcal{T}, d_{\mathcal{T}})$ , this enables us to employ the theory of horofunctions in studying Radon measures on  $\mathcal{ML}$  that are supported on uniquely ergodic laminations.

A sufficient condition for a measured lamination to be uniquely ergodic is the recurrence of the associated Teichmüller geodesic ray to a compact subset in the moduli space  $\mathcal{M} = \text{Mod}(S) \backslash \mathcal{T}$ . Namely, the *Masur criterion* [Mas92, Theorem 1.1] asserts that for a Teichmüller geodesic ray  $\gamma : [0, +\infty) \rightarrow \mathcal{T}$ , if there exist a compact subset  $K \subset \mathcal{M}$  and a sequence  $t_n \rightarrow +\infty$  such that  $\gamma(t_n) \in \mathcal{T}$  projects into  $K \subset \mathcal{M}$  for all  $n \in \mathbb{N}$ , then  $\gamma$  converges to a uniquely ergodic lamination in both the Thurston compactification and the Gardiner–Masur compactification.

**2.4. Busemann cocycles.** For  $x, y \in \mathcal{T}$ , the function  $d_{\mathcal{T}}(x, \cdot) - d_{\mathcal{T}}(y, \cdot)$  on  $\mathcal{T}$  may not continuously extend to the Thurston boundary  $\mathcal{PML}$ . Nevertheless, Miyachi showed that if  $\xi \in \mathcal{ML}$  is uniquely ergodic, then for every sequence  $\{z_n\}_{n \in \mathbb{N}} \subset \mathcal{T}$  converging to  $[\xi] \in \mathcal{UE}$  the limit

$$\beta_{\xi}(x, y) := \lim_{n \rightarrow +\infty} d_{\mathcal{T}}(x, z_n) - d_{\mathcal{T}}(y, z_n)$$

exists and is independent of the choice of  $\{z_n\}_{n \in \mathbb{N}}$  [Miy13, Corollary 1]. The function  $\beta$  is called the *Busemann cocycle*. Indeed, it satisfies the cocycle relation: for  $w, x, y \in \mathcal{T}$ ,

$$\beta_{\xi}(w, y) = \beta_{\xi}(w, x) + \beta_{\xi}(x, y).$$

Moreover, for  $g \in \text{Mod}(S)$ , it is easy to see that

$$\beta_{g\xi}(gx, gy) = \beta_{\xi}(x, y).$$

Although  $\beta_{\xi}$  only depends on the projective class  $[\xi]$ , not  $\xi$ , we use the notation  $\beta_{\xi}$  for later purposes.

## 3. HUBBARD–MASUR COORDINATES FOR MEASURED LAMINATIONS

In [HM79], Hubbard and Masur proved that given a point  $x_0 \in \mathcal{T}$ , the space  $\mathcal{Q}(S, x_0)$  of holomorphic quadratic differentials on the Riemann surface  $(S, x_0)$  is homeomorphic to  $\mathcal{ML}$ , where the homeomorphism is given by the vertical measured foliation of a given holomorphic quadratic differential on  $x_0$ .

This enables us to consider certain coordinate systems on  $\mathcal{ML}$ , which we call Hubbard–Masur coordinates, by specifying a homeomorphism

$$\mathcal{ML} \simeq \mathcal{PML} \times \mathbb{R}.$$

For this, we employ the notion of extremal lengths.

**3.1. Extremal lengths.** Given a point  $x \in \mathcal{T}$  and the isotopy class  $\alpha$  of a simple closed curve on  $S$ , the *extremal length* of  $\alpha$  on Riemann surface  $(S, x)$  is defined as

$$(3.1) \quad \text{Ext}_x(\alpha) := \sup_{\sigma} \frac{\ell_{\alpha}(\sigma)^2}{\text{Area}(\sigma)}$$

where the supremum is over all metrics  $\sigma$  conformally equivalent to  $x$ , and  $\ell_{\alpha}(\sigma)$  is the length of  $\alpha$  in the metric  $\sigma$ . The extremal length continuously extends to the function

$$\text{Ext}_x : \mathcal{ML} \rightarrow \mathbb{R}$$

such that  $\text{Ext}_x(t\xi) = t^2 \text{Ext}_x(\xi)$  for all  $\xi \in \mathcal{ML}$  and  $t > 0$ , by Kerckhoff [Ker80, Proposition 3].

Miyachi [Miy13, Corollary 2] and Walsh [Wal19, Section 6] showed that for  $[\xi] \in \mathcal{UE}$  and  $x, y \in \mathcal{T}$ , the following holds:

$$(3.2) \quad \beta_{\xi}(x, y) = \frac{1}{2} \log \frac{\text{Ext}_x(\xi)}{\text{Ext}_y(\xi)}.$$

**3.2. Hubbard–Masur coordinates.** Fixing a basepoint  $x_0 \in \mathcal{T}$ , we now define the *Hubbard–Masur coordinates* for  $\mathcal{ML}$  (with respect to  $x_0$ ) as follows:

$$\text{HM} : \mathcal{ML} \rightarrow \mathcal{PML} \times \mathbb{R}$$

$$\xi \mapsto \left( [\xi], \frac{1}{2} \log \text{Ext}_{x_0}(\xi) \right)$$

which is a homeomorphism. We define the  $\text{Mod}(S)$ -action on  $\mathcal{PML} \times \mathbb{R}$  by

$$g \cdot ([\xi], t) := \left( g[\xi], t + \frac{1}{2} \log \frac{\text{Ext}_{g^{-1}x_0}(\xi)}{\text{Ext}_{x_0}(\xi)} \right)$$

for  $g \in \text{Mod}(S)$ ,  $\xi \in \mathcal{ML}$ , and  $t \in \mathbb{R}$ . Note that this is well-defined, independently of the choice of the representative  $\xi$  of  $[\xi]$ .

By Equation (3.2), this can be rephrased in terms of the Busemann function. For a uniquely ergodic lamination  $\xi$  and a mapping class  $g$ ,

$$g \cdot ([\xi], t) = (g[\xi], t + \beta_{\xi}(g^{-1}x_0, x_0)).$$

We now show the equivariance of the action.

**Proposition 3.1.** *The map  $\text{HM} : \mathcal{ML} \rightarrow \mathcal{PML} \times \mathbb{R}$  is  $\text{Mod}(S)$ -equivariant.*

*Proof.* Let  $g \in \text{Mod}(S)$  and  $\xi \in \mathcal{ML}$ . It follows from the definition that

$$\text{HM}(g\xi) = \left( g[\xi], \frac{1}{2} \log \text{Ext}_{x_0}(g \cdot \xi) \right).$$

Since  $\text{Ext}_{x_0}(g\xi) = \text{Ext}_{g^{-1}x_0}(\xi)$ , we have

$$\text{HM}(g\xi) = \left( g[\xi], \frac{1}{2} \log \text{Ext}_{x_0}(\xi) + \frac{1}{2} \log \frac{\text{Ext}_{g^{-1}x_0}(\xi)}{\text{Ext}_{x_0}(\xi)} \right),$$

which is  $g \cdot \text{HM}(\xi)$  by definition.  $\square$

**3.3. Liu–Su and Walsh’s coordinates on the horofunction boundary.** Liu and Su proved in [LS14] and Walsh proved in [Wal19] that the horofunction boundary for  $(\mathcal{T}, d_{\mathcal{T}})$  is equal to the Gardiner–Masur boundary. In both works, fixing a basepoint  $x_0 \in \mathcal{T}$ , the authors constructed a continuous injection

$$\mathcal{T} \cup \partial^{GM} \mathcal{T} \rightarrow \{1\text{-Lipschitz functions on } \mathcal{T} \text{ vanishing at } x_0\},$$

and its restriction to the Gardiner–Masur boundary

$$\partial^{GM} \mathcal{T} \rightarrow \{\text{horofunctions on } \mathcal{T} \text{ vanishing at } x_0\}$$

is a homeomorphism. See Section 5.6 for more discussion on horofunctions.

#### 4. SQUEEZING PROPERTY OF PSEUDO-ANOSOV AXES

We now discuss elements of  $\text{Mod}(S)$ , the mapping classes of  $S$ . The celebrated Nielsen–Thurston classification ([Thu88], [FLP79]) asserts that a mapping class  $\varphi \in \text{Mod}(S)$  is either

- periodic, i.e.,  $\varphi^n = \text{id}$  for some  $n \in \mathbb{N}$ ,
- reducible, i.e., there exists a multicurve on  $S$  invariant under  $\varphi$ , or
- pseudo-Anosov, i.e., there exists a pair of transverse measured laminations  $\xi^+, \xi^- \in \mathcal{ML}$  and  $\lambda > 1$  such that

$$(4.1) \quad \varphi(\xi^+) = \lambda \cdot \xi^+ \quad \text{and} \quad \varphi(\xi^-) = \frac{1}{\lambda} \cdot \xi^-.$$

The measured laminations  $\xi^+$  and  $\xi^-$  are called the unstable and stable measured laminations of  $\varphi$ , respectively, and the constant  $\lambda > 1$  is called the stretch factor of  $\varphi$ .

Among the three categories we are particularly interested in pseudo-Anosov mapping classes. Let  $\varphi \in \text{Mod}(S)$  be a pseudo-Anosov mapping class. We summarize some standard facts:

- (1) It follows from Equation (4.1) that the unstable and stable measured laminations of  $\varphi$  give two fixed points  $[\xi^+], [\xi^-] \in \mathcal{PML}$  in the Thurston boundary. Moreover, on the Thurston compactification  $\mathcal{T} \cup \mathcal{PML}$ ,  $\varphi$  exhibits the north-south dynamics.

More precisely, for each compact subset  $K \subset (\mathcal{T} \cup \mathcal{PML}) \setminus \{[\xi^-]\}$ , we have  $\varphi^n K \rightarrow [\xi^+]$  as  $n \rightarrow +\infty$ . Similarly, for each compact subset  $K \subset (\mathcal{T} \cup \mathcal{PML}) \setminus \{[\xi^+]\}$ , we have  $\varphi^n K \rightarrow [\xi^-]$  as  $n \rightarrow -\infty$ .

- (2) The projective measured laminations  $[\xi^+], [\xi^-] \in \mathcal{PML}$  are called attracting and repelling fixed points of  $\varphi$  respectively, and they are in fact uniquely ergodic, i.e.,  $[\xi^+], [\xi^-] \in \mathcal{UE}$ .
- (3) There exists a unique bi-infinite Teichmüller geodesic  $\gamma \subset \mathcal{T}$  whose endpoints are  $[\xi^\pm] \in \mathcal{PML}$  in the Thurston compactification. Moreover,  $\gamma$  is invariant under  $\varphi$ , and the action of  $\varphi$  on  $\gamma$  is the translation by  $\log \lambda$ , where  $\lambda$  is the stretch factor of  $\varphi$ .

The invariant geodesic  $\gamma$  is called the *axis* of the pseudo-Anosov  $\varphi$ .

The first two items are part of Thurston's proof of the Nielsen–Thurston classification using the Thurston compactification. We refer the reader to Thurston's exposition [Thu88] and textbooks ([FLP79], [FM12]). The last item is proved in [FLP79], [GM91, Theorem 3.1], and [MS93, Theorem 9.2].

**4.1. Contracting property and squeezing property.** The two central dynamical notions in this paper are contracting and squeezing properties of isometries. In [Min96], Minsky proved that Teichmüller geodesics that are *precompact* in the moduli space  $\mathcal{M} = \text{Mod}(S) \setminus \mathcal{T}$  are contracting. More precisely, for each compact subset  $K \subset \mathcal{M}$ , there exists  $D = D(K) > 0$  such that every geodesic in  $\mathcal{T}$  that projects into  $K \subset \mathcal{M}$  is  $D$ -contracting.

Recall that a pseudo-Anosov mapping class has an invariant Teichmüller geodesic, called its *axis*, which descends to a closed loop in  $\mathcal{M}$ . Therefore, axes of pseudo-Anosov mapping classes are contracting. They in fact enjoy a stronger property, called the *squeezing* property, thanks to their periodicity. This essentially follows from Minsky's contraction theorem in [Min96] and Masur's stability theorem in [Mas80]. We give a proof for completeness.

**Proposition 4.1.** *The axis of a pseudo-Anosov mapping class is squeezing.*

*Proof.* Let  $\varphi \in \text{Mod}(S)$  be pseudo-Anosov and denote by  $\gamma \subset \mathcal{T}$  its axis. We fix a unit-speed parametrization  $\gamma : \mathbb{R} \rightarrow \mathcal{T}$ . Suppose to the contrary that  $\gamma$  is not squeezing. Then there exist  $\epsilon > 0$  and sequences  $\{x_n\}_{n \in \mathbb{N}}, \{y_n\}_{n \in \mathbb{N}} \subset \mathcal{T}$  and  $\{t_n\}_{n \in \mathbb{N}} \subset \mathbb{R}$  such that for each  $n \in \mathbb{N}$ ,  $\gamma(t_n - a_n) \in \pi_\gamma(x_n)$  and  $\gamma(t_n + b_n) \in \pi_\gamma(y_n)$  for some  $a_n, b_n > n$  while  $d_{\mathcal{T}}(\gamma(t_n), [x_n, y_n]) \geq \epsilon$ . Since  $\varphi$  acts on  $\gamma$  by a translation, we may assume that  $t_n$  is bounded.

By [Min96, Contraction Theorem],  $\gamma$  is contracting. Hence by applying [CCT25, Lemma 2.2] (see Lemma 5.2), there exists  $C > 0$  such that for each  $n \in \mathbb{N}$ , there exist  $w_n, z_n \in [x_n, y_n]$  satisfying

- $d_{\mathcal{T}}(w_n, \gamma(t_n - a_n)) \leq C$ ,
- $d_{\mathcal{T}}(z_n, \gamma(t_n + b_n)) \leq C$ , and
- $[w_n, z_n]$  is contained in the  $C$ -neighborhood of  $\gamma$ .

Since  $a_n, b_n \rightarrow +\infty$  and  $t_n$  is bounded, after passing to a subsequence,  $[w_n, z_n]$  converges to a bi-infinite geodesic contained in the  $2C$ -neighborhood

of  $\gamma$ . Since both endpoints of  $\gamma$  are contained in  $\mathcal{UE}$ , it follows from [KM96, Lemma 1.4.2] that the limit of  $[w_n, z_n]$  is the bi-infinite geodesic between the endpoints of  $\gamma$ , which must be  $\gamma$ .

On the other hand,  $t_n$  is bounded and  $d_{\mathcal{T}}(\gamma(t_n), [w_n, z_n]) \geq \epsilon$  for all  $n \in \mathbb{N}$ , which is a contradiction. This finishes the proof.  $\square$

The squeezing property can be considered as a version of  $\text{CAT}(-1)$  property along special directions. It is sensible to generalize some dynamical phenomena in  $\text{CAT}(-1)$  spaces to metric spaces with squeezing isometries. For example, in [Cho24] the first author studied continuity and differentiability of the drift of a random walk on the Teichmüller space using the squeezing property of pseudo-Anosov mapping classes.

## 5. CONTRACTING AND SQUEEZING ISOMETRIES

The previous sections tell us that Teichmüller space is a metric space with squeezing isometries. In this section, we further develop this perspective.

Throughout this section, let  $(X, d)$  be a proper geodesic metric space and fix a basepoint  $x_0 \in X$ . For  $x, y \in X$ , we denote by  $[x, y] \subset X$  an arbitrarily chosen geodesic connecting  $x$  to  $y$ . For  $w, z \in [x, y]$ , we implicitly assume  $[w, z]$  to be a segment of  $[x, y]$ . Every parametrization of a geodesic is of unit speed. We denote the isometry group of  $X$  by  $\text{Isom}(X)$ .

We say that two geodesics  $\gamma_1, \gamma_2 \subset X$  are  $C$ -equivalent if their Hausdorff distance is at most  $C$  and if their beginning/ending points are pairwise  $C$ -close. When two reals  $a, b \in \mathbb{R}$  differ by at most  $C$ , we write  $a =_C b$ .

**5.1. Contracting subsets.** For a closed subset  $A \subset X$ , we denote by  $\pi_A(\cdot) : X \rightarrow 2^A$  the *nearest point projection*. That is, we define

$$\pi_A(x) := \left\{ a \in A : d(x, a) = \inf_{z \in A} d(x, z) \right\}.$$

**Definition 5.1.** Let  $C \geq 0$ . We say that a closed subset  $A \subset X$  is  $C$ -contracting if for every geodesic  $\gamma \subset X$  with  $d(\gamma, A) \geq C$  we have  $\text{Diam } \pi_A(\gamma) \leq C$ . We say that  $A$  is (strongly) contracting if it is  $C$ -contracting for some  $C \geq 0$ .

By definition, if  $A \subset X$  is  $C$ -contracting, then  $\text{Diam } \pi_A(x) \leq 2C$  for all  $x \in X$ . It is also easy to see that if  $A \subset X$  is  $C$ -contracting and  $x, y \in X$  are in the  $C$ -neighborhood of  $A$ , then  $[x, y]$  is contained in the  $2.5C$ -neighborhood of  $A$ .

As an example, every geodesic in a  $\delta$ -hyperbolic space is  $C(\delta)$ -contracting for some  $C(\delta) \geq 0$  depending only on  $\delta$ . In particular, geodesics in a simplicial tree or  $\mathbb{H}^2$  are contracting with a uniform contracting constant. More generally, every  $(K, B)$ -quasigeodesic in a  $\delta$ -hyperbolic space is  $C(\delta, K, B)$ -contracting for a constant  $C(\delta, K, B) \geq 0$ .

The following is reminiscent of the Morse lemma in a Gromov hyperbolic space. We use the version in [CCT25] due to its conciseness, but we note

that similar results were already observed in ([CDP90, Proposition 10.2.1], [Sis13, Lemma 2.4, 2.5], [ACT15, Proposition 2.9, Lemma 2.10, Lemma 2.11], [Yan14, Proposition 3.1]).

**Lemma 5.2** ([CCT25, Lemma 2.2]). *Let  $\gamma \subset X$  be a  $C$ -contracting geodesic and  $x, y \in X$ . Suppose that  $\text{Diam } \pi_\gamma([x, y]) > C$ . Then there exist points  $p, q \in [x, y]$ , with  $p$  closer to  $x$  than  $q$ , such that*

- $\pi_\gamma([x, y])$  and  $[p, q]$  are  $4C$ -equivalent,
- $\text{Diam}(\pi_\gamma([x, p]) \cup \{p\}) \leq 2C$ ,
- $\text{Diam}(\pi_\gamma([q, y]) \cup \{q\}) \leq 2C$ , and,
- for all  $x' \in \pi_\gamma(x)$  and  $y' \in \pi_\gamma(y)$ ,  $[x', y']$  and  $[p, q]$  are  $10C$ -equivalent.

The following is an immediate corollary of Lemma 5.2.

**Corollary 5.3.** *For a  $C$ -contracting geodesic  $\gamma \subset X$ , the following holds.*

- (1) *The map  $\pi_\gamma(\cdot)$  is  $(1, 4C)$ -Lipschitz: for each  $x, y \in X$ ,*

$$\text{Diam } \pi_\gamma(\{x, y\}) \leq d(x, y) + 4C.$$

- (2) *Let  $x \in X$  and  $\gamma(t) \in \pi_\gamma(x)$ . Then for every  $s \in \mathbb{R}$ , we have*

$$d(x, \gamma(s)) =_{4C} d(x, \gamma(t)) + |t - s|.$$

## 5.2. Contracting isometries.

**Definition 5.4.** We say that an isometry  $g \in \text{Isom}(X)$  is *axial* if there exists a bi-infinite geodesic  $\gamma : \mathbb{R} \rightarrow X$  invariant under  $g$  such that

$$g \cdot \gamma(t) = \gamma(t + \tau_g) \quad \text{for all } t \in \mathbb{R}$$

for some  $\tau_g > 0$ . We call  $\gamma$  an *axis* of  $g$  and  $\tau_g$  the *translation length* of  $g$ .

An axial isometry  $g \in \text{Isom}(X)$  is called  *$C$ -contracting* for  $C \geq 0$  if it has a  $C$ -contracting axis.

Given an axial isometry  $g \in \text{Isom}(X)$ , note that

$$\tau_g = \lim_{n \rightarrow +\infty} \frac{d(x, g^n x)}{n} > 0 \quad \text{for each } x \in X.$$

Then we can observe the following:

$$\tau_g = \inf_{x \in X} d(x, gx) \quad \text{and} \quad \tau_{g^k} = |k| \tau_g \quad \text{for each } k \in \mathbb{Z}.$$

For each  $h \in \text{Isom}(X)$ ,  $hgh^{-1}$  is also axial and  $\tau_{hgh^{-1}} = \tau_g$ .

## 5.3. Squeezing isometries.

**Definition 5.5.** We say that a bi-infinite geodesic  $\gamma : \mathbb{R} \rightarrow X$  is *squeezing* if for each  $\epsilon > 0$  there exists  $L = L(\epsilon) > 0$  such that for each  $x, y \in X$  and  $t \in \mathbb{R}$  with  $\gamma(t - a) \in \pi_\gamma(x)$  and  $\gamma(t + b) \in \pi_\gamma(y)$  for some  $a, b \geq L$ , we have

$$d([x, y], \gamma(t)) \leq \epsilon.$$

We call an axial isometry  $g \in \text{Isom}(X)$  *squeezing* if it has a squeezing axis.

By definition, squeezing isometries are contracting. Note that both squeezing and contracting properties are invariant under conjugation. If an isometry  $g \in \text{Isom}(X)$  is squeezing, then it has a unique axis (up to reparametrization). We denote it by  $A_g$ . For  $h \in \text{Isom}(X)$ ,  $A_{hgh^{-1}} = hA_g$ .

As we will see later, squeezing geodesics are well-suited for studying horofunctions due to the following lemma.

**Lemma 5.6.** *Let  $\gamma : \mathbb{R} \rightarrow X$  be a squeezing geodesic. Fix  $\epsilon > 0$  and let  $L = L(\epsilon) > 0$  be as in Definition 5.5. Let  $x_1, x_2, y_1, y_2 \in X$  and suppose that for some  $t \in \mathbb{R}$ , we have*

$$\pi_\gamma(x_i) \cap \gamma((-\infty, t - L]) \neq \emptyset \quad \text{and} \quad \pi_\gamma(y_i) \cap \gamma([t + L, +\infty)) \neq \emptyset \quad \text{for } i = 1, 2.$$

*Then we have*

$$d(x_1, y_1) - d(x_1, y_2) =_{8\epsilon} d(x_2, y_1) - d(x_2, y_2).$$

*Proof.* Let  $i \in \{1, 2\}$ . By the squeezing property, there exist  $p \in [x_i, y_1]$  and  $q \in [x_i, y_2]$  that are  $\epsilon$ -close to  $\gamma(t)$ . By the triangle inequality, we have

$$d(x_i, y_1) - d(x_i, y_2) =_{4\epsilon} d(\gamma(t), y_1) - d(\gamma(t), y_2).$$

This gives the desired estimate.  $\square$

**5.4. Alignment.** We denote the closed  $K$ -neighborhoods by  $\mathcal{N}_K(\cdot)$ .

**Definition 5.7** (Alignment). Let  $w, x, y, z \in X$ . For a geodesic  $[x, y] \subset X$  and  $K \geq 0$ , we say that the sequence  $(w, [x, y])$  is  $K$ -aligned if

$$\pi_{[x, y]}(w) \subset \mathcal{N}_K(x).$$

Similarly, we say that the sequence  $([x, y], z)$  is  $K$ -aligned if  $(z, [y, x])$  is  $K$ -aligned.

Finally, we say that the sequence  $(w, [x, y], z)$  is  $K$ -aligned if both sequences  $(w, [x, y])$  and  $([x, y], z)$  are  $K$ -aligned. See Figure 2.

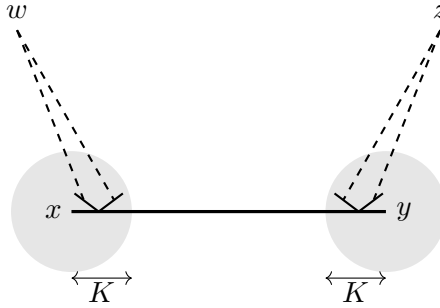


FIGURE 2. Alignment of geodesics and points.

The following is immediate.

**Lemma 5.8.** *Let  $\gamma \subset X$  be a geodesic of length  $L \geq 0$ , let  $0 \leq D \leq L$  and let  $x \in X$ . Then  $(\gamma, x)$  is not  $D$ -aligned or  $(x, \gamma)$  is not  $(L - D)$ -aligned.*

If two closed sets are within finite Hausdorff distance, the contracting property of one set implies that of the other ([ACT15, Lemma 2.8], [Yan19, Proposition 2.4(2)]). Furthermore, every subsegment of a contracting geodesic is contracting with a uniform contracting constant ([BF09, Lemma 3.2], [Yan20, Proposition 2.2(3)]). These facts have the following consequence, whose proof is included for completeness.

**Lemma 5.9.** *Let  $g \in \text{Isom}(X)$  be a contracting isometry with an axis  $\gamma : \mathbb{R} \rightarrow X$  and let  $x_0 \in X$ . Then there exists  $C = C(g, \gamma, x_0) > 0$  such that the following holds.*

- (0)  $\gamma$  is  $C$ -contracting.
- (1)  $[x_0, g^k x_0]$  is  $C$ -contracting for all  $k \in \mathbb{Z}$ .
- (2)  $d(g^k x_0, \gamma(\tau_g k)) < C$  for all  $k \in \mathbb{Z}$ .
- (3) Let  $k \in \mathbb{N}$ , let  $x \in X$ , and let  $K \geq C$ . Then
 
$$(x, [x_0, g^k x_0]) \text{ is not } K\text{-aligned} \implies \pi_\gamma(x) \subset \gamma([K - C, +\infty)).$$
- (4) Let  $k \in \mathbb{N}$ , let  $x \in X$ , and let  $0 \leq K \leq \tau_g k - C$ . Then
 
$$(x, [x_0, g^k x_0]) \text{ is } K\text{-aligned} \implies \pi_\gamma(x) \subset \gamma((-\infty, K + C]).$$

Moreover,  $C$  can be chosen so that  $C(g, \gamma, x_0) = C(g^k, \gamma, x_0)$  for all  $k \in \mathbb{N}$  and  $C(g^{-1}, \hat{\gamma}, x_0) = C(g, \gamma, x_0)$  where  $\hat{\gamma}$  is the reversal of  $\gamma$ .

We often write  $C(g) = C(g, \gamma, x_0)$  by implicitly choosing its axis  $\gamma$ .

*Proof.* By definition, there exists  $C_0 > 0$  so that  $\gamma$  is  $C_0$ -contracting. Item (0) holds for all  $C \geq C_0$ . Item (2) is also immediate for any  $C > d(x_0, \gamma(0))$ .

Let

$$D = 10(C_0 + d(x_0, \gamma(0))).$$

We will see that  $C = 100D$  plays the desired role.

We first show Item (1). By Lemma 5.2,  $[x_0, g^k x_0]$  and  $\gamma([0, \tau_g k])$  are  $D$ -equivalent for each  $k \in \mathbb{Z}$ . It suffices to consider  $k \in \mathbb{N}$  with  $\tau_g k > 98D$ . Fix such  $k \in \mathbb{N}$  and observe from Corollary 5.3(2) that for  $x \in X$ ,

- if  $\pi_\gamma(x) \cap \gamma((-\infty, 0)) \neq \emptyset$ , then  $\pi_{[x_0, g^k x_0]}(x) \subset \mathcal{N}_{5D}(\gamma(0))$ .
- if  $\pi_\gamma(x) \cap \gamma((\tau_g k, +\infty)) \neq \emptyset$ , then  $\pi_{[x_0, g^k x_0]}(x) \subset \mathcal{N}_{5D}(\gamma(\tau_g k))$ .
- if  $\pi_\gamma(x) \cap \gamma([0, \tau_g k]) \neq \emptyset$ , then  $\pi_{[x_0, g^k x_0]}(x) \subset \mathcal{N}_{5D}(\pi_\gamma(x))$ .

Let  $\eta \subset X$  be a geodesic such that  $d(\eta, [x_0, g^k x_0]) \geq 100D$ . We have two cases. First, if  $\text{Diam } \pi_\gamma(\eta) \geq D$ , then there exists a subsegment of  $\eta$  that is  $4C_0$ -equivalent to  $\pi_\gamma(\eta)$ . In this case,  $\pi_\gamma(\eta)$  cannot intersect  $\gamma([0, \tau_g k])$ ; otherwise we have  $d(\eta, \gamma([0, \tau_g k])) \leq 4C_0$  and  $d(\eta, [x_0, g^k x_0]) \leq 2D$ , a contradiction. Hence, either  $\pi_\gamma(\eta) \subset \gamma((-\infty, 0))$  or  $\pi_\gamma(\eta) \subset \gamma((\tau_g k, +\infty))$  holds. By the above observation, we have  $\text{Diam } \pi_{[x_0, g^k x_0]}(\eta) \leq 100D$ .

If  $\text{Diam } \pi_\gamma(\eta) < D$ , then we again have  $\text{Diam } \pi_{[x_0, g^k x_0]}(\eta) \leq 100D$  by the above observation. Therefore, Item (1) holds for  $C = 100D$ .

We now show Item (3). Let  $k \in \mathbb{N}$ , let  $K \geq 100D$  and suppose that  $(x, [x_0, g^k x_0])$  is not  $K$ -aligned. Recall that  $[x_0, g^k x_0]$  and  $\gamma([0, \tau_g k])$  are  $D$ -equivalent. Hence there exist  $p \in \pi_{[x_0, g^k x_0]}(x)$  and  $t_p \in [0, \tau_g k]$  such that

$$d(x_0, p) > K \quad \text{and} \quad d(p, \gamma(t_p)) \leq D.$$

Note also that  $d(x_0, \gamma(0)) \leq D$ . We then have

$$t_p \geq K - 2D.$$

If there exists  $t \in (-\infty, K - 6D]$  such that  $\gamma(t) \in \pi_\gamma(x)$ , then

$$\begin{aligned} d(x, \gamma(t_p)) &\geq d(x, \gamma(K - 6D)) + t_p - (K - 6D) - D \\ &\geq d(x, \gamma(K - 6D)) + 3D \end{aligned}$$

by Corollary 5.3(2). Since  $0 \leq K - 6D \leq t_p \leq \tau_g k$ , and since  $[x_0, g^k x_0]$  and  $\gamma([0, \tau_g k])$  are  $D$ -equivalent, we have

$$d(x, p) \geq d(x, \gamma(t_p)) - D \geq d(x, \gamma(K - 6D)) + 2D \geq d(x, [x_0, g^k x_0]) + D,$$

which contradicts the fact that  $p \in \pi_{[x_0, g^k x_0]}(x)$ . Therefore,

$$\pi_\gamma(x) \subset \gamma((K - 6D, +\infty))$$

and hence Item (3) holds for  $C = 100D$ .

For Item (4), let  $k \in \mathbb{N}$  and  $0 \leq K \leq \tau_g k - 100D$ , and suppose that  $(x, [x_0, g^k x_0])$  is  $K$ -aligned. Then  $([x_0, g^k x_0], x)$  is not  $(d(x_0, g^k x_0) - K)$ -aligned by Lemma 5.8. Note that

$$d(x_0, g^k x_0) - K \geq \tau_g k - K \geq 100D.$$

Hence, we can apply (a symmetric version of) Item (3) and deduce that

$$\pi_\gamma(x) \subset \gamma((-\infty, \tau_g k - (d(x_0, g^k x_0) - K) + 6D)) \subset \gamma((-\infty, K + 6D)).$$

Therefore, Item (4) holds for  $C = 100D$  as well.

The ‘‘Moreover’’ part is straightforward.  $\square$

**5.5. Non-elementary subgroups of isometries.** The class of subgroups we mainly consider is the following:

**Definition 5.10.** A subgroup  $\Gamma < \text{Isom}(X)$  is called *non-elementary* if

- $\Gamma$  is not virtually cyclic,
- the  $\Gamma$ -action on  $X$  is proper, and
- $\Gamma$  contains a contracting isometry.

We say that two contracting isometries  $g, h \in \text{Isom}(X)$  are *independent* if their orbits  $\{g^i x_0\}_{i \in \mathbb{Z}}$  and  $\{h^i x_0\}_{i \in \mathbb{Z}}$  have infinite Hausdorff distance. This is equivalent to saying that  $\{g^i x_0\}_{i \in \mathbb{Z}}$  and  $\{h^i x_0\}_{i \in \mathbb{Z}}$  have bounded nearest-point projections onto each other.

These two notions are related by the following well-known fact: see [BF09, Proposition 6.5], [Sis18, Corollary 4.4], [ACT15, Lemma 2.23] and [Yan19, Lemma 2.11, Lemma 2.12]

**Lemma 5.11.** *Let  $\Gamma < \text{Isom}(X)$  be a non-elementary subgroup. For a contracting isometry  $g \in \Gamma$ , there exists  $h \in \Gamma$  such that  $hgh^{-1}$  and  $g$  are independent. Moreover, there are infinitely many pairwise independent contracting isometries in  $\Gamma$ .*

The following is a variant of the so-called *extension lemma* of Yang. We include the proof of this variant for the sake of completeness.

**Lemma 5.12** (Extension lemma [Yan19, Lemma 1.13]). *Let  $\Gamma < \text{Isom}(X)$  be a non-elementary subgroup. Then for each contracting isometry  $\varphi \in \Gamma$ , there exist  $a_1, a_2, a_3 \in \Gamma$  and  $\kappa = \kappa(\varphi) > 0$  such that*

- $[x_0, \varphi^n x_0]$  is  $\kappa$ -contracting for all  $n \in \mathbb{N}$ , and
- for each  $x, y \in X$ , there exists  $a \in \{a_1, a_2, a_3\}$  that makes

$$(x, a \cdot [x_0, \varphi^n x_0], a\varphi^n a \cdot y) \quad \kappa\text{-aligned for all } n \in \mathbb{N}.$$

Moreover,  $\kappa$  can be chosen so that  $\kappa(\varphi^k) = \kappa(\varphi)$  for all  $k \in \mathbb{Z}$ .

*Proof.* Note that the “Moreover” part is straightforward. Hence we only prove the first claim. Let  $\gamma : \mathbb{R} \rightarrow X$  be an axis of  $\varphi$ , which is contracting. By [Sis18, Corollary 4.4] and [Yan19, Lemma 2.11], the set

$$E(\varphi) := \{h \in \Gamma : \{\varphi^i x_0\}_{i \in \mathbb{Z}} \text{ and } h\{\varphi^i x_0\}_{i \in \mathbb{Z}} \text{ have finite Hausdorff distance}\}$$

is a finite extension of  $\langle \varphi \rangle$ . Furthermore,  $\pi_\gamma(h\gamma)$  has finite diameter for every  $h \notin E(\varphi)$ . Since  $\Gamma$  is non-elementary, it is not the union of finitely many virtually cyclic subgroups. Hence, there exist  $h, h' \in \Gamma$  such that  $\pi_{u\gamma}(v\gamma)$  has finite diameter for every pair of distinct elements  $u, v \in \{\text{id}, h, h'\}$ .

Let  $C = C(\varphi, \gamma, x_0)$  be as in Lemma 5.9. We choose  $\kappa \geq 100C$  such that

- $\text{Diam}\{x_0, hx_0, h'x_0\} < 0.01\kappa$ , and
- $\pi_{u\gamma}(v\gamma) \subset u\gamma([-0.01\kappa, 0.01\kappa])$  for distinct  $u, v \in \{\text{id}, h, h'\}$ .

Now let  $x \in X$ . Suppose  $(x, [x_0, \varphi^m x_0])$  is not  $\kappa$ -aligned for some  $m \in \mathbb{N}$ .

**Claim.** *Then  $(x, h[x_0, \varphi^n x_0])$  is  $\kappa$ -aligned for all  $n \in \mathbb{N}$ .*

To show this, suppose to the contrary that  $(x, h[x_0, \varphi^n x_0])$  is also not  $\kappa$ -aligned for some  $n \in \mathbb{N}$ . Then  $\pi_{h[x_0, \varphi^n x_0]}(x)$  is at least  $0.9\kappa$ -far from  $hx_0$ , whereas  $\pi_{h[x_0, \varphi^n x_0]}(x_0)$  is  $2d(x_0, hx_0)$ -close to  $hx_0$ , i.e., it is  $0.02\kappa$ -close to  $hx_0$ . Lemma 5.2 tells us that  $[x_0, x]$  has a subsegment  $[p, q]$  that is contained in the  $0.04\kappa$ -neighborhood of  $h[x_0, \varphi^n x_0]$ , and such that  $d(p, q) \geq 0.8\kappa$  and  $d(x_0, p) \leq d(x_0, hx_0) + 0.02\kappa \leq 0.05\kappa$ . Similarly,  $[x_0, x]$  has a subsegment  $[p', q']$  that is contained in the  $0.04\kappa$ -neighborhood of  $[x_0, \varphi^m x_0]$ , and such that  $d(x_0, q') \geq 0.8\kappa$  and  $d(x_0, p') \leq 0.05\kappa$ .

Then the two subsegments  $[p, q]$  and  $[p', q']$  of  $[x_0, x]$  have an overlap of length at least  $0.7\kappa$ . Hence, there exist points  $P \in [x_0, \varphi^m x_0]$  and  $Q \in h[x_0, \varphi^n x_0]$  that are  $0.65\kappa$ -far from  $x_0$  and are  $0.08\kappa$ -close to each other. We can take  $P' \in \gamma$  that is  $0.01\kappa$ -close to  $P$  and  $Q' \in h\gamma$  that is  $0.01\kappa$ -close to  $Q$ . Then  $d(P', Q') \leq 0.1\kappa$  and  $\pi_\gamma(Q')$  is  $0.2\kappa$ -close to  $P'$ . Since  $d(x_0, P') \geq 0.64\kappa$ ,  $\pi_\gamma(h\gamma)$  contains a point that is  $0.44\kappa$ -far from  $x_0$ .

On the other hand,  $\pi_\gamma(h\gamma) \subset \gamma([-0.01\kappa, 0.01\kappa])$  and  $d(x_0, \gamma(0)) \leq 0.01\kappa$ . This is a contradiction. Therefore,  $(x, h[x_0, \varphi^n x_0])$  is  $\kappa$ -aligned for all  $n \in \mathbb{N}$ .

Similarly, we conclude that  $(x, h'[x_0, \varphi^n x_0])$  is  $\kappa$ -aligned for all  $n \in \mathbb{N}$ . The same argument applies after replacing  $id$  with  $h$  or  $h'$ . We conclude that there exist at least two elements  $a \in \{id, h, h'\}$  such that  $(x, a[x_0, \varphi^n x_0])$  is  $\kappa$ -aligned for all  $n \in \mathbb{N}$ . Likewise, for every  $y \in X$ ,  $(a^{-1}[\varphi^{-n} x_0, x_0], y)$  is  $\kappa$ -aligned for all  $n \in \mathbb{N}$  for at least two elements  $a \in \{id, h, h'\}$ . Hence, we can choose  $a \in \{id, h, h'\}$  that works for both  $x$  and  $y$ .  $\square$

**5.6. Horofunctions and cocycles.** Let  $\text{Lip}^1(X)$  be the space of  $\mathbb{R}$ -valued 1-Lipschitz functions on  $X$  and let  $\text{Lip}_{x_0}^1(X)$  be its subspace vanishing at the basepoint  $x_0$ , i.e.,

$$\begin{aligned}\text{Lip}^1(X) &:= \{f : X \rightarrow \mathbb{R} : f \text{ is 1-Lipschitz}\}, \\ \text{Lip}_{x_0}^1(X) &:= \{f \in \text{Lip}^1(X) : f(x_0) = 0\},\end{aligned}$$

equipped with the compact-open topology. Note that  $\text{Lip}_{x_0}^1(X)$  is closed in  $\text{Lip}^1(X)$ .

Recall that  $X$  is separable since it is a proper metric space. Therefore,  $\text{Lip}_{x_0}^1(X)$  is compact, Hausdorff, and second countable [MT18, Proposition 3.1]. Hence, it is completely metrizable and is Polish. We identify  $\text{Lip}^1(X)$  and  $\text{Lip}_{x_0}^1(X) \times \mathbb{R}$  via the homeomorphism

$$(5.1) \quad f \in \text{Lip}^1(X) \mapsto (f - f(x_0), f(x_0)).$$

The group  $\text{Isom}(X)$  naturally acts on  $\text{Lip}^1(X)$  by  $g \cdot f := f \circ g^{-1}$  for  $g \in \text{Isom}(X)$  and  $f \in \text{Lip}^1(X)$ . However, this action does not leave  $\text{Lip}_{x_0}^1(X)$  invariant.

Due to this subtlety, we identify  $\text{Lip}_{x_0}^1(X)$  with the space of  $\mathbb{R}$ -valued 1-Lipschitz cocycles on  $X$ , i.e.,  $c : X \times X \rightarrow \mathbb{R}$  such that  $|c(x, y)| \leq d(x, y)$  and  $c(x, z) = c(x, y) + c(y, z)$  for all  $x, y, z \in X$ . For each  $f \in \text{Lip}^1(X)$ , we define the associated cocycle  $\beta_f : X \times X \rightarrow \mathbb{R}$  by

$$\beta_f(x, y) = f(x) - f(y).$$

Its restriction to  $\text{Lip}_{x_0}^1(X)$  gives the homeomorphism between  $\text{Lip}_{x_0}^1(X)$  and the space of all  $\mathbb{R}$ -valued 1-Lipschitz cocycles. Then the identification  $\text{Lip}^1(X) \simeq \text{Lip}_{x_0}^1(X) \times \mathbb{R}$  in Equation (5.1) can be rephrased as

$$f \mapsto (\beta_f, f(x_0)).$$

The  $\text{Isom}(X)$ -action on  $\text{Lip}^1(X)$  is now given as follows: for  $g \in \text{Isom}(X)$  and  $f \in \text{Lip}^1(X)$ ,

$$g \cdot (\beta_f, f(x_0)) = (\beta_{g \cdot f}, f(x_0) + \beta_f(g^{-1}x_0, x_0)).$$

Note that on the first component, which corresponds to  $\text{Lip}_{x_0}^1(X)$ , we have  $\beta_f \mapsto \beta_{g \cdot f}$ .

There is a natural embedding  $\iota : X \hookrightarrow \text{Lip}_{x_0}^1(X)$ , defined by

$$\iota : z \in X \mapsto [f_z(\cdot) := d(\cdot, z) - d(x_0, z)].$$

The closure of  $\iota(X) \subset \text{Lip}_{x_0}^1(X)$  is called the *horofunction compactification* of  $X$  and is denoted by  $\overline{X}^h$ . The complement  $\overline{X}^h \setminus \iota(X)$  is called the *horofunction boundary* (or *horoboundary*) of  $X$  and is denoted by  $\partial^h X$ .

As the space  $\text{Lip}_{x_0}^1(X)$  is identified with the space of 1-Lipschitz cocycles, we also regard elements of  $\partial^h X$  as *Busemann cocycles*, by identifying

$$f \in \partial^h X \quad \longleftrightarrow \quad \beta_f(\cdot, \cdot).$$

Using this identification, the  $\text{Isom}(X)$ -action on  $\partial^h X$  is given by

$$\beta_f(\cdot, \cdot) \mapsto \beta_{g \cdot f}(\cdot, \cdot) \quad \text{for } g \in \text{Isom}(X) \text{ and } f \in \partial^h X.$$

This is the continuous extension of the isometric action of  $\text{Isom}(X)$  on  $X$  in the following sense. Let  $\{z_i\}_{i \in \mathbb{N}} \subset X$  be a sequence such that  $f_{z_i} \rightarrow f \in \text{Lip}_{x_0}^1(X)$ . Then for every  $g \in \text{Isom}(X)$ , we have

$$\begin{aligned} d(\cdot, gz_i) - d(x_0, gz_i) &= d(g^{-1}(\cdot), z_i) - d(x_0, z_i) + d(x_0, z_i) - d(g^{-1}x_0, z_i) \\ &\rightarrow f(g^{-1}(\cdot)) - f(g^{-1}x_0). \end{aligned}$$

This implies

$$\beta_{f_{gz_i}}(\cdot, \cdot) \rightarrow \beta_{g \cdot f}(\cdot, \cdot).$$

In terms of the identification  $\text{Lip}^1(X) \simeq \text{Lip}_{x_0}^1(X) \times \mathbb{R}$ , the subspace of  $\text{Lip}^1(X)$  corresponding to  $\partial^h X$  is the space

$$(5.2) \quad \mathcal{H} := \partial^h X \times \mathbb{R},$$

which is  $\text{Isom}(X)$ -invariant. As a subspace of  $\text{Lip}^1(X)$ ,  $\mathcal{H}$  does not depend on the choice of the basepoint  $x_0 \in X$ . What depends on the choice of  $x_0$  is the description of  $\text{Isom}(X)$ -action on  $\mathcal{H}$  in terms of the identification  $\mathcal{H} = \partial^h X \times \mathbb{R}$ .

We call elements of  $\partial^h X \times \mathbb{R}$  *horofunctions*. Horofunctions that differ by an additive constant correspond to the same Busemann cocycle.

Both  $\partial^h X$  and  $\mathcal{H} = \partial^h X \times \mathbb{R}$  are Polish. Hence, every locally finite Borel measure on these spaces is Radon, i.e., it is both inner and outer regular on Borel subsets.

We now extend the notion of alignment to horofunctions.

**Definition 5.13.** Let  $\xi \in \partial^h X$  and  $\gamma \subset X$  be a compact geodesic. For  $K \geq 0$ , we say that  $(\xi, \gamma)$  is *K-aligned* if for every sequence  $\{z_i\}_{i \in \mathbb{N}} \subset X$  converging to  $\xi$ ,  $(z_i, \gamma)$  is *K-aligned* eventually (i.e., for all large  $i \in \mathbb{N}$ ). We define the alignment for  $(\gamma, \xi')$  and  $(\xi, \gamma, \xi')$  similarly for  $\xi' \in \overline{X}^h$ .

By abuse of notation, for a Busemann cocycle  $\beta_\xi$  that corresponds to  $\xi \in \partial^h X$ , we say that  $(\beta_\xi, \gamma)$  is *K-aligned* when  $(\xi, \gamma)$  is *K-aligned*. Lastly, for an element  $f \in \mathcal{H}$ , we say that  $(f, \gamma)$  is *K-aligned* if  $(\beta_f, \gamma)$  is *K-aligned*. We define the alignment for triples similarly.

For contracting geodesics, we observe the following:

**Lemma 5.14.** *Let  $\gamma \subset X$  be a  $C$ -contracting compact geodesic for  $C > 0$ . Let  $\xi \in \partial^h X$  and  $\{z_i\}_{i \in \mathbb{N}} \subset X$  be a sequence converging to  $\xi$ . Then we have*

$$\limsup_{N \rightarrow +\infty} \text{Diam} \bigcup_{k \geq N} \pi_\gamma(z_k) \leq 9C.$$

*Proof.* Since  $\gamma$  is compact and  $z_i \rightarrow \xi$ , there exists  $N > 0$  such that

$$d(x, z_k) - d(y, z_k) =_{0.1C} d(x, z_l) - d(y, z_l) \quad \text{for all } k, l > N \text{ and } x, y \in \gamma.$$

Now suppose to the contrary that  $\text{Diam}(\pi_\gamma(z_k) \cup \pi_\gamma(z_l)) > 9C$  for some  $k, l > N$ . Then by Lemma 5.2, there exist  $p, q \in [z_k, z_l]$  such that

$$\text{Diam}(\{p\} \cup \pi_\gamma(z_k)) \leq 2C \quad \text{and} \quad \text{Diam}(\{q\} \cup \pi_\gamma(z_l)) \leq 2C.$$

Note that  $d(p, q) > 5C$ . Let  $p' \in \pi_\gamma(z_k)$  and  $q' \in \pi_\gamma(z_l)$  be arbitrary points. We then have

$$d(z_k, p') \leq d(z_k, p) + 2C \leq (d(z_k, q) - 5C) + 2C \leq d(z_k, q') - C.$$

For a similar reason, we have  $d(z_l, q') \leq d(z_l, p') - C$ . This contradicts the condition for  $k, l > N$ .  $\square$

The following version of extension lemma can easily be deduced from Lemma 5.12 and Lemma 5.14:

**Lemma 5.15** (Extension lemma). *Let  $\Gamma < \text{Isom}(X)$  be a non-elementary subgroup. Let  $\varphi \in \Gamma$  be a contracting isometry, and let  $\kappa = \kappa(\varphi) > 0$  and  $a_1, a_2, a_3 \in \Gamma$  be as given in Lemma 5.12. Then for each  $\xi, \xi' \in \overline{X}^h$ , there exists  $a \in \{a_1, a_2, a_3\}$  such that*

$$(\xi, a \cdot [x_0, \varphi^n x_0], a\varphi^n a \cdot \xi') \text{ is } 10\kappa\text{-aligned for all } n \in \mathbb{N}.$$

*Remark 5.16.* There is a way to extend the nearest-point projection  $\pi_\gamma(\cdot)$  to  $\overline{X}^h$  (cf. [Cou24, Definition 3.8]). For  $\xi \in \partial^h X$ , we can define

$$\pi_\gamma(\xi) := \left\{ x \in \gamma : \beta_\xi(x, x_0) = \inf_{y \in \gamma} \beta_\xi(y, x_0) \right\}.$$

Then one can show that if  $\{z_i\}_{i \in \mathbb{N}} \subset X$  is a sequence converging to  $\xi \in \partial^h X$ , then  $\pi_\gamma(z_i) \rightarrow \pi_\gamma(\xi)$  up to a uniform error. One can define alignment in terms of this extended nearest-point projection as well.

**5.7. Conical limit sets.** We finish this section by defining conical limit sets, which are also called radial limit sets.

**Definition 5.17.** Let  $\Gamma < \text{Isom}(X)$  be a subgroup acting properly on  $X$ . A point  $\xi \in \partial^h X$  is called a *conical limit point* of  $\Gamma$  if there exist  $K > 0$  and an infinite sequence  $\{g_n\}_{n \in \mathbb{N}} \subset \Gamma$  such that

$$\beta_\xi(x_0, g_n x_0) \geq d(x_0, g_n x_0) - K \quad \text{for all } n \in \mathbb{N}.$$

We denote the *conical limit set* by  $\Lambda_c(\Gamma) \subset \partial^h X$ .

For example, given a geodesic ray  $\gamma \subset X$  whose  $K$ -neighborhood contains infinitely many points in a  $\Gamma$ -orbit, any horofunction obtained as a limit point of  $\gamma$  is conical. One can see that  $\Lambda_c(\Gamma)$  is  $\Gamma$ -invariant.

## 6. PATTERSON–SULLIVAN THEORY WITH SQUEEZING ISOMETRIES

In this section, we review Coulon’s and Yang’s extensions of the Patterson–Sullivan theory of conformal densities ([Cou24], [Yan24]). Continuing the setting of Section 5, we let  $(X, d)$  be a proper geodesic metric space. Given a non-elementary subgroup  $\Gamma < \text{Isom}(X)$  with a squeezing isometry, we focus on a certain subset of the conical limit set of  $\Gamma$  and study its properties from the viewpoint of the theories of Coulon and Yang.

In the rest of this section, we fix a basepoint  $x_0 \in X$ . The notion of conformal density plays an important role.

**Definition 6.1.** For  $\Gamma < \text{Isom}(X)$  and  $\delta \geq 0$ , a family of Borel measures  $\{\nu_x\}_{x \in X}$  on  $\partial^h X$  is called a  $\delta$ -dimensional conformal density of  $\Gamma$  if

- ( $\Gamma$ -invariance) for every  $g \in \Gamma$  and  $x \in X$ ,

$$g_*\nu_x = \nu_{gx},$$

- ( $\delta$ -conformality) for every  $x, y \in X$ , the two measures  $\nu_x$  and  $\nu_y$  are in the same measure class and

$$\frac{d\nu_x}{d\nu_y}(\xi) = e^{-\delta \cdot \beta_\xi(x, y)} \quad \text{a.e., and}$$

- (normalization)  $\nu_{x_0}(\partial^h X) = 1$ .

Note that the normalization is only for convenience, in order to have uniqueness of the conformal density in certain cases.

In our setting, Coulon and Yang constructed conformal densities, extending the construction of Patterson [Pat76] and Sullivan [Sul79] in the case where  $X$  is a real hyperbolic space. For a subgroup  $\Gamma < \text{Isom}(X)$  that acts properly on  $X$ , the *critical exponent*  $\delta_\Gamma \geq 0$  of  $\Gamma$  is defined as the abscissa of convergence of the Poincaré series  $s \mapsto \sum_{g \in \Gamma} e^{-sd(x, gx)}$ , for (any)  $x \in X$ .

**Proposition 6.2** ([Cou24, Proposition 4.3], [Yan24, Lemma 6.3]). *Let  $\Gamma < \text{Isom}(X)$  be a non-elementary subgroup such that  $\delta_\Gamma < +\infty$ . Then there exists a  $\delta_\Gamma$ -dimensional conformal density of  $\Gamma$ .*

**6.1. Guided limit sets.** The following notion is a variant of Coulon’s contracting limit sets [Cou24] and Yang’s  $(L, \mathcal{F})$ -limit sets [Yan24].

**Definition 6.3.** Let  $\Gamma < \text{Isom}(X)$  be a non-elementary subgroup. Let  $\varphi \in \Gamma$  be a contracting isometry, let  $C(\varphi) > 0$  be as in Lemma 5.9 and let  $K \geq C(\varphi)$ . We say that  $\xi \in \partial^h X$  is a  $(\varphi, K)$ -guided limit point of  $\Gamma$  if for each sufficiently large  $n \in \mathbb{N}$ , there exists  $h \in \Gamma$  such that

$$(x_0, h[x_0, \varphi^n x_0], \xi) \quad \text{is } K\text{-aligned.}$$

The collection of  $(\varphi, K)$ -guided limit points of  $\Gamma$  is called the  $(\varphi, K)$ -guided limit set of  $\Gamma$ . We denote it by  $\Lambda_{\varphi, K}(\Gamma)$ .

Later,  $\varphi \in \Gamma$  is often assumed to be squeezing. In such cases, we call  $(\varphi, K)$ -guided limit points and the  $(\varphi, K)$ -guided limit set,  $(\varphi, K)$ -squeezed limit points and the  $(\varphi, K)$ -squeezed limit set, respectively.

We first discuss some properties of guided limit sets.

**Lemma 6.4.** *Let  $\Gamma < \text{Isom}(X)$  be a non-elementary subgroup. Let  $\varphi \in \Gamma$  be a contracting isometry and let  $C = C(\varphi) > 0$  be as in Lemma 5.9. Then for each  $K > C$ ,*

$$\Lambda_{\varphi, K}(\Gamma) = \Lambda_{\varphi, C}(\Gamma).$$

*Moreover,  $\Lambda_{\varphi, C}(\Gamma)$  is  $\Gamma$ -invariant.*

*Proof.* Let  $\gamma : \mathbb{R} \rightarrow X$  be an axis of  $\varphi$  chosen for the constant  $C = C(\varphi)$  in Lemma 5.9. Fix  $K > C$ . We then set

$$N = \lceil (K + 100C)/\tau_\varphi \rceil.$$

Now pick an arbitrary  $\xi \in \Lambda_{\varphi, K}(\Gamma)$  and let  $\{z_i\}_{i \in \mathbb{N}} \subset X$  be a sequence converging to  $\xi$ . Since  $\xi$  is  $(\varphi, K)$ -guided, for each large enough  $n \in \mathbb{N}$  there exists  $h \in \Gamma$  such that  $(x_0, h[x_0, \varphi^{n+2N}x_0], z_i)$  is  $K$ -aligned for all large  $i \in \mathbb{N}$ . Since  $d(x_0, \varphi^{n+2N}x_0) \geq (n + 2N)\tau_\varphi > K + C$ , Lemma 5.9(4) tells us that

$$\begin{aligned} \pi_{h\gamma}(x_0) &\subset h\gamma((-\infty, K + C]) \quad \text{and} \\ \pi_{h\gamma}(z_i) &\subset h\gamma\left([(n + 2N)\tau_\varphi - K - C, +\infty)\right) \quad \text{for all large } i \in \mathbb{N}. \end{aligned}$$

We now show that

$$(x_0, h\varphi^N[x_0, \varphi^n x_0], \xi) \text{ is } C\text{-aligned.}$$

Suppose to the contrary that  $(x_0, h\varphi^N[x_0, \varphi^n x_0])$  is not  $C$ -aligned. Then by Lemma 5.9(3), we have  $\pi_{h\gamma}(x_0) \subset h\gamma([\tau_\varphi N, +\infty))$ . Since  $K + C < \tau_\varphi N$ , this is a contradiction. Therefore,  $(x_0, h\varphi^N[x_0, \varphi^n x_0])$  is  $C$ -aligned.

Similarly, using  $(n + 2N)\tau_\varphi - K - C > (n + N)\tau_\varphi$ , we deduce that  $(h\varphi^N[x_0, \varphi^n x_0], z_i)$  is  $C$ -aligned for all large  $i \in \mathbb{N}$ . Since this is the case for an arbitrary sequence  $\{z_i\}_{i \in \mathbb{N}} \subset X$  converging to  $\xi$ ,  $(x_0, h\varphi^N[x_0, \varphi^n x_0], \xi)$  is  $C$ -aligned. We conclude that  $\xi \in \Lambda_{\varphi, C}(\Gamma)$ , proving the first statement.

We now show that  $\Lambda_{\varphi, C}(\Gamma)$  is  $\Gamma$ -invariant. Fix  $\xi \in \Lambda_{\varphi, C}(\Gamma)$  and  $g \in \Gamma$ . Then for each sufficiently large  $n \in \mathbb{N}$ , there exists  $h \in \Gamma$  such that  $(x_0, h[x_0, \varphi^n x_0], \xi)$  is  $C$ -aligned. It is clear that  $(gx_0, gh[x_0, \varphi^n x_0], g\xi)$  is  $C$ -aligned. By Corollary 5.3(1), this implies that  $(x_0, gh[x_0, \varphi^n x_0], g\xi)$  is  $(5C + d(x_0, gx_0))$ -aligned. Hence,  $g\xi$  is a  $(\varphi, 5C + d(x_0, gx_0))$ -guided limit point of  $\Gamma$ , and therefore  $g\xi \in \Lambda_{\varphi, C}(\Gamma)$  by the first statement. This shows the desired  $\Gamma$ -invariance.  $\square$

**Definition 6.5.** We say that a non-elementary subgroup  $\Gamma < \text{Isom}(X)$  is of *divergence type* if  $\delta_\Gamma < +\infty$  and its Poincaré series diverges at  $\delta_\Gamma$ , i.e., we have  $\sum_{g \in \Gamma} e^{-\delta_\Gamma d(x_0, gx_0)} = +\infty$ .

As a part of their generalizations of Hopf–Tsuji–Sullivan dichotomy, Coulon and Yang proved the following:

**Proposition 6.6** ([Yan24, Theorem 1.14], [Cou24, Theorem 1.4]). *Let  $\Gamma < \text{Isom}(X)$  be a non-elementary subgroup of divergence type. Let  $\varphi \in \Gamma$  be a contracting isometry and let  $C = C(\varphi) > 0$  be as in Lemma 5.9. Then for*

every  $\delta_\Gamma$ -dimensional conformal density  $\{\nu_x\}_{x \in X}$  of  $\Gamma$  and for every  $x \in X$ ,  $\Lambda_{\varphi, C}(\Gamma)$  is  $\nu_x$ -conull.

**6.2. Squeezed limit sets and ergodic properties.** As mentioned above, we call the guided limit sets for squeezing isometries the squeezed limit sets.

**Definition 6.7.** Let  $\Gamma < \text{Isom}(X)$  be a non-elementary subgroup containing a squeezing isometry  $\varphi \in \Gamma$ . Let  $C(\varphi) > 0$  be as in Lemma 5.9 and let  $K \geq C(\varphi)$ . We say that  $\xi \in \partial^h X$  is a  $(\varphi, K)$ -squeezed limit point of  $\Gamma$  if for each sufficiently large  $n \in \mathbb{N}$ , there exists  $h \in \Gamma$  such that

$$(x_0, h[x_0, \varphi^n x_0], \xi) \text{ is } K\text{-aligned.}$$

The collection of  $(\varphi, K)$ -squeezed limit points of  $\Gamma$  is called the  $(\varphi, K)$ -squeezed limit set of  $\Gamma$ . We use the same notation  $\Lambda_{\varphi, K}(\Gamma)$  for this.

This notion of squeezed limit sets plays a key role in this paper. Importantly, the following observation allows us to establish the ergodicity of conformal densities on  $\partial^h X$ , as we will see below. In the following, we regard points in  $\partial^h X$  as horofunctions  $X \rightarrow \mathbb{R}$  vanishing at  $x_0$ .

**Lemma 6.8.** Let  $\Gamma < \text{Isom}(X)$  be a non-elementary subgroup containing a squeezing isometry  $\varphi \in \Gamma$ . Let  $C = C(\varphi) > 0$  be as in Lemma 5.9. For  $\xi, \zeta \in \partial^h X$ , if  $\xi \in \Lambda_{\varphi, C}(\Gamma)$  and  $\|\xi - \zeta\|_\infty < +\infty$ , then  $\xi = \zeta$ .

*Proof.* Let  $\gamma : \mathbb{R} \rightarrow X$  be the unique axis of  $\varphi$ . Let  $\xi, \zeta \in \partial^h X$  be given as in the statement, regarding them as horofunctions vanishing at  $x_0$ . Let  $B := \|\xi - \zeta\|_\infty < +\infty$ . This implies that for every  $x \in X$  we have

$$(6.1) \quad \xi(x) =_B \zeta(x).$$

We fix an arbitrary  $x \in X$  and an arbitrary  $\epsilon > 0$ . Our goal is to prove that  $\xi(x) =_\epsilon \zeta(x)$ . Let  $L = L(0.01\epsilon) > 0$  be as in Definition 5.5 for  $\gamma$ , and let  $n \in \mathbb{N}$  be an integer such that

$$n > \frac{100(L + C + B + d(x_0, x))}{\tau_\varphi}.$$

Now, let  $\{z_i\}_{i \in \mathbb{N}} \subset X$  be a sequence converging to  $\xi$ . Since  $\xi \in \Lambda_{\varphi, C}(\Gamma)$ , there exists  $h \in \Gamma$  such that  $(x_0, h[x_0, \varphi^n x_0], z_i)$  is  $C$ -aligned for all large  $i \in \mathbb{N}$ . Since  $n\tau_\varphi - C > C$ , Lemma 5.9(4) tells us that

$$(6.2) \quad \begin{aligned} \pi_{h\gamma}(x_0) &\subset h\gamma((-\infty, 2C]) \quad \text{and} \\ \pi_{h\gamma}(z_i) &\subset h\gamma([n\tau_\varphi - 2C, +\infty)) \quad \text{for all large } i \in \mathbb{N}. \end{aligned}$$

Since  $\pi_{h\gamma}(\cdot)$  is  $(1, 4C)$ -Lipschitz by Corollary 5.3(1), we have

$$\pi_{h\gamma}(x) \subset h\gamma((-\infty, 6C + d(x_0, x)]).$$

By Corollary 5.3(2), Equation (6.2) implies that

$$(6.3) \quad d(z_i, h\gamma(n\tau_\varphi - 2C)) - d(z_i, h\gamma(n\tau_\varphi - 19C - 2B)) =_{8C} -(2B + 17C).$$

Now let  $\{z'_i\}_{i \in \mathbb{N}} \subset X$  be a sequence converging to  $\zeta$ .

**Claim.** *We have*

$$(6.4) \quad \pi_{h\gamma}(z'_i) \subset h\gamma([n\tau_\varphi - 20C - 2B, +\infty)) \quad \text{for all large } i \in \mathbb{N}.$$

Suppose to the contrary that, passing to a subsequence, we have

$$(6.5) \quad \pi_{h\gamma}(z'_i) \subset h\gamma((-\infty, n\tau_\varphi - 18C - 2B]) \quad \text{for all } i \in \mathbb{N}.$$

Here, we used the fact that  $\text{Diam } \pi_{h\gamma}(z'_i) \leq 2C$ . By Corollary 5.3(2), Equation (6.5) implies that

$$d(z'_i, h\gamma(n\tau_\varphi - 2C)) - d(z'_i, h\gamma(n\tau_\varphi - 18C - 2B)) =_{8C} 2B + 16C.$$

Comparing this with Equation (6.3), we observe that  $\xi(a) - \xi(b)$  and  $\zeta(a) - \zeta(b)$  differ by more than  $4B$  for the two points  $a := h\gamma(n\tau_\varphi - 2C)$  and  $b := h\gamma(n\tau_\varphi - 19C - 2B)$ . This contradicts Equation (6.1). The claim follows.

Hence, Equation (6.4) holds. Therefore, for each large enough  $i$  we have

$$\begin{aligned} \pi_{h\gamma}(x_0), \pi_{h\gamma}(x) &\subset h\gamma((-\infty, 6C + d(x_0, x)]) \quad \text{and} \\ \pi_{h\gamma}(z_i), \pi_{h\gamma}(z'_i) &\subset h\gamma([n\tau_\varphi - 20C - 2B, +\infty)). \end{aligned}$$

Since  $n\tau_\varphi - 20C - 2B \geq 6C + d(x_0, x) + 2L$ , we conclude from Lemma 5.6 that  $d(x_0, z_i) - d(x, z_i) =_\epsilon d(x_0, z'_i) - d(x, z'_i)$ . Since both  $\xi$  and  $\zeta$  are horofunctions vanishing at  $x_0$ , taking the limit  $i \rightarrow +\infty$  yields  $\xi(x) =_\epsilon \zeta(x)$ .  $\square$

To discuss ergodicity of conformal densities, Coulon [Cou24] considered conformal densities restricted to the *reduced horoboundary* instead of the usual horoboundary. Yang [Yan24] considered another notion encompassing the reduced horoboundary, namely, the *reduced convergence boundary*.

Instead of defining these objects precisely, let us point out that Coulon's and Yang's ergodicity results are formulated in terms of saturated Borel subsets of  $\partial^h X$ , that is,  $E \subset \partial^h X$  with the property that if  $\xi \in E$  and  $\zeta \in \partial^h X$  satisfy  $\|\xi - \zeta\|_\infty < +\infty$ , then  $\zeta \in E$ . Hence, by Proposition 6.6 and Lemma 6.8, their ergodicity results can be stated as follows:

**Proposition 6.9** ([Yan24, Theorem 1.16], [Cou24, Theorem 1.5]). *Let  $\Gamma < \text{Isom}(X)$  be a non-elementary subgroup containing a squeezing isometry, and let  $\{\nu_x\}_{x \in X}$  be a  $\delta_\Gamma$ -dimensional conformal density of  $\Gamma$ . If  $\Gamma$  is of divergence type, then the  $\Gamma$ -action on  $(\partial^h X, \nu_x)$  is ergodic for all  $x \in X$ .*

The above ergodicity implies the uniqueness of the  $\delta_\Gamma$ -dimensional conformal density when  $\Gamma$  is as in Proposition 6.9. Indeed, if  $\{\nu_x\}_{x \in X}$  and  $\{\nu'_x\}_{x \in X}$  are  $\delta_\Gamma$ -dimensional conformal densities of  $\Gamma$ , then  $\{(\nu_x + \nu'_x)/2\}_{x \in X}$  is also a  $\delta_\Gamma$ -dimensional conformal density of  $\Gamma$ . Since  $\nu_x$  is absolutely continuous with respect to  $(\nu_x + \nu'_x)/2$  for each  $x \in X$ , the ergodicity applies to their Radon–Nikodym derivatives which are  $\Gamma$ -invariant due to the conformality. Therefore,  $\nu_x = \nu'_x$  for all  $x \in X$ .

## 7. RIGIDITY OF ERGODIC INVARIANT RADON MEASURES

Let  $(X, d)$  be a proper, geodesic metric space. In this section, we prove the rigidity of ergodic, invariant Radon measures (Theorem 1.13).

**7.1. Candidates for measures.** We first define a Radon measure which will be the unique measure with the desired properties in our rigidity theorem. To do this, we fix a basepoint  $x_0 \in X$  and use the identification  $\mathcal{H} = \partial^h X \times \mathbb{R}$  as in Equation (5.2). Recall that the  $\text{Isom}(X)$ -action on  $\mathcal{H}$  is written as follows: for  $g \in \text{Isom}(X)$  and  $(\xi, t) \in \mathcal{H}$ , we have

$$g(\xi, t) = (g\xi, t + \beta_\xi(g^{-1}x_0, x_0)).$$

Via this identification, we define a Radon measure on  $\mathcal{H}$  as follows:

**Definition 7.1.** Let  $\Gamma < \text{Isom}(X)$  be a non-elementary subgroup containing a squeezing isometry and let  $\nu := \{\nu_x\}_{x \in X}$  be a  $\delta_\Gamma$ -dimensional conformal density of  $\Gamma$ . We define a Radon measure  $\mu_\nu$  on  $\mathcal{H} = \partial^h X \times \mathbb{R}$  by setting

$$d\mu_\nu(\xi, t) := e^{\delta_\Gamma \cdot t} \cdot d\nu_{x_0}(\xi) dt.$$

When, in addition,  $\Gamma$  is of divergence type, we write

$$\mu_\Gamma := \mu_\nu.$$

*Remark 7.2.* It follows from the conformality of  $\nu$  that  $\mu_\nu$  is  $\Gamma$ -invariant. Moreover, considering  $\mathcal{H}$  as a subspace of  $\text{Lip}^1(X)$ , the measure  $\mu_\Gamma$  does not depend on the choice of  $x_0 \in X$ . When  $\Gamma$  is of divergence type, there exists a unique  $\delta_\Gamma$ -dimensional conformal density  $\{\nu_x\}_{x \in X}$  of  $\Gamma$  by Proposition 6.2 and Proposition 6.9. This is the reason for writing  $\mu_\Gamma = \mu_\nu$  in this case.

To present a precise statement, we also consider the following notion concerning the distribution of translation lengths of squeezing isometries.

**Definition 7.3.** For a subgroup  $\Gamma < \text{Isom}(X)$ , its *squeezing spectrum* is

$$\text{Spec}_{sq}(\Gamma) := \{\tau_g \in \mathbb{R} : g \in \Gamma \text{ is a squeezing isometry}\}.$$

We say that the squeezing spectrum is *non-arithmetic* if it generates a dense additive subgroup of  $\mathbb{R}$ .

**7.2. Rigidity of measures.** Let us restate Theorem 1.13, our main rigidity theorem. Recall the notion of the conical limit set from Definition 5.17.

**Theorem 7.4.** *Let  $\Gamma < \text{Isom}(X)$  be a non-elementary subgroup with non-arithmetic squeezing spectrum. Suppose that there exists a non-zero  $\Gamma$ -invariant Radon measure  $\mu$  on  $\mathcal{H}$  supported on  $\Lambda_c(\Gamma) \times \mathbb{R}$ . Then  $\Gamma$  is of divergence type and  $\mu$  is a constant multiple of  $\mu_\Gamma$ .*

The rest of this section is devoted to the proof of Theorem 7.4. We prove the theorem by establishing a robust relationship between invariant Radon measures and squeezed limit sets. Note that by the ergodic decomposition, it suffices to consider invariant *ergodic* measures.

**7.3. Concentration on squeezed limit sets.** We first show that invariant ergodic Radon measures on  $\mathcal{H}$  are charged on squeezed limit sets. We emphasize that in the following, we consider an arbitrary Radon measure on  $\mathcal{H}$ , not necessarily induced by a conformal density.

**Theorem 7.5.** *Let  $\Gamma < \text{Isom}(X)$  be a non-elementary subgroup, let  $\varphi$  be a contracting isometry in  $\Gamma$  and let  $C = C(\varphi)$  be as in Lemma 5.9. Let  $\mu$  be a  $\Gamma$ -invariant ergodic Radon measure on  $\mathcal{H}$  supported on  $\Lambda_c(\Gamma) \times \mathbb{R}$ . Then the measure  $\mu$  is supported on*

$$\Lambda_{\varphi, C}(\Gamma) \times \mathbb{R} \subset \mathcal{H}.$$

*Proof.* Let  $\kappa(\varphi) > 0$  and  $a_1, a_2, a_3 \in \Gamma$  be as in Lemma 5.12 for  $\Gamma$ . Let  $C(\varphi) > 0$  be as in Lemma 5.9 for  $g = \varphi$ . We set  $C_0 := 10(\kappa(\varphi) + C(\varphi))$ .

For each  $K > 0$  let

$$\mathcal{H}_K := \left\{ (\xi, t) \in \mathcal{H} : \begin{array}{l} \exists \text{ an infinite sequence } \{g_i\}_{i \in \mathbb{N}} \subset \Gamma \text{ s.t.} \\ \beta_\xi(x_0, g_i x_0) \geq d(x_0, g_i x_0) - K \text{ for all } i \in \mathbb{N} \end{array} \right\}.$$

Then  $\Gamma \cdot \mathcal{H}_K \subset \mathcal{H}$  is  $\Gamma$ -invariant. Moreover,

$$\Lambda_c(\Gamma) \times \mathbb{R} = \bigcup_{K > 0} \Gamma \cdot \mathcal{H}_K.$$

Since  $\Lambda_c(\Gamma) \times \mathbb{R}$  has positive  $\mu$ -value,

$$\Gamma \cdot \mathcal{H}_K \text{ has positive } \mu\text{-value for all large } K > 0.$$

We fix such  $K > 100C_0 + 2 \sum_{i=1}^3 d(x_0, a_i x_0)$ . Then it follows from the  $\Gamma$ -invariance of  $\mu$  that  $\mu(\mathcal{H}_K) > 0$ . For each  $R > 0$ , we set  $\mathcal{H}_{K,R} := \{(\xi, t) \in \mathcal{H}_K : -R \leq t \leq R\}$ . Since  $\mathcal{H}_K = \bigcup_{R=1}^{\infty} \mathcal{H}_{K,R}$ ,

$$\mu(\mathcal{H}_{K,R}) > 0 \text{ for all large } R > 0.$$

We fix such  $R > 0$ .

Now we pick  $n > 100(C_0 + K)/\tau_\varphi$  and  $k \in \mathbb{N}$ . We define a map

$$F = F_{n,k} : \mathcal{H}_{K,R} \rightarrow \mathcal{H}$$

as follows. For each  $\mathfrak{X} \in \mathcal{H}_{K,R}$ , there exists  $g \in \Gamma$  such that

$$(7.1) \quad d(x_0, gx_0) > k \quad \text{and} \quad \beta_{\mathfrak{X}}(x_0, gx_0) \geq d(x_0, gx_0) - K.$$

Among all such  $g$ 's, take the one with minimal  $d(x_0, gx_0)$  and call it  $g_{\mathfrak{X}}$ .<sup>4</sup> Then the map  $\mathfrak{X} \in \mathcal{H}_{K,R} \mapsto g_{\mathfrak{X}}$  is Borel measurable. By Lemma 5.15, there exists  $a_{\mathfrak{X}} \in \{a_1, a_2, a_3\}$  such that<sup>5</sup>

$$(7.2) \quad (x_0, g_{\mathfrak{X}} \cdot a_{\mathfrak{X}}[x_0, \varphi^n x_0], g_{\mathfrak{X}} \cdot a_{\mathfrak{X}} \varphi^n a_{\mathfrak{X}} \cdot g_{\mathfrak{X}}^{-1} \mathfrak{X}) \text{ is } C_0\text{-aligned}.$$

This map  $\mathfrak{X} \mapsto a_{\mathfrak{X}}$  is also Borel measurable. We now set

$$F(\mathfrak{X}) := g_{\mathfrak{X}} \cdot a_{\mathfrak{X}} \varphi^n a_{\mathfrak{X}} \cdot g_{\mathfrak{X}}^{-1} \mathfrak{X}.$$

<sup>4</sup>A technicality arises when several candidates tie. An easy remedy is to first enumerate  $\Gamma = \{g^{(1)}, g^{(2)}, \dots\}$ , and choose the earliest whenever there is a tie.

<sup>5</sup>Again, when more than one of  $\{a_1, a_2, a_3\}$  do the job we choose the earliest.

Let

$$D := 100 \left( C_0 + \tau_\varphi n + \sum_{i=1}^3 d(x_0, a_i x_0) \right).$$

**Claim.** *We have that*

(7.3)  *$F$  is at most  $M$ -to-one, where  $M := 3 \cdot \#\{g \in \Gamma : d(x_0, gx_0) \leq D\}$ .*

To prove this claim, suppose that we have  $\mathfrak{X}, \mathfrak{X}' \in \mathcal{H}_{K,R}$  with the same image  $F(\mathfrak{X}) = F(\mathfrak{X}') =: (\xi_0, t_0) \in \partial^h X \times \mathbb{R}$ .

Let  $\{u_i\}_{i \in \mathbb{N}} \subset X$  be a sequence converging to  $\xi_0$ . Up to a subsequence,

$$(x_0, g_{\mathfrak{X}} a_{\mathfrak{X}}[x_0, \varphi^n x_0], u_i) \quad \text{is } C_0\text{-aligned for all } i \in \mathbb{N}.$$

Here, note that  $d(x_0, \varphi^n x_0) \geq \tau_\varphi n \geq 100C_0$ . It follows from Lemma 5.2 that for each  $i \in \mathbb{N}$ , there exists  $P \in [x_0, u_i]$  such that  $d(g_{\mathfrak{X}} a_{\mathfrak{X}} x_0, P) \leq 3C_0$ , and similarly, there exists  $Q \in [x_0, u_i]$  such that  $d(g_{\mathfrak{X}'} a_{\mathfrak{X}'} x_0, Q) \leq 3C_0$ . We have three cases.

- (1) If  $d(x_0, g_{\mathfrak{X}} x_0) < d(x_0, g_{\mathfrak{X}'} x_0) - 0.5D$ : In this case, along  $[x_0, u_i]$ ,  $P$  is closer to  $x_0$  than  $Q$  is, and  $d(P, Q) \geq 0.5D - 6C_0$ . For convenience we write  $\eta := g_{\mathfrak{X}} a_{\mathfrak{X}}[x_0, \varphi^n x_0]$ . Note that

$$\begin{aligned} d(u_i, \eta) &\geq d(u_i, g_{\mathfrak{X}} a_{\mathfrak{X}} x_0) - d(g_{\mathfrak{X}} a_{\mathfrak{X}} x_0, g_{\mathfrak{X}} a_{\mathfrak{X}} \varphi^n x_0) \\ &\geq d(u_i, P) - 3C_0 - d(x_0, \varphi^n x_0) \\ &\geq d(u_i, Q) + 0.2D \\ &> d(u_i, g_{\mathfrak{X}'} a_{\mathfrak{X}'} x_0) + d(a_{\mathfrak{X}'} x_0, x_0) + C_0 \\ &\geq d(u_i, g_{\mathfrak{X}'} x_0) + C_0. \end{aligned}$$

This implies that  $d(\eta, [u_i, g_{\mathfrak{X}'} x_0]) \geq C_0$ . Since  $\eta$  is  $C_0$ -contracting,  $\text{Diam } \pi_\eta([u_i, g_{\mathfrak{X}'} x_0]) \leq C_0$ . As  $(x_0, \eta, u_i)$  is  $C_0$ -aligned, we also have that

$$(7.4) \quad (x_0, \eta, g_{\mathfrak{X}'} x_0) \quad \text{is } 2C_0\text{-aligned}.$$

Denote by  $\xi' \in \partial^h X$  the Busemann cocycle component of  $\mathfrak{X}'$  and let  $\{z'_i\}_{i \in \mathbb{N}} \subset X$  be a sequence converging to  $\xi'$ . We claim that

$$(7.5) \quad (\eta, z'_i) \quad \text{is } (K + 30C_0)\text{-aligned for all large } i \in \mathbb{N}.$$

Suppose to the contrary that, passing to a subsequence,  $(\eta, z'_i)$  is never  $(K + 30C_0)$ -aligned. We choose

$$p \in \pi_\eta(x_0), \quad q \in \pi_\eta(z'_i), \quad \text{and} \quad r \in \pi_\eta(g_{\mathfrak{X}'} x_0).$$

By Equation (7.4) and the assumption that  $(\eta, z'_i)$  is not  $(K + 30C_0)$ -aligned,  $r$  is closer to  $g_{\mathfrak{X}} a_{\mathfrak{X}} \varphi^n x_0$  than  $p$  and  $q$  are. Moreover, we have  $d(p, r), d(q, r) \geq K + 25C_0$ . Now, it follows from Lemma 5.2 that

$$\begin{aligned} d(x_0, z'_i) &\leq d(x_0, \eta) + d(p, q) + d(\eta, z'_i), \\ d(g_{\mathfrak{X}'} x_0, z'_i) &=_{8C_0} d(g_{\mathfrak{X}'} x_0, \eta) + d(r, q) + d(\eta, z'_i), \quad \text{and} \\ d(x_0, g_{\mathfrak{X}'} x_0) &=_{8C_0} d(x_0, \eta) + d(p, r) + d(\eta, g_{\mathfrak{X}'} x_0). \end{aligned}$$

Then we have

$$\begin{aligned} d(x_0, g_{\mathfrak{X}'}x_0) + d(g_{\mathfrak{X}'}x_0, z'_i) - d(x_0, z'_i) \\ \geq 2d(g_{\mathfrak{X}'}x_0, \eta) + 2\min(d(p, r), d(q, r)) - 16C_0 \\ \geq 2K + 34C_0. \end{aligned}$$

In particular,  $d(x_0, z'_i) - d(g_{\mathfrak{X}'}x_0, z'_i) \leq d(x_0, g_{\mathfrak{X}'}x_0) - K - C_0$ . Taking the limit  $i \rightarrow +\infty$ , we have

$$\beta_{\mathfrak{X}'}(x_0, g_{\mathfrak{X}'}x_0) \leq d(x_0, g_{\mathfrak{X}'}x_0) - K - C_0.$$

This contradicts the definition of  $g_{\mathfrak{X}'}$ , and therefore Equation (7.5) follows.

By Lemma 5.2, it follows from Equation (7.4) and Equation (7.5) that  $[x_0, z'_i]$  passes through the  $4C_0$ -neighborhood of  $g_{\mathfrak{X}}a_{\mathfrak{X}}x_0$ , for all large  $i \in \mathbb{N}$ . This implies

$$\beta_{\mathfrak{X}'}(x_0, g_{\mathfrak{X}}a_{\mathfrak{X}}x_0) \geq d(x_0, g_{\mathfrak{X}}a_{\mathfrak{X}}x_0) - 8C_0.$$

Since  $K > 100C_0 + 2\sum_{i=1}^3 d(x_0, a_i x_0)$ , we have

$$\begin{aligned} \beta_{\mathfrak{X}'}(x_0, g_{\mathfrak{X}}x_0) &= \beta_{\mathfrak{X}'}(x_0, g_{\mathfrak{X}}a_{\mathfrak{X}}x_0) + \beta_{\mathfrak{X}'}(g_{\mathfrak{X}}a_{\mathfrak{X}}x_0, g_{\mathfrak{X}}x_0) \\ &\geq d(x_0, g_{\mathfrak{X}}a_{\mathfrak{X}}x_0) - 8C_0 - d(a_{\mathfrak{X}}x_0, x_0) \\ &\geq d(x_0, g_{\mathfrak{X}}x_0) - 8C_0 - 2d(a_{\mathfrak{X}}x_0, x_0) \\ &> d(x_0, g_{\mathfrak{X}}x_0) - K. \end{aligned}$$

Meanwhile, we also have  $k < d(x_0, g_{\mathfrak{X}}x_0) < d(x_0, g_{\mathfrak{X}'}x_0) - 0.5D$ . This contradicts the definition of  $g_{\mathfrak{X}'}$ , according to which  $d(x_0, g_{\mathfrak{X}'}x_0)$  is minimal among the elements of  $\Gamma$  satisfying Equation (7.1).

- (2) If  $d(x_0, g_{\mathfrak{X}'}x_0) < d(x_0, g_{\mathfrak{X}}x_0) - 0.5D$ : In this case, one obtains a contradiction to that in (1).
- (3) If  $d(x_0, g_{\mathfrak{X}}x_0) =_{0.5D} d(x_0, g_{\mathfrak{X}'}x_0)$ : Recall that for each fixed  $i \in \mathbb{N}$ , we have  $P, Q \in [x_0, u_i]$  such that  $d(g_{\mathfrak{X}}a_{\mathfrak{X}}x_0, P), d(g_{\mathfrak{X}'}a_{\mathfrak{X}'}x_0, Q) \leq 3C_0$ . Hence, we have

$$d(g_{\mathfrak{X}}x_0, P), d(g_{\mathfrak{X}'}x_0, Q) \leq 0.1D.$$

Since both  $P$  and  $Q$  belong to the geodesic  $[x_0, u_i]$ , this, together with  $d(x_0, g_{\mathfrak{X}}x_0) =_{0.5D} d(x_0, g_{\mathfrak{X}'}x_0)$ , implies

$$d(g_{\mathfrak{X}}x_0, g_{\mathfrak{X}'}x_0) \leq D.$$

Now when  $\mathfrak{X}$  is given (and hence  $a_{\mathfrak{X}}, g_{\mathfrak{X}}$  are given as well),

$$\mathfrak{X}' = g_{\mathfrak{X}'}a_{\mathfrak{X}'}^{-1}\varphi^{-n}a_{\mathfrak{X}'}^{-1}(g_{\mathfrak{X}}^{-1}g_{\mathfrak{X}})a_{\mathfrak{X}}\varphi^n a_{\mathfrak{X}}g_{\mathfrak{X}}^{-1}\mathfrak{X}$$

is determined by  $g_{\mathfrak{X}'}^{-1}g_{\mathfrak{X}}$  and  $a_{\mathfrak{X}'} \in \{a_1, a_2, a_3\}$ . The number of these choices is at most  $M$ .

Therefore, Equation (7.3) follows.

Then we have

$$\begin{aligned}
\mu(F(\mathcal{H}_{K,R})) &= \mu \left( \bigcup_{g \in \Gamma, a \in \{a_1, a_2, a_3\}} F(\{\mathfrak{X} \in \mathcal{H}_{K,R} : g\mathfrak{X} = g, a_{\mathfrak{X}} = a\}) \right) \\
&\geq \frac{1}{M} \sum_{g \in \Gamma, a \in \{a_1, a_2, a_3\}} \mu(F(\{\mathfrak{X} \in \mathcal{H}_{K,R} : g\mathfrak{X} = g, a_{\mathfrak{X}} = a\})) \\
&= \frac{1}{M} \sum_{g \in \Gamma, a \in \{a_1, a_2, a_3\}} \mu(ga\varphi^n ag^{-1}\{\mathfrak{X} \in \mathcal{H}_{K,R} : g\mathfrak{X} = g, a_{\mathfrak{X}} = a\}) \\
&= \frac{1}{M} \sum_{g \in \Gamma, a \in \{a_1, a_2, a_3\}} \mu(\{\mathfrak{X} \in \mathcal{H}_{K,R} : g\mathfrak{X} = g, a_{\mathfrak{X}} = a\}) \\
&= \frac{1}{M} \mu(\mathcal{H}_{K,R}).
\end{aligned}$$

To describe the image of  $F$ , let  $\mathfrak{X} = (\xi, t) \in \mathcal{H}_{K,R}$ . For simplicity, write  $g := g_{\mathfrak{X}}$  and  $a := a_{\mathfrak{X}}$ . Then

$$F(\mathfrak{X}) = (ga\varphi^n ag^{-1}\xi, t + \beta_{\xi}((ga\varphi^n ag^{-1})^{-1}x_0, x_0)).$$

For each sequence  $\{u_i\}_{i \in \mathbb{N}} \subset X$  converging to  $ga\varphi^n ag^{-1}\xi \in \partial^h X$ , we have

$$\begin{aligned}
\beta_{\xi}((ga\varphi^n ag^{-1})^{-1}x_0, x_0) &= \beta_{ga\varphi^n ag^{-1}\xi}(x_0, ga\varphi^n ag^{-1}x_0) \\
&= \lim_{i \rightarrow +\infty} d(x_0, u_i) - d(ga\varphi^n ag^{-1}x_0, u_i).
\end{aligned}$$

By Equation (7.2) and Lemma 5.2, we have for all large  $i \in \mathbb{N}$  that

$$\begin{aligned}
\beta_{\xi}((ga\varphi^n ag^{-1})^{-1}x_0, x_0) &=_{15C_0} d(x_0, gax_0) + d(x_0, \varphi^n x_0) + d(ga\varphi^n x_0, u_i) \\
&\quad - d(ga\varphi^n ag^{-1}x_0, u_i) \\
&=_{C_0} d(x_0, gax_0) + d(x_0, \varphi^n x_0) \\
&\quad + \beta_{ga\varphi^n ag^{-1}\xi}(ga\varphi^n x_0, ga\varphi^n ag^{-1}x_0) \\
&=_{d(x_0, ax_0)} d(x_0, gx_0) + d(x_0, \varphi^n x_0) \\
&\quad + \beta_{\xi}(ga^{-1}x_0, gx_0) + \beta_{\xi}(gx_0, x_0) \\
&=_{d(x_0, ax_0)} d(x_0, gx_0) + d(x_0, \varphi^n x_0) + \beta_{\xi}(gx_0, x_0) \\
&=_K d(x_0, \varphi^n x_0).
\end{aligned}$$

Therefore,

$$t + \beta_{\xi}((ga\varphi^n ag^{-1})^{-1}x_0, x_0) \in [-R - D, R + D].$$

In addition, by Equation (7.2), we have  $d(x_0, gax_0) > k - \sum_{i=1}^3 d(x_0, a_i x_0)$ , and moreover,  $(x_0, ga[x_0, \varphi^n x_0], F(\mathfrak{X}))$  is  $C_0$ -aligned.

This implies that  $F(\mathcal{H}_{K,R})$  is contained in

$$B_{k;n} := \left\{ (\zeta, s) \in \mathcal{H} : \begin{array}{l} -R - D \leq s \leq R + D \text{ and } \exists h \in \Gamma \text{ such that} \\ d(x_0, hx_0) > k - \sum_{i=1}^3 d(x_0, a_i x_0) \text{ and} \\ (x_0, h[x_0, \varphi^n x_0], \zeta) \text{ is } C_0\text{-aligned} \end{array} \right\}.$$

Hence, we have

$$\mu(B_{k;n}) \geq \mu(\mathcal{H}_{K,R})/M > 0.$$

Note that the set  $B_{k;n}$  is decreasing in  $k$ . Since  $\mu$  is a Radon measure and  $B_{k;n} \subset \partial^h X \times [-R-D, R+D]$  which is *compact*, we have  $\mu(B_{k;n}) < +\infty$ . Therefore, setting

$$B_n := \bigcap_{k>0} B_{k;n},$$

we have

$$\mu(B_n) = \lim_{k \rightarrow +\infty} \mu(B_{k;n}) \geq \mu(\mathcal{H}_{K,R})/M > 0.$$

Now,  $\Gamma \cdot B_n$  is a  $\Gamma$ -invariant set of positive  $\mu$ -measure. Hence, by the  $\Gamma$ -ergodicity of  $\mu$ , we have that  $\Gamma \cdot B_n$  is  $\mu$ -conull, and therefore

$$\bigcap_n \Gamma \cdot B_n \text{ is } \mu\text{-conull.}$$

We then show that for each  $(\zeta, s) \in \bigcap_n \Gamma \cdot B_n$ , we have  $\zeta \in \Lambda_{\varphi, 2C_0}(\Gamma)$ . This finishes the proof by Lemma 6.4.

Let  $(\zeta, s) \in \bigcap_n \Gamma \cdot B_n$ . Then for each large enough  $n \in \mathbb{N}$ , there exists  $h_0 \in \Gamma$  such that

$$(x_0, h[x_0, \varphi^n x_0], h_0^{-1}\zeta) \text{ is } C_0\text{-aligned for infinitely many } h \in \Gamma.$$

In other words,

$$(h_0 x_0, h_0 h[x_0, \varphi^n x_0], \zeta) \text{ is } C_0\text{-aligned for infinitely many } h \in \Gamma.$$

Among the infinitely many such  $h \in \Gamma$ , we can choose one such that

$$d(h_0 x_0, h_0 h[x_0, \varphi^n x_0]) > d(x_0, h_0 x_0) + C_0$$

and hence

$$d([h_0 x_0, x_0], h_0 h[x_0, \varphi^n x_0]) > C_0.$$

Since  $h_0 h[x_0, \varphi^n x_0]$  is  $C_0$ -contracting, we now have that  $(x_0, h_0 h[x_0, \varphi^n x_0])$  is  $2C_0$ -aligned. Therefore,

$$(x_0, h_0 h[x_0, \varphi^n x_0], \zeta) \text{ is } 2C_0\text{-aligned.}$$

Since this holds for all large  $n \in \mathbb{N}$ , we conclude that  $\zeta \in \Lambda_{\varphi, 2C_0}(\Gamma)$ .  $\square$

Theorem 7.5 applies to each contracting isometry of  $\Gamma$ . We thus define:

**Definition 7.6.** Let  $\Gamma < \text{Isom}(X)$  be a non-elementary subgroup. For each contracting isometry  $\varphi \in \Gamma$ , let  $C(\varphi) > 0$  as in Lemma 5.9. We then define

$$\begin{aligned} \Lambda_{\text{Gui}}(\Gamma) &:= \bigcap_{\varphi \in \Gamma, \text{ contracting}} \Lambda_{\varphi, C(\varphi)}(\Gamma) \\ \Lambda_{\text{Sq}}(\Gamma) &:= \bigcap_{\varphi \in \Gamma, \text{ squeezing}} \Lambda_{\varphi, C(\varphi)}(\Gamma) \end{aligned}$$

and call them the *guided limit set* and the *squeezed limit set* of  $\Gamma$ , respectively. In other words, every point in  $\Lambda_{\text{Gui}}(\Gamma)$  is a  $(\varphi, C(\varphi))$ -guided limit point of  $\Gamma$  for all contracting  $\varphi \in \Gamma$ .

*Remark 7.7.* The set  $\Lambda_{\text{Gui}}(\Gamma)$  is closely related to the *Myrberg limit set*. A point  $\xi \in \partial^h X$  is called a *Myrberg limit point* of  $\Gamma$  if there exists  $c > 0$  such that for each  $g \in \Gamma$ , there exists  $h \in \Gamma$  such that

$[x_0, \xi)$  passes through the  $c$ -neighborhoods of  $hx_0$  and  $hgx_0$ , in this order.

The Myrberg limit set is the set of all Myrberg limit points of  $\Gamma$ .

When  $\Gamma < \text{Isom}(X)$  is non-elementary, Yang proved in [Yan24, Corollary 4.17] that the Myrberg limit set of  $\Gamma$  is equal to the intersection of  $(\varphi, C(\varphi))$ -guided limit sets for all contracting isometries  $\varphi \in \Gamma$ .

In view of Lemma 6.4, Theorem 7.5 implies the following:

**Corollary 7.8.** *Let  $\Gamma < \text{Isom}(X)$  be a non-elementary subgroup. Let  $\mu$  be a  $\Gamma$ -invariant ergodic Radon measure on  $\mathcal{H}$  supported on  $\Lambda_c(\Gamma) \times \mathbb{R}$ . Then the measure  $\mu$  is supported on*

$$\Lambda_{\text{Gui}}(\Gamma) \times \mathbb{R} \subset \mathcal{H}.$$

**7.4. Neighborhoods in squeezed limit sets.** We have seen that the squeezed limit sets are the relevant region for invariant ergodic measures. Let us now introduce a notion of neighborhoods of squeezed limit points. For  $g, \varphi \in \text{Isom}(X)$ ,  $C > 0$ , and  $n \in \mathbb{N}$ , we set

$$U_C(g; \varphi, n) := \left\{ \xi \in \partial^h X : (x_0, g[x_0, \varphi^n x_0], \xi) \text{ is } C\text{-aligned} \right\}.$$

The case of interest is when  $\varphi$  is a squeezing isometry.

**Lemma 7.9.** *Let  $\Gamma < \text{Isom}(X)$  be a non-elementary subgroup containing a squeezing isometry  $\varphi \in \Gamma$ , and let  $C = C(\varphi) > 0$  be as in Lemma 5.9. Then*

$$\{U_C(g; \varphi, n) : g \in \Gamma, n \in \mathbb{N}\}$$

*forms a basis for the topology of  $\Lambda_{\varphi, C}(\Gamma) \subset \partial^h X$ .*

*In other words, for each  $\xi \in \Lambda_{\varphi, C}(\Gamma)$ , for each open set  $O \subset \partial^h X$  with  $\xi \in O$  and for each  $N \in \mathbb{N}$ , there exist  $g \in \Gamma$ ,  $n > N$ , and an open set  $V \subset \partial^h X$  such that*

$$\xi \in V \subset U_C(g; \varphi, n) \subset O.$$

*Proof.* Let  $\gamma : \mathbb{R} \rightarrow X$  be the unique axis of  $\varphi$ . We fix  $\xi \in \Lambda_{\varphi, C}(\Gamma)$ , an open set  $O \subset \partial^h X$  containing  $\xi$ , and  $N \in \mathbb{N}$ .

Let us first recall that  $\partial^h X$  carries the compact-open topology. Hence, the sets of the form

$$O_{R, \epsilon} := \{\zeta \in \partial^h X : |\xi(x) - \zeta(x)| < \epsilon \text{ for all } x \in \mathcal{N}_R(x_0)\} \quad \text{for } R, \epsilon > 0$$

form a local basis at  $\xi$ . Fix  $R, \epsilon > 0$  such that  $O_{R, \epsilon} \subset O$ . We may assume that  $\epsilon < C$ .

Let  $L = L(0.01\epsilon) > 0$  be as in Definition 5.5 for  $\gamma$ . Let  $R' := \lceil \frac{R+100C}{\tau_\varphi} \rceil$  and  $L' := \lceil \frac{2L+100C}{\tau_\varphi} \rceil$ , and take  $k \in \mathbb{N}$  such that

$$k > R' + L' + \lceil 100C/\tau_\varphi \rceil + N.$$

Let  $\{z_i\}_{i \in \mathbb{N}} \subset X$  be a sequence converging to  $\xi$ . Since  $\xi \in \Lambda_{\varphi, C}(\Gamma)$ , there exists  $h \in \Gamma$  such that  $\xi \in U_C(h; \varphi, k)$ , i.e.,

$$(x_0, h[x_0, \varphi^k x_0], z_i) \text{ is } C\text{-aligned for all large } i \in \mathbb{N}.$$

Since  $\tau_\varphi k > 2C$ , Lemma 5.9(4) tells us that

$$(7.6) \quad \begin{aligned} \pi_{h\gamma}(x_0) &\subset h\gamma((-\infty, 2C]) \quad \text{and} \\ \pi_{h\gamma}(z_i) &\subset h\gamma([\tau_\varphi k - 2C, +\infty)) \quad \text{for all large } i \in \mathbb{N}. \end{aligned}$$

Since  $\pi_{h\gamma}(\cdot)$  is  $(1, 4C)$ -Lipschitz by Corollary 5.3(1), we have

$$(7.7) \quad \pi_{h\gamma}(x) \subset h\gamma((-\infty, 6C + R]) \quad \text{for all } x \in \mathcal{N}_R(x_0).$$

In view of Equation (7.6) and Lemma 5.9(3),  $(x_0, h\varphi^{R'}[x_0, \varphi^{L'}x_0])$  is  $C$ -aligned. Similarly,  $(h\varphi^{R'}[x_0, \varphi^{L'}x_0], z_i)$  is  $C$ -aligned eventually. Therefore,

$$\xi \in U_C(h\varphi^{R'}; \varphi, L').$$

Now let  $\zeta \in U_C(h\varphi^{R'}; \varphi, L')$  and take a sequence  $\{z'_i\}_{i \in \mathbb{N}} \subset X$  converging to  $\zeta$ . Then by Lemma 5.9(4),

$$\pi_{h\gamma}(z'_i) \subset h\gamma([(R' + L')\tau_\varphi - 2C, +\infty)) \quad \text{for all large } i \in \mathbb{N}.$$

Combining this with Equation (7.6) and Equation (7.7), we can apply Lemma 5.6 and conclude that for each  $x \in \mathcal{N}_R(x_0)$ ,

$$d(x, z_i) - d(x_0, z_i) =_{\epsilon/2} d(x, z'_i) - d(x_0, z'_i) \quad \text{for all large } i \in \mathbb{N}.$$

This implies that  $|\xi(x) - \zeta(x)| < \epsilon$  for all  $x \in \mathcal{N}_R(x_0)$ , and therefore

$$\xi \in U_C(h\varphi^{R'}; \varphi, L') \subset O_{R, \epsilon} \subset O.$$

Now, set

$$V := \{\zeta \in \partial^h X : |\xi(x) - \zeta(x)| < \epsilon \text{ for all } x \in h\gamma([0, \tau_\varphi k])\}$$

which is an open neighborhood of  $\xi$ . We now show that  $V \subset U_C(h\varphi^{R'}; \varphi, L')$ , which finishes the proof.

Let  $\zeta \in V$  and  $\{z'_i\}_{i \in \mathbb{N}} \subset X$  be a sequence converging to  $\zeta$ .

**Claim.** *We have*

$$(7.8) \quad \pi_{h\gamma}(z'_i) \subset h\gamma([\tau_\varphi k - 20C, +\infty)) \quad \text{for all large } i \in \mathbb{N}.$$

If not, then after passing to a subsequence,

$$d(z'_i, h\gamma(\tau_\varphi k - 2C)) - d(z'_i, h\gamma(\tau_\varphi k - 20C)) =_{8C} 18C \quad \text{for all } i \in \mathbb{N}$$

by Corollary 5.3(2). On the other hand, by Equation (7.6), it follows from Corollary 5.3(2) that after passing to a subsequence,

$$d(z_i, h\gamma(\tau_\varphi k - 20C)) - d(z_i, h\gamma(\tau_\varphi k - 2C)) =_{8C} 18C \quad \text{for all } i \in \mathbb{N}.$$

These imply that

$$\begin{aligned} \beta_\zeta(h\gamma(\tau_\varphi k - 2C), h\gamma(\tau_\varphi k - 20C)) &=_{8C} 18C \quad \text{and} \\ \beta_\xi(h\gamma(\tau_\varphi k - 20C), h\gamma(\tau_\varphi k - 2C)) &=_{8C} 18C. \end{aligned}$$

Equivalently, by  $\beta_\xi(x, y) = \xi(x) - \xi(y)$ ,

$$\begin{aligned}\zeta(h\gamma(\tau_\varphi k - 2C)) - \zeta(h\gamma(\tau_\varphi k - 20C)) &=_{8C} 18C \quad \text{and} \\ \xi(h\gamma(\tau_\varphi k - 2C)) - \xi(h\gamma(\tau_\varphi k - 20C)) &=_{8C} -18C.\end{aligned}$$

This contradicts the fact that  $|\xi(x) - \zeta(x)| < \epsilon < C$  for all  $x \in h\gamma([0, \tau_\varphi k])$ .

Hence, Equation (7.8) holds, and Lemma 5.9(3) tells us that

$$(x_0, h\varphi^{R'}[x_0, \varphi^{L'}x_0], \zeta) \quad \text{is } C\text{-aligned.}$$

This shows  $\zeta \in U_C(h\varphi^{R'}; \varphi, L')$ , and therefore  $V \subset U_C(h\varphi^{R'}; \varphi, L')$  as desired.  $\square$

**7.5. Quasi-invariance under translations.** For  $a \in \mathbb{R}$ , consider the map  $T_a : \mathcal{H} \rightarrow \mathcal{H}$  given by  $(\xi, t) \mapsto (\xi, t + a)$ . For a Radon measure  $\mu$  on  $\mathcal{H}$ , we consider its pullback measure  $T_a^*\mu$ : for each Borel subset  $E \subset \mathcal{H}$ ,

$$T_a^*\mu(E) := \mu(T_a E).$$

For a contracting  $g \in \text{Isom}(X)$ , we simply write  $T_g := T_{\tau_g}$ . We show that invariant ergodic measures on  $\mathcal{H}$  are quasi-invariant under these translations.

**Theorem 7.10.** *Let  $\Gamma < \text{Isom}(X)$  be a non-elementary subgroup containing a squeezing isometry. Let  $\mu$  be a  $\Gamma$ -invariant ergodic Radon measure on  $\mathcal{H}$  supported on  $\Lambda_c(\Gamma) \times \mathbb{R}$ . Then for each squeezing isometry  $\varphi \in \Gamma$ , there exists  $\lambda \geq 0$  such that*

$$\frac{dT_\varphi^*\mu}{d\mu} = e^\lambda \quad \text{a.e.}$$

*Proof.* Let  $\varphi \in \Gamma$  be a squeezing isometry and let  $C = C(\varphi) > 0$  be as in Lemma 5.9, with a chosen of axis  $\gamma : \mathbb{R} \rightarrow X$ . Note that for every  $n \in \mathbb{N}$ ,  $\tau_{\varphi^n} = n \cdot \tau_\varphi$  and  $C(\varphi^n) = C(\varphi)$ . Since  $T_\varphi$  commutes with the  $\Gamma$ -action on  $\mathcal{H}$ ,  $T_{\varphi^{-n}}^*\mu$  is also  $\Gamma$ -invariant and ergodic. Hence, if  $\frac{dT_{\varphi^n}^*(T_{\varphi^{-n}}^*\mu)}{dT_{\varphi^{-n}}^*\mu} = e^{\lambda_1}$  and  $\frac{dT_{\varphi^{n+1}}^*(T_{\varphi^{-n}}^*\mu)}{dT_{\varphi^{-n}}^*\mu} = e^{\lambda_2}$  for some  $\lambda_1, \lambda_2 \in \mathbb{R}$ , then  $\frac{dT_\varphi^*\mu}{d\mu} = e^{\lambda_2 - \lambda_1}$ . Therefore, it suffices to consider the case where

$$\tau_\varphi > 100C.$$

We first aim to show that

$$(7.9) \quad (T_\varphi^*\mu)(E) \geq \mu(E)$$

for each Borel subset  $E \subset \mathcal{H}$ . By Corollary 7.8,  $\mu$  is supported on  $\Lambda_{\varphi, C}(\Gamma) \times \mathbb{R}$ .

**Step 1.** First consider the case where  $E = K \times I$  for a compact subset  $K \subset \Lambda_{\varphi, C}(\Gamma)$  and a compact interval  $I \subset \mathbb{R}$ .

We fix some open subset  $O \subset \partial^h X$  such that  $K \subset O$  and  $\epsilon > 0$ . Let  $L = L(0.001\epsilon) > 0$  be as in Definition 5.5 for  $\gamma$ . By Lemma 7.9, for each

$\xi \in K$ , there exist an element  $g(\xi) \in \Gamma$  and  $n(\xi) > (2L + 100C)/\tau_\varphi + 4$  such that

$$\xi \in U_C(g(\xi); \varphi, n(\xi)) \subset O.$$

Let  $\mathcal{U} := \{U_C(g(\xi); \varphi, n(\xi)) : \xi \in K\}$ , which is a countable collection of sets as  $\Gamma$  is countable. For convenience, let us enumerate  $\mathcal{U}$  based on their distance from  $x_0$ , i.e., let

$$\mathcal{U} = \{U_1, U_2, \dots\}$$

where  $U_i := U_C(g_i; \varphi, n_i)$  for each  $i \in \mathbb{N}$  so that

$$d(x_0, g_1 \varphi^{n_1} x_0) \leq d(x_0, g_2 \varphi^{n_2} x_0) \leq \dots$$

We will now define a subcollection

$$\mathcal{V} := \{U_{i(1)}, U_{i(2)}, \dots\} \subset \mathcal{U}$$

by inductively defining  $i(1), i(2), \dots$ . We let  $i(1) = 1$ . Now, having defined  $i(1), \dots, i(N)$ , define  $i(N+1)$  as the smallest  $j \in \mathbb{N}$  such that  $U_j$  is disjoint from  $U_{i(1)} \cup \dots \cup U_{i(N)}$ .

For each  $l \in \mathbb{N}$ , we set

$$(7.10) \quad C_l := U_{i(l)} \cup \bigcup \{U_k : k \geq i(l), U_k \cap U_{i(l)} \neq \emptyset\}.$$

Then  $\{C_l : l \in \mathbb{N}\}$  is a covering of  $K$  contained in  $O$ . Indeed, for  $k \geq 1$ , if  $k \neq i(l)$  for all  $l \in \mathbb{N}$ , then  $U_k$  intersects  $U_{i(1)} \cup \dots \cup U_{i(l_0)}$  where  $i(l_0)$  is the maximal index among  $i(1), i(2), \dots$  that is less than  $k$ .

**Claim.** For each  $l \in \mathbb{N}$ ,

$$(7.11) \quad C_l \subset U_C(g_{i(l)}; \varphi, n_{i(l)} - 1).$$

To see this claim, let  $k \geq i(l)$  be such that  $U_k = U_C(g_k; \varphi, n_k)$  and  $U_{i(l)}$  have a common element  $\xi$ . Then for every  $z \in X$  close enough to  $\xi$  in  $\overline{X}^h$ ,

$$(7.12) \quad (x_0, g_{i(l)}[x_0, \varphi^{n_{i(l)}} x_0], z) \quad \text{is } C\text{-aligned.}$$

By Lemma 5.2, there exists  $q \in [x_0, z]$  that is  $3C$ -close to  $g_{i(l)} \varphi^{n_{i(l)}} x_0$ . Similarly, by the condition  $\xi \in U_k$ ,  $[x_0, z]$  contains a point  $p$  that is  $3C$ -close to  $g_k \varphi^{n_k} x_0$ . Since  $k \geq i(l)$ , our enumeration convention tells us that

$$d(x_0, p) \geq d(x_0, g_k \varphi^{n_k} x_0) - 3C \geq d(x_0, g_{i(l)} \varphi^{n_{i(l)}} x_0) - 3C \geq d(x_0, q) - 6C.$$

In other words,

$$d(z, p) \leq d(z, q) + 6C.$$

If  $d(z, p) \leq 10C$ , then we have  $d(z, g_k \varphi^{n_k} x_0) \leq 13C$ . Since  $\pi_{g_{i(l)}[x_0, \varphi^{n_{i(l)}} x_0]}(\cdot)$  is  $(1, 4C)$ -Lipschitz by Corollary 5.3(1), we have

$$\text{Diam } \pi_{g_{i(l)}[x_0, \varphi^{n_{i(l)}} x_0]}(\{z, g_k \varphi^{n_k} x_0\}) \leq 17C.$$

If  $d(z, p) > 10C$ , then we take a point  $p^\dagger \in [z, p]$  such that  $d(p, p^\dagger) = 10C$ . Then  $\text{Diam}[p^\dagger, z] = d(p, z) - 10C$ , and hence

$$\begin{aligned} d(g_{i(l)}[x_0, \varphi^{n_{i(l)}}x_0], [p^\dagger, z]) &\geq d(g_{i(l)}[x_0, \varphi^{n_{i(l)}}x_0], z) - \text{Diam}[p^\dagger, z] \\ &\geq d(z, q) - 3C - d(z, p) + 10C \\ &\geq C. \end{aligned}$$

By the  $C$ -contracting property of  $g_{i(l)}[x_0, \varphi^{n_{i(l)}}x_0]$ , we have

$$\text{Diam } \pi_{g_{i(l)}[x_0, \varphi^{n_{i(l)}}x_0]}(\{z, p^\dagger\}) \leq C.$$

Since  $d(p, g_k \varphi^{n_k} x_0) \leq 3C$ , we have  $d(p^\dagger, g_k \varphi^{n_k} x_0) \leq 13C$ , and hence

$$\text{Diam } \pi_{g_{i(l)}[x_0, \varphi^{n_{i(l)}}x_0]}(\{z, g_k \varphi^{n_k} x_0\}) \leq 18C$$

by Corollary 5.3(1).

Hence, in either case, combining with Equation (7.12), we obtain that

$$(7.13) \quad (x_0, g_{i(l)}[x_0, \varphi^{n_{i(l)}}x_0], g_k \varphi^{n_k} x_0) \text{ is } 19C\text{-aligned.}$$

Now let  $\zeta \in U_k$  be arbitrary. Then for every  $z' \in X$  close to  $\zeta$  in  $\overline{X}^h$ ,

$$(x_0, g_k[x_0, \varphi^{n_k}x_0], z') \text{ is } C\text{-aligned}$$

and hence  $[x_0, z']$  passes through the  $3C$ -neighborhood of  $g_k \varphi^{n_k} x_0$  as in the argument above. Hence, we have

$$\begin{aligned} \text{Diam } \pi_{g_{i(l)}[x_0, \varphi^{n_{i(l)}}x_0]}([x_0, z']) &\geq d(g_{i(l)}x_0, g_{i(l)}\varphi^{n_{i(l)}}x_0) \\ &\quad - \text{Diam} \left( \pi_{g_{i(l)}[x_0, \varphi^{n_{i(l)}}x_0]}(x_0) \cup \{g_{i(l)}x_0\} \right) \\ &\quad - (d([x_0, z'], g_k \varphi^{n_k} x_0) + 4C) \\ &\quad - \text{Diam} \left( \pi_{g_{i(l)}[x_0, \varphi^{n_{i(l)}}x_0]}(g_k \varphi^{n_k} x_0) \cup \{g_{i(l)}\varphi^{n_{i(l)}}x_0\} \right) \\ &\geq 100C - C - (3C + 4C) - 19C \\ &\geq 70C \end{aligned}$$

where we applied Corollary 5.3(1) in the first inequality, and Equation (7.12), Equation (7.13), and the fact that  $d([x_0, z'], g_k \varphi^{n_k} x_0) \leq 3C$  in the second.

We then apply Lemma 5.2 and obtain two points  $u, v \in [x_0, z']$  such that

- (1)  $[u, v]$  and  $\pi_{g_{i(l)}[x_0, \varphi^{n_{i(l)}}x_0]}([x_0, z'])$  are within Hausdorff distance  $4C$ ,
- (2)  $\text{Diam} \left( \pi_{g_{i(l)}[x_0, \varphi^{n_{i(l)}}x_0]}([x_0, u]) \cup \{u\} \right) \leq 2C$ ,
- (3)  $\text{Diam} \left( \pi_{g_{i(l)}[x_0, \varphi^{n_{i(l)}}x_0]}([v, z']) \cup \{v\} \right) \leq 2C$ , and
- (4) for each  $u' \in \pi_{g_{i(l)}[x_0, \varphi^{n_{i(l)}}x_0]}(x_0)$  and  $v' \in \pi_{g_{i(l)}[x_0, \varphi^{n_{i(l)}}x_0]}(z')$ ,  $[u', v']$  and  $[u, v]$  are within Hausdorff distance  $10C$ .

Since  $d([x_0, z'], g_k \varphi^{n_k} x_0) \leq 3C$ , it follows from Equation (7.13) and Corollary 5.3(1) that there exists  $w \in [x_0, z']$  such that

$$(7.14) \quad \text{Diam} \left( \pi_{g_{i(l)}[x_0, \varphi^{n_{i(l)}}x_0]}(w) \cup \{g_{i(l)}\varphi^{n_{i(l)}}x_0\} \right) \leq 26C.$$

By condition (2) above, Equation (7.13), and  $d(x_0, \varphi^{n_{i(l)}}x_0) > 100C$ , we must have  $w \notin [x_0, u]$ , and hence  $w \in [v, z']$  or  $w \in [u, v]$ .

- If  $w \in [v, z']$ , then it follows from (3) above that

$$\text{Diam} \left( \pi_{g_{i(l)}[x_0, \varphi^{n_{i(l)}}x_0]}(z') \cup \{g_{i(l)}\varphi^{n_{i(l)}}x_0\} \right) \leq 28C.$$

- If  $w \in [u, v]$ , then it follows from (4) above that for each pair of points  $u' \in \pi_{g_{i(l)}[x_0, \varphi^{n_{i(l)}}x_0]}(x_0)$  and  $v' \in \pi_{g_{i(l)}[x_0, \varphi^{n_{i(l)}}x_0]}(z')$ , there exists  $w' \in [u', v']$  such that  $d(w, w') \leq 10C$ . We then have

$$\text{Diam} \left( \pi_{g_{i(l)}[x_0, \varphi^{n_{i(l)}}x_0]}(w) \cup \{w'\} \right) \leq 20C.$$

Together with Equation (7.14),  $\text{Diam}\{w', g_{i(l)}\varphi^{n_{i(l)}}x_0\} \leq 46C$ . Since  $v'$  lies between  $w'$  and  $g_{i(l)}\varphi^{n_{i(l)}}x_0$  on  $g_{i(l)}[x_0, \varphi^{n_{i(l)}}x_0]$ , and  $v' \in \pi_{g_{i(l)}[x_0, \varphi^{n_{i(l)}}x_0]}(z')$  is arbitrary, we have

$$\text{Diam} \left( \pi_{g_{i(l)}[x_0, \varphi^{n_{i(l)}}x_0]}(z') \cup \{g_{i(l)}\varphi^{n_{i(l)}}x_0\} \right) \leq 46C.$$

Therefore,  $(g_{i(l)}[x_0, \varphi^{n_{i(l)}}x_0], z')$  is  $46C$ -aligned.

By Lemma 5.9(4), we have

$$(7.15) \quad \begin{aligned} \pi_{g_{i(l)}\gamma}(z') &\subset g_{i(l)}\gamma \left( [\tau_\varphi n_{i(l)} - 47C, +\infty) \right) \\ &\subset g_{i(l)}\gamma \left( [\tau_\varphi(n_{i(l)} - 1) + 3C, +\infty) \right). \end{aligned}$$

By Lemma 5.9(3), this implies that  $(g_{i(l)}[x_0, \varphi^{n_{i(l)}-1}x_0], z')$  is  $C$ -aligned. This is the case for every  $z'$  close to  $\zeta \in U_k$ , so we conclude

$$\zeta \in U_C(g_{i(l)}; \varphi, n_{i(l)} - 1).$$

The claim is now established.

For each  $l \in \mathbb{N}$  we now define a map  $F_l : C_l \times I \rightarrow \mathcal{H}$ . For  $g = g_{i(l)}$  we let

$$(7.16) \quad F_l : \mathfrak{X} \mapsto g\varphi g^{-1}\mathfrak{X}.$$

We have  $\mu(F_l(C_l \times I)) = \mu(C_l \times I)$  as  $\mu$  is  $\Gamma$ -invariant.

**Claim.** *We have*

$$(7.17) \quad F_l(C_l \times I) \subset U_{i(l)} \times (\epsilon\text{-neighborhood of } I + \tau_\varphi).$$

To see this, we simply write  $g = g_{i(l)}$  and  $n = n_{i(l)} - 1$ . We then fix  $\mathfrak{X} = (\xi, t) \in C_l \times I$ . Note that

$$F_l(\mathfrak{X}) = (g\varphi g^{-1}\xi, t + \beta_\xi(g\varphi^{-1}g^{-1}x_0, x_0)).$$

Let  $\{z_i\}_{i \in \mathbb{N}} \subset X$  be a sequence converging to  $\xi$ .

We first show that  $g\varphi g^{-1}\xi \in U_{i(l)}$ , which follows once we show that

$$(x_0, g[x_0, \varphi^{n+1}x_0], g\varphi g^{-1}z_i) \text{ is } C\text{-aligned for all large } i \in \mathbb{N}.$$

Since  $U_{i(l)} \neq \emptyset$ , we already have that  $(x_0, g[x_0, \varphi^{n+1}x_0])$  is  $C$ -aligned. Now suppose to the contrary that  $(g[x_0, \varphi^{n+1}x_0], g\varphi g^{-1}z_i)$  is not  $C$ -aligned for infinitely many  $i \in \mathbb{N}$ . By Lemma 5.9(3), we have

$$\pi_{g\gamma}(z_i) \subset g\gamma((-\infty, n\tau_\varphi]).$$

This contradicts Equation (7.15). Therefore,  $g\varphi g^{-1}\xi \in U_{i(l)}$ .

For the second component, it suffices to show

$$(7.18) \quad |\beta_\xi(g\varphi^{-1}g^{-1}x_0, x_0) - \tau_\varphi| < \epsilon.$$

Note that

$$\beta_\xi(g\varphi^{-1}g^{-1}x_0, x_0) = \lim_{i \rightarrow +\infty} d(g\varphi^{-1}g^{-1}x_0, z_i) - d(x_0, z_i).$$

By Equation (7.11),

$$(x_0, g[x_0, \varphi^n x_0], z_i) \text{ is } C\text{-aligned for all large } i \in \mathbb{N}.$$

Hence, for all large  $i \in \mathbb{N}$ , it follows from Lemma 5.9(4) that

$$(7.19) \quad \pi_{g\gamma}(x_0) \subset g\gamma((-\infty, 2C]) \quad \text{and} \quad \pi_{g\gamma}(z_i) \subset g\gamma([n\tau_\varphi - 2C, +\infty)).$$

Since  $n\tau_\varphi - 4C > 2L$  and  $g\gamma$  is squeezing (Definition 5.5), Equation ?? implies that there exists  $p \in [x_0, z_i]$  such that

$$d(p, g\gamma(n\tau_\varphi/2)) \leq 0.001\epsilon.$$

Meanwhile, note that  $(g\varphi^{-1}g^{-1}x_0, g[x_0, \varphi^n x_0])$  is also  $C$ -aligned; otherwise, we have  $\pi_{g\gamma}(g\varphi^{-1}g^{-1}x_0) \subset g\gamma([0, +\infty))$  by Lemma 5.9(3), and therefore  $\pi_{g\gamma}(x_0) \subset g\gamma([\tau_\varphi, +\infty))$  which contradicts Equation (7.19). Hence, it follows from Lemma 5.9(4) that

$$\pi_{g\gamma}(g\varphi^{-1}g^{-1}x_0) \subset g\gamma((-\infty, 2C]).$$

Together with Equation (7.19) and  $n\tau_\varphi - 4C > 2L + 2\tau_\varphi$ , the squeezing property of  $g\gamma$  implies that there exist  $q_1, q_2 \in [g\varphi g^{-1}x_0, z_i]$ , with  $q_1$  coming earlier than  $q_2$ , such that

$$d(q_1, g\gamma(n\tau_\varphi/2 - \tau_\varphi)), d(q_2, g\gamma(n\tau_\varphi/2)) < 0.001\epsilon.$$

Now we have

$$\begin{aligned} d(g\varphi^{-1}g^{-1}x_0, z_i) - d(x_0, z_i) &= (d(g\varphi^{-1}g^{-1}x_0, q_1) + d(q_1, q_2) + d(q_2, z_i)) \\ &\quad - (d(x_0, p) + d(p, z_i)) \\ &=_{0.006\epsilon} d(g\varphi^{-1}g^{-1}x_0, g\gamma(n\tau_\varphi/2 - \tau_\varphi)) \\ &\quad + d(g\gamma(n\tau_\varphi/2 - \tau_\varphi), g\gamma(n\tau_\varphi/2)) \\ &\quad + d(g\gamma(n\tau_\varphi/2), z_i) \\ &\quad - d(x_0, g\gamma(n\tau_\varphi/2)) - d(g\gamma(n\tau_\varphi/2), z_i) \\ &= d(g\gamma(n\tau_\varphi/2 - \tau_\varphi), g\gamma(n\tau_\varphi/2)) = \tau_\varphi. \end{aligned}$$

Taking the limit  $i \rightarrow +\infty$ , Equation (7.18) follows. This completes the proof of the claim.

Now by the above claim and disjointness of  $U_{i(l)}$ 's, we have

$$\begin{aligned} \mu(O \times (\epsilon\text{-neighborhood of } I + \tau_\varphi)) &\geq \mu\left(\bigcup_l F_l(C_l \times I)\right) \\ &= \sum_l \mu(F_l(C_l \times I)) \\ &= \sum_l \mu(C_l \times I) \\ &\geq \mu(K \times I). \end{aligned}$$

Note that  $\mu(O \times (\epsilon\text{-neighborhood of } I + \tau_\varphi)) < +\infty$  since  $\mu$  is Radon. Since  $\epsilon > 0$  and the open set  $O \supset K$  were arbitrary, we have

$$(T_\varphi^* \mu)(K \times I) = \mu(K \times (I + \tau_\varphi)) \geq \mu(K \times I).$$

**Step 2.** Consider the case where  $E = A \times B$  for Borel  $A \subset \partial^h X$  and an interval  $B \subset \mathbb{R}$ . Since  $\mu$  is supported on  $\Lambda_{\varphi, C}(\Gamma) \times \mathbb{R}$ , we may assume that  $A \subset \Lambda_{\varphi, C}(\Gamma)$ . By the inner regularity of  $\mu$  and  $T_\varphi^* \mu$ , there exist compact subsets  $E_1, E_2 \subset E$  such that

$$|\mu(E) - \mu(E_1)| < \epsilon \quad \text{and} \quad |(T_\varphi^* \mu)(E) - (T_\varphi^* \mu)(E_2)| < \epsilon.$$

Considering projections of  $E_1 \cup E_2$  to  $A$  and  $B$ , we obtain compact subsets  $K \subset A$  and  $I \subset B$  such that

$$|\mu(E) - \mu(K \times I)| < \epsilon \quad \text{and} \quad |(T_\varphi^* \mu)(E) - (T_\varphi^* \mu)(K \times I)| < \epsilon.$$

By taking the convex hull of  $I$  (recall that  $B$  is an interval), we may assume that  $I$  is a compact interval. Applying Step 1 to  $K \times I$ , we have

$$(T_\varphi^* \mu)(E) \geq \mu(E) - 2\epsilon.$$

Since  $\epsilon > 0$  is arbitrary,  $(T_\varphi^* \mu)(E) \geq \mu(E)$  follows.

**Step 3.** When  $E \subset \mathcal{H}$  is a finite union of open sets of the form  $O_1 \times O_2$  for open sets  $O_1 \subset \partial^h X$  and open intervals  $O_2 \subset \mathbb{R}$ ,  $E$  is a disjoint union of finitely many Borel subsets of the form  $A \times B$ , where  $A \subset \partial^h X$  is Borel and  $B \subset \mathbb{R}$  is an interval. Hence,  $(T_\varphi^* \mu)(E) \geq \mu(E)$  follows from Step 2.

**Step 4.** When  $E \subset \mathcal{H}$  is an open set,  $E$  is a countable union of open sets of the form  $O_1 \times O_2$  for open sets  $O_1 \subset \partial^h X$  and open intervals  $O_2 \subset \mathbb{R}$ . Hence,  $(T_\varphi^* \mu)(E) \geq \mu(E)$  follows from Step 3.

**Step 5.** Finally, suppose that  $E \subset \mathcal{H}$  is a Borel subset. Then it follows from Step 4 and the outer regularity of  $\mu$  and  $T_\varphi^* \mu$  that

$$(T_\varphi^* \mu)(E) \geq \mu(E).$$

Now we have shown Equation (7.9). Hence, we can consider the Radon–Nikodym derivative  $\frac{d\mu}{dT_\varphi^* \mu}$ . Since both  $\mu$  and  $T_\varphi^* \mu$  are  $\Gamma$ -invariant,  $\frac{d\mu}{dT_\varphi^* \mu}$  is  $\Gamma$ -invariant as well. By  $\Gamma$ -ergodicity of  $T_\varphi^* \mu$ ,  $\frac{d\mu}{dT_\varphi^* \mu}$  is constant  $T_\varphi^* \mu$ -a.e., and this

constant must be positive. Hence, there exists  $\lambda \in \mathbb{R}$  such that  $\frac{dT_\varphi^* \mu}{d\mu} = e^\lambda$   $\mu$ -a.e., and moreover,  $\lambda \geq 0$  by Equation (7.9). This completes the proof.  $\square$

**7.6. Proof of the rigidity.** Let us now prove Theorem 7.4.

*Proof of Theorem 7.4.* By ergodic decomposition, it suffices to consider a  $\Gamma$ -invariant ergodic Radon measure  $\mu$  on  $\mathcal{H}$  supported on  $\Lambda_c(\Gamma) \times \mathbb{R}$ .

Let

$$A := \left\{ a \in \mathbb{R} : \exists \lambda(a) \in \mathbb{R} \text{ such that } \frac{dT_a^* \mu}{d\mu} = e^{\lambda(a)} \text{ a.e.} \right\}.$$

It is straightforward to check that  $A$  is an additive subgroup of  $\mathbb{R}$  and  $\lambda : A \rightarrow \mathbb{R}$  is an additive homomorphism. Moreover, by Theorem 7.10,

$$\text{Spec}_{sq}(\Gamma) \subset A.$$

Hence, it follows from non-arithmeticity of  $\text{Spec}_{sq}(\Gamma)$  that  $A \subset \mathbb{R}$  is dense.

**Claim.** *There exists  $\delta \geq 0$  such that*

$$\lambda(a) = \delta \cdot a \quad \text{for all } a \in A.$$

To see this claim, choose a nonzero  $a \in A$  and set  $\delta := \lambda(a)/a$ . By Theorem 7.10, we can choose  $a \in A$  so that  $\delta \geq 0$ . It suffices to show that for every nonzero  $a' \in A$ ,  $\lambda(a')/a' = \delta$  as well. There are two cases.

- (1) If  $a$  and  $a'$  are commensurable, i.e.,  $ma = na'$  for some  $m, n \in \mathbb{Z}$ , then the conclusion follows from

$$m\lambda(a) = \lambda(ma) = \lambda(na') = n\lambda(a').$$

- (2) If  $a$  and  $a'$  are not commensurable, then suppose to the contrary that  $\delta' := \lambda(a')/a'$  is distinct from  $\delta$ .

Let  $R > 0$  be large enough so that  $O := \partial^h X \times (-R, R)$  satisfies  $0 < \mu(O) < +\infty$ . Let  $K \subset O$  be an arbitrary compact subset. Then there exists  $\epsilon > 0$  such that  $K \subset \partial^h X \times (-R - \epsilon, R + \epsilon)$ .

Since  $a$  and  $a'$  are not commensurable and  $\delta \neq \delta'$ , there exist  $N, M \in \mathbb{Z}$  such that  $N > \frac{\epsilon\delta+1}{|a'|\delta-\delta'}$  and  $|Ma - Na'| < \epsilon$ . Then setting  $s := Ma - Na' \in A$ , we have

$$\begin{aligned} |\lambda(s)| &= |M\lambda(a) - N\lambda(a')| \\ &= |(Ma - Na')\delta + (\delta - \delta')Na'| \\ &\geq |\delta - \delta'| |a'| N - \epsilon\delta > 1. \end{aligned}$$

Replacing  $s$  with  $-s$  if necessary, we may assume that  $\lambda(s) > 1$ . Then we have

$$\mu(O) \geq \mu(T_s K) = (T_s^* \mu)(K) = e^{\lambda(s)} \mu(K) > e\mu(K).$$

Since  $K \subset O$  is arbitrary, we have  $\mu(O) \geq e\mu(O)$ , which is absurd.

Therefore, the claim follows.

Now consider an arbitrary  $a \in \mathbb{R}$ . Since  $A \subset \mathbb{R}$  is dense, there exists a sequence  $\{a_i\}_{i \in \mathbb{N}} \subset A$  converging to  $a$ . Let  $U \subset \mathcal{H}$  be an open subset and  $K \subset U$  a compact subset. Then for all large  $i \in \mathbb{N}$ ,  $T_{a_i}K \subset T_aU$ , and hence

$$\mu(T_aU) \geq \mu(T_{a_i}K) = e^{\lambda(a_i)}\mu(K) = e^{\delta a_i}\mu(K).$$

Taking the limit  $i \rightarrow +\infty$ , we have  $\mu(T_aU) \geq e^{\delta a}\mu(K)$ . Since  $K \subset U$  is arbitrary, this implies  $\mu(T_aU) \geq e^{\delta a}\mu(U)$ . By the same argument, we also have  $\mu(U) \geq e^{-\delta a}\mu(T_aU)$ . Hence, we have  $(T_a^*\mu)(U) = e^{\delta a}\mu(U)$ . Since this holds for any open subset  $U \subset \mathcal{H}$ , we have

$$T_a^*\mu = e^{\delta a} \cdot \mu.$$

This implies that there exists a finite Borel measure  $\nu_0$  on  $\partial^h X$  such that  $\mu$  is decomposed on  $\mathcal{H} = \partial^h X \times \mathbb{R}$  as follows:

$$d\mu(\xi, t) = e^{\delta t} \cdot d\nu_0(\xi) dt.$$

By the  $\Gamma$ -invariance of  $\mu$ , it is easy to see that for each  $g \in \Gamma$ ,

$$\frac{dg_*\nu_0}{d\nu_0}(\xi) = e^{-\delta\beta_\xi(gx_0, x_0)} \quad \text{for } \nu_0\text{-a.e. } \xi \in \partial^h X.$$

Then for  $x \in X$ , define the measure  $\nu_x$  on  $\partial^h X$  by setting

$$d\nu_x(\xi) := \frac{e^{-\delta\beta_\xi(x, x_0)}}{\nu_0(\partial^h X)} d\nu_0(\xi).$$

This is well defined, and moreover the family  $\{\nu_x\}_{x \in X}$  is a  $\delta$ -dimensional conformal density of  $\Gamma$ . Since  $\{\nu_x\}_{x \in X}$  is supported on  $\Lambda_c(\Gamma)$ ,  $\delta = \delta_\Gamma$  and  $\Gamma$  is of divergence type as a result of the generalized Hopf–Tsuji–Sullivan dichotomy ([Cou24, Corollary 4.25], [Yan24, Theorem 1.14]). Therefore,

$$\mu = \frac{1}{\nu_0(\partial^h X)} \cdot \mu_\Gamma,$$

which completes the proof.  $\square$

## 8. EXISTENCE OF ERGODIC INVARIANT RADON MEASURES

We continue with the setting of Section 7. In this section, we prove the ergodicity of the invariant Radon measure defined in Definition 7.1. This was stated as Theorem 1.11 in the introduction.

**Theorem 8.1.** *Let  $\Gamma < \text{Isom}(X)$  be a non-elementary subgroup with non-arithmetic squeezing spectrum. If  $\Gamma$  is of divergence type, then*

*the  $\Gamma$ -action on  $(\mathcal{H}, \mu_\Gamma)$  is ergodic.*

*Moreover,  $\mu_\Gamma$  is supported on  $\Lambda_c(\Gamma) \times \mathbb{R} \subset \mathcal{H}$ . Furthermore, up to scaling,  $\mu_\Gamma$  is the unique  $\Gamma$ -invariant Radon measure on  $\mathcal{H}$  that is supported on  $\Lambda_c(\Gamma) \times \mathbb{R}$ .*

Note that the fact that  $\mu_\Gamma$  is supported on  $\Lambda_c(\Gamma) \times \mathbb{R}$  is due to Coulon [Cou24] and Yang [Yan24] (Proposition 6.6). In addition, the unique ergodicity follows from Theorem 7.4, once we show the ergodicity. Hence, it suffices to show that  $\mu_\Gamma$  is  $\Gamma$ -ergodic. Together with Proposition 6.9, this is a special case of the following theorem.

**Theorem 8.2.** *Let  $\Gamma < \text{Isom}(X)$  be a non-elementary subgroup of divergence type. Let  $\Gamma_0 \triangleleft \Gamma$  be a normal subgroup such that*

- $\text{Spec}_{sq}(\Gamma_0)$  is non-arithmetic and
- the  $\Gamma_0$ -action on  $\partial^h X$  is ergodic with respect to the  $\delta_\Gamma$ -dimensional conformal density of  $\Gamma$ .

*Then,*

*the  $\Gamma_0$ -action on  $(\mathcal{H}, \mu_\Gamma)$  is ergodic.*

To prove the ergodicity, we consider the notion of essential subgroups, which was introduced by Schmidt [Sch77] and studied further by Roblin [Rob03]. For a conformal density  $\nu = \{\nu_x\}_{x \in X}$ , all measures in the family  $\nu$  are in the same measure class. Hence, when discussing whether a Borel subset has positive measure, we simply use write  $\nu$ .

**Definition 8.3.** Let  $\Gamma < \text{Isom}(X)$  and let  $\nu$  be a conformal density of  $\Gamma$ . We define the subset  $E_\nu(\Gamma) \subset \mathbb{R}$  as follows:  $a \in E_\nu(\Gamma)$  if for each  $\epsilon > 0$  and each Borel subset  $E \subset \partial^h X$  with  $\nu(E) > 0$ , there exists  $g \in \Gamma$  such that

$$\nu(E \cap gE \cap \{\xi \in \partial^h X : |\beta_\xi(x_0, gx_0) - a| < \epsilon\}) > 0.$$

It is easy to see that  $E_\nu(\Gamma)$  is a closed subgroup of  $\mathbb{R}$ . We call  $E_\nu(\Gamma)$  the *essential subgroup* for  $\Gamma$  and  $\nu$ .

The size of the essential subgroup serves as a criterion for the ergodicity of actions on  $\mathcal{H}$ . The following was proved in [Sch77] for abstract measurable dynamical systems. A more direct proof for the particular case of  $\text{CAT}(-1)$  spaces was given in [Rob03]. The same proof works in our setting as well.

**Proposition 8.4** ([Sch77], [Rob03, Proposition 2.1]). *Let  $\Gamma < \text{Isom}(X)$  and let  $\nu$  be a conformal density of  $\Gamma$ . Then the  $\Gamma$ -action on  $(\mathcal{H}, \mu_\nu)$  is ergodic if and only if the  $\Gamma$ -action on  $(\partial^h X, \nu)$  is ergodic and  $E_\nu(\Gamma) = \mathbb{R}$ .*

From this perspective, the following is the main step in the proof of Theorem 8.2, which was proved by Roblin [Rob03] when  $X$  is  $\text{CAT}(-1)$ . The  $\text{CAT}(-1)$  property was crucially used in [Rob03] to obtain a nice visual metric on the boundary that guarantees a Vitali covering relation of a specific form. In our setting, the lack of Gromov hyperbolicity is an obstacle to constructing such a visual metric, and hence we present another proof that does not require metrizing the boundary.

**Lemma 8.5.** *Let  $\Gamma < \text{Isom}(X)$  be a non-elementary subgroup of divergence type and  $\nu$  a  $\delta_\Gamma$ -dimensional conformal density of  $\Gamma$ . Let  $\varphi \in \Gamma$  be a squeezing*

isometry. Then for each  $\epsilon > 0$  and each Borel subset  $E \subset \partial^h X$  with  $\nu(E) > 0$ , there exists  $g \in \Gamma$  such that

$$\nu \left( E \cap g\varphi g^{-1}E \cap \{\xi \in \partial^h X : |\beta_\xi(x_0, g\varphi g^{-1}x_0) - \tau_\varphi| < \epsilon\} \right) > 0.$$

In particular, if  $\Gamma_0 \triangleleft \Gamma$  is a normal subgroup, then

$$\text{Spec}_{sq}(\Gamma_0) \subset E_\nu(\Gamma_0).$$

*Proof.* Let  $C = C(\varphi) > 0$  be as in Lemma 5.9. By Proposition 6.6,  $\nu$  is supported on  $\Lambda_{\varphi, C}(\Gamma)$ . Combined with the inner regularity of  $\nu$ , this means that it suffices to consider compact subsets of  $\Lambda_{\varphi, C}(\Gamma)$ .

Let  $K \subset \Lambda_{\varphi, C}(\Gamma)$  be a compact subset and let  $0 < \epsilon < \tau_\varphi$ . Suppose that for each  $g \in \Gamma$ ,

$$\nu \left( K \cap g\varphi g^{-1}K \cap \{\xi \in \partial^h X : |\beta_\xi(x_0, g\varphi g^{-1}x_0) - \tau_\varphi| < \epsilon\} \right) = 0.$$

Our goal is to show  $\nu(K) = 0$ .

To do this, let  $O \subset \partial^h X$  be an open subset containing  $K$ . We will then construct a Borel subset  $E(O) \subset O$  such that

$$(8.1) \quad \nu(K \cap E(O)) = 0 \quad \text{and} \quad \nu(E(O)) \geq e^{-2\delta_\Gamma \tau_\varphi} \cdot \nu(K).$$

Before we proceed, let us see how this leads to our goal. Suppose that we have constructed  $E(O)$ . Since  $\nu(E(O) \setminus K) \geq e^{-2\delta_\Gamma \tau_\varphi} \cdot \nu(K)$ , the inner regularity of  $\nu$  yields a compact subset  $K(O) \subset E(O) \setminus K$  such that

$$\nu(K(O)) \geq 0.5e^{-2\delta_\Gamma \tau_\varphi} \cdot \nu(K).$$

In particular,  $K(O)$  is disjoint from both  $K$  and  $\partial^h X \setminus O$ . Now we inductively define

$$K_1 := K \left( \partial^h X \right) \quad \text{and} \quad K_i := K \left( \partial^h X \setminus (K_1 \cup \cdots \cup K_{i-1}) \right) \quad \text{for } i \geq 2,$$

where each argument of  $K(\cdot)$  is open since the  $K_j$ 's are compact. Then the  $K_i$ 's are pairwise disjoint subsets with  $\nu(K_i) \geq e^{-2\delta_\Gamma \tau_\varphi} \cdot \nu(K)$  for all  $i \in \mathbb{N}$ . Since  $\nu$  is finite, we must have  $\nu(K) = 0$ .

Hence, it remains to find a set  $E(O) \subset O$  satisfying Equation (8.1). We revisit the proof of Theorem 7.10, considering the cover  $\mathcal{U}$  and its subcollection  $\mathcal{V}$  for  $K$  and  $O$ . In particular, for  $l \in \mathbb{N}$ , we consider  $C_l \subset O$  in Equation (7.10) and the restriction  $F_l = g_{i(l)}\varphi g_{i(l)}^{-1} : C_l \rightarrow \partial^h X$  of the map in Equation (7.16), where  $g_{i(l)} \in \Gamma$  is given there.

We now show that

$$E(O) := \bigcup_{l \in \mathbb{N}} F_l(C_l \cap K)$$

satisfies Equation (8.1). First, note that  $\bigcup_{l \in \mathbb{N}} F_l(C_l \cap K) \subset \bigcup_{l \in \mathbb{N}} U_{i(l)} \subset \bigcup_{l \in \mathbb{N}} C_l \subset O$  by Equation (7.17) and Equation (7.10).

It follows from (7.18) that for each  $l \in \mathbb{N}$ ,

$$(8.2) \quad F_l(C_l) \subset \left\{ \xi \in \partial^h X : |\beta_\xi(x_0, g_{i(l)}\varphi g_{i(l)}^{-1}x_0) - \tau_\varphi| < \epsilon \right\}.$$

We then have

$$\begin{aligned} & K \cap F_l(C_l \cap K) \\ & \subset K \cap g_{i(l)} \varphi g_{i(l)}^{-1} K \cap \left\{ \xi \in \partial^h X : |\beta_\xi(x_0, g_{i(l)} \varphi g_{i(l)}^{-1} x_0) - \tau_\varphi| < \epsilon \right\} \end{aligned}$$

and hence our hypothesis on  $K$  implies  $\nu(K \cap F_l(C_l \cap K)) = 0$ . Therefore,

$$\nu \left( K \cap \bigcup_{l \in \mathbb{N}} F_l(C_l \cap K) \right) = 0,$$

showing the first claim in Equation (8.1).

Now it remains to estimate  $\nu \left( \bigcup_{l \in \mathbb{N}} F_l(C_l \cap K) \right)$ . By Equation (8.2), the fact that  $\epsilon < \tau_\varphi$ , and the conformality of  $\nu$ , we have for each  $l \in \mathbb{N}$  that

$$\nu(F_l(C_l \cap K)) = \int_{C_l \cap K} e^{-\delta_\Gamma \beta_\xi(g_{i(l)} \varphi^{-1} g_{i(l)}^{-1} x_0, x_0)} d\nu(\xi) \geq e^{-2\delta_\Gamma \tau_\varphi} \nu(C_l \cap K).$$

By Equation (7.17), we also have that  $F_l(C_l \cap K)$ 's are pairwise disjoint. Therefore,

$$\begin{aligned} \nu \left( \bigcup_{l \in \mathbb{N}} F_l(C_l \cap K) \right) &= \sum_{l \in \mathbb{N}} \nu(F_l(C_l \cap K)) \\ &\geq e^{-2\delta_\Gamma \tau_\varphi} \sum_{l \in \mathbb{N}} \nu(C_l \cap K) \geq e^{-2\delta_\Gamma \tau_\varphi} \cdot \nu \left( \bigcup_{l \in \mathbb{N}} (C_l \cap K) \right). \end{aligned}$$

Since  $K \subset \bigcup_{l \in \mathbb{N}} C_l$  as in Equation (7.10), this implies

$$\nu \left( \bigcup_{l \in \mathbb{N}} F_l(C_l \cap K) \right) \geq e^{-2\delta_\Gamma \tau_\varphi} \cdot \nu(K).$$

Therefore, the second claim in Equation (8.1) follows.  $\square$

*Proof of Theorem 8.2.* Let  $\nu$  be the  $\delta_\Gamma$ -dimensional conformal density of  $\Gamma$ . By Lemma 8.5,  $\text{Spec}_{sq}(\Gamma_0) \subset \mathbf{E}_\nu(\Gamma_0)$ . Since  $\text{Spec}_{sq}(\Gamma_0)$  is non-arithmetic and  $\mathbf{E}_\nu(\Gamma_0)$  is a closed subgroup of  $\mathbb{R}$ , we have  $\mathbf{E}_\nu(\Gamma_0) = \mathbb{R}$ . By the assumption that the  $\Gamma_0$ -action on  $(\partial^h X, \nu)$  is ergodic, it follows from Proposition 8.4 that the  $\Gamma_0$ -action on  $(\mathcal{H}, \mu_\nu)$  is ergodic. Since  $\mu_\nu = \mu_\Gamma$  by definition, this completes the proof.  $\square$

## 9. SUBGROUPS OF MAPPING CLASS GROUPS AND MEASURE CLASSIFICATION

In the rest of this paper, let  $S$  be a connected orientable surface of genus  $g$  with  $p$  punctures, where  $3g - 3 + p \geq 1$ . We apply the results of the previous sections to the case where  $X$  is the Teichmüller space  $\mathcal{T} = \mathcal{T}(S)$ . In this section, we deduce Theorem 1.2, Theorem 1.3, and Theorem 1.8.

**9.1. Non-elementary subgroups of the mapping class group.** A subgroup  $\Gamma < \text{Mod}(S)$  is non-elementary if and only if  $\Gamma$  contains two pseudo-Anosov mapping classes with disjoint sets of fixed points in the Thurston boundary  $\mathcal{PML}$ . Since the axis of a pseudo-Anosov mapping class is squeezing as in Proposition 4.1, a non-elementary subgroup  $\Gamma < \text{Mod}(S)$  indeed satisfies the conditions of Definition 5.10 for its action on  $\mathcal{T}(S)$ .

In fact, it follows from the Nielsen–Thurston classification that pseudo-Anosov mapping classes are precisely squeezing isometries in  $\text{Mod}(S)$ . Hence, Theorem 1.4, which we prove in Appendix A as Theorem A.1, gives the *non-arithmeticity* of the squeezing spectra.

The notion of divergence type is defined similarly. Fixing a basepoint  $x_0 \in \mathcal{T}$ , the *Poincaré series* of a non-elementary subgroup  $\Gamma < \text{Mod}(S)$  is

$$\mathcal{P}_\Gamma(s) := \sum_{g \in \Gamma} e^{-s d_{\mathcal{T}}(x_0, g x_0)}.$$

Since  $\text{Mod}(S)$  has at most exponential growth [KM96, Theorem 1.3.2] and contains a free subgroup [McC85, Theorem B], there exists  $0 < \delta_\Gamma < +\infty$  such that  $\mathcal{P}_\Gamma(s)$  diverges for  $s < \delta_\Gamma$  and converges for  $s > \delta_\Gamma$ .

We call  $\delta_\Gamma$  the *critical exponent* of  $\Gamma$ . If  $\mathcal{P}_\Gamma(\delta_\Gamma) = +\infty$ , we say that  $\Gamma$  is of *divergence type*. Otherwise,  $\Gamma$  is of *convergence type*.

**9.2. Ergodicity and unique ergodicity.** We first deduce Theorem 1.2 and Theorem 1.3 from our theory in Section 5, Section 6, Section 7, and Section 8, by setting  $(X, d) = (\mathcal{T}, d_{\mathcal{T}})$ . We continue to fix a basepoint  $x_0 \in \mathcal{T}$ .

Given a non-elementary subgroup  $\Gamma < \text{Mod}(S)$ , recall the notion of the *recurrence locus* for  $\Gamma$  from the introduction:

$$\mathcal{R}_\Gamma := \left\{ \xi \in \mathcal{ML} : \begin{array}{l} \text{Teichmüller geodesic ray given by } q_\xi \in \mathcal{Q}(S, x_0) \\ \text{recurs to a compact subset in } \Gamma \backslash \mathcal{T} \end{array} \right\}$$

where  $q_\xi \in \mathcal{Q}(S, x_0)$  is the holomorphic quadratic differential corresponding to  $\xi \in \mathcal{ML}$ , given by the Hubbard–Masur theorem.

In terms of the Hubbard–Masur coordinates we introduced in Section 3.2, it follows from the Masur criterion [Mas92, Theorem 1.1] that

$$\text{HM}(\mathcal{R}_\Gamma) \subset \mathcal{UE} \times \mathbb{R}.$$

Recall that  $\mathcal{UE}$  sits in both  $\mathcal{PML}$  and  $\partial^{GM}\mathcal{T}$ , with the same topology [Miy13, Theorem 2]. Hence,  $\mathcal{UE} \times \mathbb{R}$  sits in both  $\mathcal{PML} \times \mathbb{R} \simeq \mathcal{ML}$  via the Hubbard–Masur coordinates and  $\partial^{GM}\mathcal{T} \times \mathbb{R} \simeq \partial^h\mathcal{T} \times \mathbb{R}$  via the coordinates in Section 3.3. That is, the identification

$$\iota : \mathcal{UE} \times \mathbb{R} \subset \mathcal{ML} \rightarrow \mathcal{UE} \times \mathbb{R} \subset \partial^h\mathcal{T} \times \mathbb{R} = \mathcal{H}$$

is a homeomorphism. Hence, a Borel measure  $\mu$  on  $\mathcal{ML}$  supported on  $\mathcal{UE} \times \mathbb{R}$  can be viewed as a Borel measure on  $\partial^h\mathcal{T} \times \mathbb{R}$  supported on  $\mathcal{UE} \times \mathbb{R}$ . Note

that  $\iota$  preserves the  $\mathbb{R}$ -coordinate. Combining these observations, we have that

$$(9.1) \quad (\iota \circ \text{HM})(\mathcal{R}_\Gamma) = \Lambda_c(\Gamma) \times \mathbb{R} \subset \mathcal{H}.$$

Furthermore, if  $\mu$  is locally finite on  $\mathcal{ML}$ , then  $\iota_*\mu$  is also locally finite on  $\partial^h\mathcal{T} \times \mathbb{R}$ . Indeed, for a compact subset  $K \subset \partial^h\mathcal{T} \times \mathbb{R}$ , the set  $\{t : (\xi, t) \in K \cap (\mathcal{UE} \times \mathbb{R})\}$  is bounded. This implies that  $\iota^{-1}(K \cap (\mathcal{UE} \times \mathbb{R}))$  is precompact, as it sits in  $\mathcal{PML} \times [-R, R]$  for some large  $R > 0$ . If  $\mu(\iota^{-1}(K \cap (\mathcal{UE} \times \mathbb{R})))$  is assumed to be finite, then  $\iota_*\mu(K)$  is finite as well. In conclusion, locally finite measures on  $\mathcal{ML}$  supported on  $\mathcal{UE} \times \mathbb{R}$  are also locally finite when viewed on  $\partial^h\mathcal{T} \times \mathbb{R}$ . Since  $\partial^h\mathcal{T} \times \mathbb{R}$  is Polish, such measures are Radon.

Hence, if  $\Gamma$  is of divergence type, then we can pull back the measure  $\mu_\Gamma$  on  $\Lambda_c(\Gamma) \times \mathbb{R}$  defined in Definition 7.1, via the embedding  $\iota \circ \text{HM}$ . This gives the measure on  $\mathcal{R}_\Gamma$  which coincides with the one constructed in Section 1.4; by abuse of notation, we also denote it by  $\mu_\Gamma$ .

Therefore, together with the non-arithmeticity in Theorem 1.4 (Theorem A.1), the ergodicity (Theorem 1.2) and the unique ergodicity (Theorem 1.3) follow from Theorem 8.1 and Theorem 7.4 respectively. Note that as in Theorem 8.2, an analogous ergodicity theorem for normal subgroups can also be deduced.

**9.3. Convex cocompact subgroups of the mapping class group.** In [FM02], Farb and Mosher introduced the following notion:

**Definition 9.1.** A finitely generated subgroup  $\Gamma < \text{Mod}(S)$  is called *convex cocompact* if  $\Gamma$  has a quasi-convex orbit in  $\mathcal{T}$ .

Some important features of convex cocompact subgroups are as follows:

**Theorem 9.2** ([FM02, Theorem 1.1]). *Let  $\Gamma < \text{Mod}(S)$  be a convex cocompact subgroup. Then*

- (1)  $\Gamma$  is a hyperbolic group,
- (2) there exists a  $\Gamma$ -equivariant embedding

$$\partial\Gamma \xrightarrow{\iota} \mathcal{UE} \subset \mathcal{PML}$$

where  $\partial\Gamma$  denotes the Gromov boundary of  $\Gamma$ , and,

- (3)  $\Gamma$  acts cocompactly on its weak hull  $\text{WH}(\Gamma) \subset \mathcal{T}$ , the union of all bi-infinite Teichmüller geodesics with endpoints in  $\Lambda(\Gamma) := \iota(\partial\Gamma)$ .

We now discuss the divergence type of non-elementary convex cocompact subgroups. Theorem 9.2 asserts that every non-elementary convex cocompact subgroup  $\Gamma < \text{Mod}(S)$  is Gromov hyperbolic with respect to either a word metric or the Teichmüller metric on its orbit. By [Coo93, Théorème 7.2],  $\Gamma$  has *purely exponential growth*, i.e., there exists  $C > 1$  (depending on the choice of the basepoint  $x_0$ ) such that

$$\frac{1}{C}e^{\delta_\Gamma r} \leq \#\{g \in \Gamma : d_\mathcal{T}(x_0, gx_0) < r\} \leq Ce^{\delta_\Gamma r} \quad \text{for all } r > C.$$

This implies that  $\Gamma$  is of divergence type.

In addition, it follows from the work of McCarthy–Papadopoulos [MP89] that the set  $\Lambda(\Gamma) \subset \mathcal{PML}$  is the unique  $\Gamma$ -minimal subset of  $\mathcal{PML}$ , and moreover the  $\Gamma$ -action on the complement  $\mathcal{PML} \setminus \Lambda(\Gamma)$  is properly discontinuous. In this regard, the set  $\Lambda(\Gamma) \subset \mathcal{PML}$  can be viewed as the *limit set* of  $\Gamma$ .

Furthermore, the cocompactness in Theorem 9.2(3) and the embedding in Equation (9.1) give a characterization of the recurrence locus for  $\Gamma$ :

$$(9.2) \quad \mathcal{R}_\Gamma = \{\xi \in \mathcal{ML} : [\xi] \in \Lambda(\Gamma)\}.$$

Hence, the proper discontinuity of the  $\Gamma$ -action on  $\mathcal{PML} \setminus \Lambda(\Gamma)$  implies that the  $\Gamma$ -action on  $\mathcal{ML} \setminus \mathcal{R}_\Gamma$  is properly discontinuous as well. Thus any  $\Gamma$ -invariant ergodic Radon measure on  $\mathcal{ML} \setminus \mathcal{R}_\Gamma$  is a counting measure on a single  $\Gamma$ -orbit, up to a constant multiple. Since  $\Gamma$  is of divergence type, Theorem 1.8 follows from Theorem 1.3.

## 10. CLASSIFICATION OF ORBIT CLOSURES

In this section, we prove the following classification of orbit closures:

**Theorem 10.1.** *Let  $\Gamma < \text{Mod}(S)$  be a non-elementary convex cocompact subgroup. Then for each  $\xi \in \mathcal{ML}$ , either*

$$\Gamma \cdot \xi \text{ is discrete} \quad \text{or} \quad \overline{\Gamma \cdot \xi} = \mathcal{R}_\Gamma.$$

*More precisely,*

- *if  $\xi \notin \mathcal{R}_\Gamma$ , then  $\Gamma \cdot \xi$  is discrete.*
- *if  $\xi \in \mathcal{R}_\Gamma$ , then  $\overline{\Gamma \cdot \xi} = \mathcal{R}_\Gamma$ .*

As mentioned earlier, the  $\Gamma$ -action on  $\mathcal{PML} \setminus \Lambda(\Gamma)$  is properly discontinuous [MP89]. Hence, in view of the characterization of  $\mathcal{R}_\Gamma$  in Equation (9.2), it suffices to show that the  $\Gamma$ -action on  $\mathcal{R}_\Gamma$  is minimal.

We do this by adjusting a standard argument for classifying horospherical orbit closures for Kleinian groups, which is based on the nice shape of horospheres in hyperbolic spaces. As we do not have such well-behaved horospheres in the Teichmüller space, we use a recent theory of expanding coarse cocycles for convergence group actions by Blayac–Canary–Zhu–Zimmer [BCZZ24] and an interpretation of the action of convex cocompact subgroups as expanding coarse cocycles given by the second author and Zimmer [KZ25].

From the viewpoint of the characterization in Equation (9.2) and the discussion in Section 5.6, the minimality of the  $\Gamma$ -action on  $\mathcal{R}_\Gamma$  follows once we show that the  $\Gamma$ -action on  $\Lambda(\Gamma) \times \mathbb{R}$  given by

$$g \cdot (\xi, t) = (g\xi, t + \beta_\xi(g^{-1}x_0, x_0))$$

is minimal, where  $x_0 \in \mathcal{T}$  is the fixed basepoint.

Given a pseudo-Anosov  $g \in \text{Mod}(S)$ , we denote by  $g^+, g^- \in \mathcal{PML}$  its attracting and repelling fixed points, respectively.

**Lemma 10.2.** *For each  $(\xi, t) \in \Lambda(\Gamma) \times \mathbb{R}$  and each pseudo-Anosov  $g \in \Gamma$ , there exists  $s \in \mathbb{R}$  such that  $(g^+, s) \in \overline{\Gamma \cdot (\xi, t)}$  or  $(g^-, s) \in \overline{\Gamma \cdot (\xi, t)}$ .*

*Proof.* By Theorem 9.2(3), there exists a sequence  $\{g_n\}_{n \in \mathbb{N}} \subset \Gamma$  such that  $g_n x_0 \rightarrow \xi$  while staying in a bounded neighborhood of a Teichmüller geodesic ray towards  $\xi \in \mathcal{UE}$ . This implies

$$(10.1) \quad \beta_\xi(g_n x_0, x_0) \rightarrow -\infty \quad \text{as } n \rightarrow +\infty.$$

This also implies the conical convergence in the Cayley graph of the hyperbolic group  $\Gamma$ , where  $\partial\Gamma$  and  $\Lambda(\Gamma) = \iota(\partial\Gamma)$  are identified by Theorem 9.2(2). After passing to a subsequence, we may assume that  $g_n^{-1}\xi$  converges to some  $\zeta \in \Lambda(\Gamma)$ . Suppose first that  $g^- \neq \zeta$ .

Fix an open neighborhood  $U \subset \Lambda(\Gamma)$  of  $\zeta$  such that  $g^- \notin \overline{U}$ . Then there exists a compact subset  $K$  in the Cayley graph of  $\Gamma$  such that every bi-infinite geodesic with one endpoint  $g^- \in \partial\Gamma$  and the other in  $\overline{U}$  intersects  $K$ . Passing to a subsequence, we may assume that  $g_n^{-1}\xi \in U$  for all  $n \in \mathbb{N}$ . This implies

$$(10.2) \quad \sup_{k, n \in \mathbb{N}} |\beta_{g_n^{-1}\xi}(g^{-k}x_0, x_0) - d(g^{-k}x_0, x_0)| < +\infty$$

(see [KZ25, Proposition 12.6] and [BCZZ24, Section 5] for details).

Observe that for  $k, n \in \mathbb{N}$ ,

$$g^k g_n^{-1}(\xi, t) = (g^k g_n^{-1}\xi, t + \beta_\xi(g_n x_0, x_0) + \beta_{g_n^{-1}\xi}(g^{-k}x_0, x_0)).$$

Then by Equation (10.2), for each  $n \in \mathbb{N}$ , we can choose  $k_n \in \mathbb{N}$  such that the sequence

$$t + \beta_\xi(g_n x_0, x_0) + \beta_{g_n^{-1}\xi}(g^{-k_n}x_0, x_0) \quad \text{is bounded in } n.$$

By Equation (10.1), we also have  $k_n \rightarrow +\infty$  as  $n \rightarrow +\infty$ .

After passing to a subsequence, we can set

$$s := \lim_{n \rightarrow +\infty} (t + \beta_\xi(g_n x_0, x_0) + \beta_{g_n^{-1}\xi}(g^{-k_n}x_0, x_0)).$$

Since  $g_n^{-1}\xi \in U$  and  $g^- \notin \overline{U}$ , we have

$$g^{k_n} g_n^{-1}\xi \rightarrow g^+ \quad \text{as } n \rightarrow +\infty.$$

This finishes the proof in this case.

If  $g^- = \zeta$ , then we have  $g^+ \neq \zeta$ . Hence, we can apply the same argument replacing  $g$  with  $g^{-1}$  and this completes the proof.  $\square$

*Proof of Theorem 10.1.* For each  $s \in \mathbb{R}$ , consider the map

$$\begin{aligned} a_s : \Lambda(\Gamma) \times \mathbb{R} &\rightarrow \Lambda(\Gamma) \times \mathbb{R} \\ (\xi, t) &\mapsto (\xi, t + s) \end{aligned}$$

which gives an  $\mathbb{R}$ -action on  $\Lambda(\Gamma) \times \mathbb{R}$ , commuting with the  $\Gamma$ -action. Hence, we regard it as a right  $\mathbb{R}$ -action. Note that this was denoted by  $T_s$  in Section 7.5; we use the new notation to emphasize that it acts on the right.

To prove the desired minimality, we fix  $(\xi, t) \in \Lambda(\Gamma) \times \mathbb{R}$  and  $\epsilon > 0$ . By the non-arithmeticity in Theorem 1.4 (Theorem A.1), there exist pseudo-Anosovs  $g_1, \dots, g_k \in \Gamma$  such that the additive subgroup  $\langle \tau_{g_1}, \dots, \tau_{g_k} \rangle \subset \mathbb{R}$  generated by their  $d_{\mathcal{T}}$ -translation lengths is  $\epsilon$ -dense.

By Lemma 10.2, after replacing  $g_1$  with  $g_1^{-1}$  if necessary, we have

$$(g_1^+, s_1) \in \overline{\Gamma \cdot (\xi, t)} \quad \text{for some } s_1 \in \mathbb{R}.$$

Then for every  $j \in \mathbb{Z}$ ,

$$g_1^j(g_1^+, s_1) = (g_1^+, s_1 + \beta_{g_1^+}(g_1^{-j}x_0, x_0)) = (g_1^+, s_1 + j\tau_{g_1}) = (g_1^+, s_1)a_{j\tau_{g_1}}$$

belongs to  $\Gamma \cdot (g_1^+, s_1) \subset \overline{\Gamma \cdot (\xi, t)}$ . This implies

$$\overline{\Gamma \cdot (g_1^+, s_1)a_{\langle \tau_{g_1} \rangle}} \subset \overline{\Gamma \cdot (\xi, t)}.$$

Applying the same argument to  $\overline{\Gamma \cdot (g_1^+, s_1)}$ , we obtain inductively that for some  $s_k \in \mathbb{R}$ ,

$$\overline{\Gamma \cdot (g_k^+, s_k)a_{\langle \tau_{g_1}, \dots, \tau_{g_k} \rangle}} \subset \overline{\Gamma \cdot (\xi, t)}.$$

Now let  $(\zeta, s) \in \Lambda(\Gamma) \times \mathbb{R}$  be arbitrary. Since the  $\Gamma$ -action on  $\Lambda(\Gamma)$  is minimal, there exists a sequence  $\{h_n\}_{n \in \mathbb{N}} \subset \Gamma$  such that  $h_n g_k^+ \rightarrow \zeta$  as  $n \rightarrow +\infty$ . Since  $\langle \tau_{g_1}, \dots, \tau_{g_k} \rangle$  is  $\epsilon$ -dense, for each  $n \in \mathbb{N}$ , there exists  $t_n \in \langle \tau_{g_1}, \dots, \tau_{g_k} \rangle$  such that

$$|s - (s_k + \beta_{g_k^+}(h_n^{-1}x_0, x_0) + t_n)| < \epsilon.$$

Hence, after passing to a subsequence, the sequence  $h_n(g_k^+, s_k)a_{t_n}$  converges to  $(\zeta, s') \in \overline{\Gamma \cdot (\xi, t)}$  for some  $s' \in \mathbb{R}$  with  $|s - s'| \leq \epsilon$ . Since this holds for each  $\epsilon > 0$ , we have  $(\zeta, s) \in \overline{\Gamma \cdot (\xi, t)}$ . This finishes the proof.  $\square$

#### APPENDIX A. NON-ARITHMETICITY OF TEICHMÜLLER LENGTH SPECTRA OF NON-ELEMENTARY SUBGROUPS OF $\text{Mod}(S)$

In this appendix, we prove Theorem 1.4. For each  $g \in \text{Mod}(S)$ , we denote its *Teichmüller translation length* by

$$\text{len}(g) := \lim_{n \rightarrow +\infty} \frac{d_{\mathcal{T}}(o, g^n o)}{n}, \quad o \in \mathcal{T}$$

which is well defined and independent of the choice of  $o \in \mathcal{T}$ .

For a subgroup  $\Gamma < \text{Mod}(S)$ , we call

$$\text{len}(\Gamma) := \{\text{len}(g) : g \in \Gamma \text{ pseudo-Anosov}\}$$

the *Teichmüller length spectrum* of  $\Gamma$ . The quotient space  $\text{Mod}(S) \backslash \mathcal{T}$  is the moduli space of Riemann surface structures on  $S$ , and the Teichmüller length spectrum  $\text{len}(\text{Mod}(S))$  coincides with the set of lengths of all closed geodesics in the moduli space  $\text{Mod}(S) \backslash \mathcal{T}$ , with respect to the induced metric on it. Similarly, the Teichmüller length spectrum  $\text{len}(\Gamma)$  of  $\Gamma < \text{Mod}(S)$  coincides with the set of lengths of all closed geodesics in the associated cover  $\Gamma \backslash \mathcal{T}$  of the moduli space.

We say that  $\Gamma < \text{Mod}(S)$  has *non-arithmetic* Teichmüller length spectrum if the additive subgroup

$$\langle \text{len}(\Gamma) \rangle < \mathbb{R}$$

generated by its Teichmüller length spectrum is dense. We also call  $\text{len}(\Gamma)$  non-arithmetic in this case.

We now restate our non-arithmeticity result.

**Theorem A.1** (Non-arithmeticity). *Let  $\Gamma < \text{Mod}(S)$  be a non-elementary subgroup. Then its Teichmüller length spectrum  $\text{len}(\Gamma)$  is non-arithmetic.*

Note that the non-elementary hypothesis cannot be dropped; if  $\Gamma$  is a cyclic subgroup generated by a single pseudo-Anosov mapping class, then  $\langle \text{len}(\Gamma) \rangle \subset \mathbb{R}$  is a scaled copy of  $\mathbb{Z}$ .

**A.1. Measured foliations and pseudo-Anosov mapping classes.** Before proving Theorem A.1, let us briefly review some basic facts on measured foliations and pseudo-Anosov mapping classes. For more comprehensive overview, we refer to [FLP79] and [FM12].

We denote by  $\mathcal{MF} = \mathcal{MF}(S)$  the space of equivalence classes of (singular) measured foliations on  $S$ , where the equivalence is generated by isotopy and Whitehead moves (see [FLP79, Exposé 5] for details). Each element of  $\mathcal{MF}$  can be represented by a pair  $(\mathcal{F}, \mu)$  of (singular) foliation  $\mathcal{F}$  on  $S$  and a transverse measure  $\mu$ .

The mapping class group  $\text{Mod}(S)$  acts naturally on  $\mathcal{MF}$  by homeomorphisms. Following the Nielsen–Thurston classification ([Nie44], [Thu88], cf. [FLP79, Exposé 9]), a mapping class  $g \in \text{Mod}(S)$  is called *pseudo-Anosov* if there exist transverse measured foliations  $(\mathcal{F}_u, \mu_u), (\mathcal{F}_s, \mu_s) \in \mathcal{MF}$  and a real number  $\lambda_g > 1$  such that

$$g \cdot (\mathcal{F}_u, \mu_u) = (\mathcal{F}_u, \lambda_g \cdot \mu_u) \quad \text{and} \quad g \cdot (\mathcal{F}_s, \mu_s) = (\mathcal{F}_s, \lambda_g^{-1} \cdot \mu_s).$$

We call  $(\mathcal{F}_u, \mu_u)$  and  $(\mathcal{F}_s, \mu_s)$  the *unstable and stable measured foliations* of  $g$  respectively, and they are uniquely determined up to scaling of the transverse measures. The real number  $\lambda_g$  is called the *stretch factor* of  $g$ , and it satisfies

$$\text{len}(g) = \log \lambda_g$$

([Ber78], [Thu88]). Note that  $\lambda_{g^{-1}} = \lambda_g^{-1}$ .

The space  $\mathcal{PMF} = \mathcal{PMF}(S)$  of projective measured foliations on  $S$  is defined by

$$\mathcal{PMF} := \mathcal{MF} / \sim$$

where the projectivization is given by scaling transverse measures. Thurston compactified the Teichmüller space  $\mathcal{T}$  using  $\mathcal{PMF}$  as its boundary. The compactification  $\mathcal{T} \sqcup \mathcal{PMF}$  is now referred to as the Thurston compactification, and hence  $\mathcal{PMF}$  is also referred to as the Thurston boundary. Thurston also showed that the  $\text{Mod}(S)$ -action on  $\mathcal{T}$  and the induced  $\text{Mod}(S)$ -action on  $\mathcal{PMF}$  glue together to give the  $\text{Mod}(S)$ -action on  $\mathcal{T} \sqcup \mathcal{PMF}$  by homeomorphisms ([Thu88], [Thu97]).

For a pseudo-Anosov  $g \in \text{Mod}(S)$ , we choose unstable and stable measured foliations

$$g^+, g^- \in \mathcal{MF}$$

of  $g$  respectively; these are unique up to scaling, and we fix a choice. It follows from the above discussion that their projective classes

$$[g^+], [g^-] \in \mathcal{PMF}$$

are distinct fixed points of  $g$  acting on  $\mathcal{T} \sqcup \mathcal{PMF}$ . Moreover,  $g$  exhibits the north-south dynamics: for  $x \in \mathcal{T} \sqcup \mathcal{PMF} \setminus \{[g^\pm]\}$ ,

$$g^n x \rightarrow [g^+] \quad \text{as } n \rightarrow +\infty \quad \text{and} \quad g^n x \rightarrow [g^-] \quad \text{as } n \rightarrow -\infty$$

and the convergence is uniform on compact subsets [McC85, Section 4] (cf. [Iva92, Theorem 3.5]). Note also that  $[(g^{-1})^+] = [g^-]$  and  $[(g^{-1})^-] = [g^+]$ .

Two pseudo-Anosov mapping classes  $g, h \in \text{Mod}(S)$  have either the same fixed points (i.e.,  $\{[g^\pm]\} = \{[h^\pm]\}$ ) or disjoint fixed points (i.e.,  $\{[g^\pm]\} \cap \{[h^\pm]\} = \emptyset$ ) [MP89, Lemma 2.5]. We say that they are *independent* when they have disjoint sets of fixed points in  $\mathcal{PMF}$ . A subgroup  $\Gamma < \text{Mod}(S)$  is called *non-elementary* if there exist independent pseudo-Anosov elements  $g, h \in \Gamma$ .

**A.2. Cross-ratios on  $\mathcal{MF}$  and  $\mathcal{PMF}$ .** We prove Theorem A.1 by introducing the notion of cross-ratios on  $\mathcal{MF}$  and  $\mathcal{PMF}$ .

For two isotopy classes  $\alpha, \beta$  of simple closed curves on  $S$ , we denote by  $i(\alpha, \beta) \in \mathbb{R}_{\geq 0}$  their *geometric intersection number*, i.e., the minimal number of intersection points of their representatives. Isotopy classes of (essential) simple closed curves on  $S$  can be regarded as measured foliations whose transverse measures are given by geometric intersection numbers with them, by foliating their annular neighborhoods with closed leaves isotopic to those simple closed curves. In this regard, the function  $i(\cdot, \cdot)$  continuously extends to

$$i : \mathcal{MF} \times \mathcal{MF} \rightarrow \mathbb{R}_{\geq 0}$$

which we also call the geometric intersection number ([Thu22, Section 9.3], [Ree81, Corollary 1.11]). Then  $i(\cdot, \cdot)$  is invariant under the  $\text{Mod}(S)$ -action and homogeneous under scaling: for  $x, y \in \mathcal{MF}$ ,  $g \in \text{Mod}(S)$ , and  $t, s > 0$ , we have

$$i(g \cdot x, g \cdot y) = i(x, y) \quad \text{and} \quad i(t \cdot x, s \cdot y) = ts \cdot i(x, y)$$

where  $t \cdot x$  and  $s \cdot y$  are obtained by scaling transverse measures of  $x$  and  $y$  by  $t$  and  $s$  respectively.

For a pseudo-Anosov mapping class  $g \in \text{Mod}(S)$ , we have

$$i(x, g^-) > 0 \quad \text{for every } x \in \mathcal{MF} \text{ with } [x] \neq [g^-]$$

and the same property also holds for  $g^+$  ([FLP79, Théorème 1, Exposé 12], [FLP79, Lemme 6, Lemme 16, Exposé 9], [Ree81, Theorem 1.12]).

**Definition A.2** (Cross-ratios). For  $x, y, z, w \in \mathcal{MF}$ , we define their *cross-ratio*

$$[x, y, z, w] = \frac{i(x, w)i(y, z)}{i(x, z)i(y, w)} \in \mathbb{R}_{\geq 0}$$

whenever  $i(x, z)i(y, w) \neq 0$ . Note that this is invariant under scaling transverse measures, and hence the cross-ratio is well defined on  $\mathcal{PMF}$  as well.

In Proposition A.4 below, we relate the cross-ratio to dynamics of pseudo-Anosov mapping classes. We begin with the following lemma, which might be standard to experts; we present a proof for the sake of completeness. With an extra element “0”, we give a topology on  $\mathcal{MF} \sqcup \{0\}$  by declaring that a sequence  $\{x_n\}_{n \in \mathbb{N}} \subset \mathcal{MF}$  converges to 0 if the associated sequence of transverse measures converges to the zero measure.

**Lemma A.3.** *Let  $g \in \text{Mod}(S)$  be pseudo-Anosov. Then the following hold:*

(1) *For any  $x \in \mathcal{MF}$ , we have*

$$\lambda_g^{-n} g^n x \rightarrow \frac{i(x, g^-)}{i(g^+, g^-)} \cdot g^+ \in \mathcal{MF} \sqcup \{0\} \quad \text{as } n \rightarrow +\infty.$$

(2) *Let  $\{x_n\}_{n \in \mathbb{N}} \subset \mathcal{MF}$  be a sequence such that  $x_n \rightarrow x \in \mathcal{MF}$  as  $n \rightarrow +\infty$  where  $i(x, g^-) > 0$ . Then*

$$\lambda_g^{-n} g^n x_n \rightarrow \frac{i(x, g^-)}{i(g^+, g^-)} \cdot g^+ \quad \text{as } n \rightarrow +\infty.$$

(3) *The convergence in (1) and (2) is uniform on compact subsets of  $\mathcal{MF} \setminus \mathbb{R}_{>0} \cdot g^-$  in the sense that for any compact  $Q \subset \mathcal{MF} \setminus \mathbb{R}_{>0} \cdot g^-$  and any open neighborhood  $\mathcal{U} \subset \mathcal{MF}$  of  $\frac{i(Q, g^-)}{i(g^+, g^-)} \cdot g^+ \subset \mathcal{MF}$ ,*

$$\lambda_g^{-n} g^n Q \subset \mathcal{U} \quad \text{for all large } n \in \mathbb{N}.$$

*Proof.* We first prove (2). Let  $\{x_n\}_{n \in \mathbb{N}} \subset \mathcal{MF}$  be a sequence such that  $x_n \rightarrow x \in \mathcal{MF}$  as  $n \rightarrow +\infty$ , where  $i(x, g^-) > 0$ . Since  $i(x, g^-) > 0$ , we have  $[x] \neq [g^-] \in \mathcal{PMF}$ . Then by the north-south dynamics of  $g$  on  $\mathcal{PMF}$ , we have

$$[g^n x_n] \rightarrow [g^+] \in \mathcal{PMF} \quad \text{as } n \rightarrow +\infty.$$

Hence, there exists a sequence  $\{t_n\}_{n \in \mathbb{N}} \subset \mathbb{R}_{>0}$  such that  $t_n g^n x_n \rightarrow g^+$  as  $n \rightarrow +\infty$ .

As  $n \rightarrow +\infty$ , we have

$$i(\lambda_g^{-n} g^n x_n, g^-) = \lambda_g^{-n} i(x_n, g^- g^-) = i(x_n, g^-) \rightarrow i(x, g^-) > 0.$$

Note that  $\lambda_g^{-n} g^n x_n = (\lambda_g^{-n} t_n^{-1}) t_n g^n x_n$  for all  $n \in \mathbb{N}$ , and that  $t_n g^n x_n \rightarrow g^+$  as  $n \rightarrow +\infty$ . It follows from the above computation that

$$\lambda_g^{-n} t_n^{-1} = \frac{i(\lambda_g^{-n} g^n x_n, g^-)}{i(t_n g^n x_n, g^-)} \rightarrow \frac{i(x, g^-)}{i(g^+, g^-)} \quad \text{as } n \rightarrow +\infty.$$

Therefore,

$$\lambda_g^{-n} g^n x_n = (\lambda_g^{-n} t_n^{-1}) t_n g^n x_n \rightarrow \frac{i(x, g^-)}{i(g^+, g^-)} \cdot g^+ \quad \text{as } n \rightarrow +\infty,$$

as desired.

We now prove (1). By (2), it suffices to consider the case that  $[x] = [g^-] \in \mathcal{PMF}$ . Then

$$\lambda_g^{-n} g^n x = \lambda_g^{-2n} x \rightarrow 0 \quad \text{as } n \rightarrow +\infty.$$

Since  $i(x, g^-) = 0$  in this case, the desired convergence holds.

Finally, (3) follows from (2) by the sequential compactness of  $Q$ .  $\square$

We now deduce the following relation between cross-ratios and stretch factors from Lemma A.3. For rank-one symmetric spaces, this was proved in [Kim01, Theorem 1]. Note that in the following,  $g^n h^n$  is pseudo-Anosov for all large  $n \in \mathbb{N}$  since  $g$  and  $h$  are independent ([McC85], [Iva92]).

**Proposition A.4.** *Let  $g, h \in \text{Mod}(S)$  be independent pseudo-Anosov mapping classes. Then*

$$\frac{\lambda_{g^n h^n}}{\lambda_g^n \lambda_h^n} \rightarrow [g^+, h^+, g^-, h^-] \quad \text{as } n \rightarrow +\infty.$$

*Proof.* For each  $n \in \mathbb{N}$ , let  $f_n := g^n h^n \in \text{Mod}(S)$ , which is pseudo-Anosov when  $n$  is large enough. For such  $n \in \mathbb{N}$ , we choose  $f_n^\pm \in \mathcal{MF}$  such that  $i(f_n^+, f_n^-) = 1$ . It then follows from the north-south dynamics of  $g$  and  $h$  that

$$[f_n^+] \rightarrow [g^+] \in \mathcal{PMF} \quad \text{and} \quad [f_n^-] \rightarrow [h^-] \in \mathcal{PMF} \quad \text{as } n \rightarrow +\infty.$$

Hence, by rescaling transverse measures of  $f_n^\pm \in \mathcal{MF}$ , we may assume that the sequences  $\{f_n^\pm\}_{n \in \mathbb{N}} \subset \mathcal{MF}$  are convergent. We denote their limits by  $x, y \in \mathcal{MF}$ , i.e.,

$$f_n^+ \rightarrow x \quad \text{and} \quad f_n^- \rightarrow y \quad \text{as } n \rightarrow +\infty.$$

Since  $i(f_n^+, f_n^-) = 1$  for all large  $n \in \mathbb{N}$ , we have

$$\lambda_{f_n} = i(f_n f_n^+, f_n^-) = i(g^n h^n f_n^+, f_n^-),$$

and hence

$$(A.1) \quad \frac{\lambda_{g^n h^n}}{\lambda_g^n \lambda_h^n} = i((\lambda_g^{-n} g^n) (\lambda_h^{-n} h^n) f_n^+, f_n^-) = i(\lambda_h^{-n} h^n f_n^+, \lambda_g^{-n} g^{-n} f_n^-).$$

Since  $[x] = [g^+]$  and  $[y] = [h^-]$ , the independence of  $g$  and  $h$  yields

$$i(x, h^-) > 0 \quad \text{and} \quad i(y, g^+) > 0.$$

Hence, by Lemma A.3(2), we have as  $n \rightarrow +\infty$  that

$$\lambda_h^{-n} h^n f_n^+ \rightarrow \frac{i(x, h^-)}{i(h^+, h^-)} h^+ \quad \text{and} \quad \lambda_g^{-n} g^{-n} f_n^- \rightarrow \frac{i(y, g^+)}{i(g^+, g^-)} g^-.$$

Now since  $x = tg^+$  and  $y = sh^-$  for some  $t, s > 0$ , it follows from the above convergences and Equation (A.1) that

$$\frac{\lambda_{g^n h^n}}{\lambda_g^n \lambda_h^n} \rightarrow \frac{i(x, h^-)i(y, g^+)i(h^+, g^-)}{i(h^+, h^-)i(g^+, g^-)} = ts \cdot \frac{i(g^+, h^-)^2 i(h^+, g^-)}{i(g^+, g^-)i(h^+, h^-)} \quad \text{as } n \rightarrow +\infty.$$

Since  $1 = i(f_n^+, f_n^-) \rightarrow i(x, y) = ts \cdot i(g^+, h^-)$  as  $n \rightarrow +\infty$ , we also have

$$ts = \frac{1}{i(g^+, h^-)}.$$

Therefore,

$$\frac{\lambda_{g^n h^n}}{\lambda_g^n \lambda_h^n} \rightarrow \frac{i(g^+, h^-)i(h^+, g^-)}{i(g^+, g^-)i(h^+, h^-)} = [g^+, h^+, g^-, h^-] \quad \text{as } n \rightarrow +\infty.$$

This finishes the proof.  $\square$

**A.3. Proof of Theorem A.1.** We are now ready to prove Theorem A.1. Let  $\Gamma < \text{Mod}(S)$  be a non-elementary subgroup and suppose to the contrary that its Teichmüller length spectrum  $\text{len}(\Gamma)$  is arithmetic, i.e., for some  $c \geq 0$  we have

$$\langle \text{len}(\Gamma) \rangle \subset c \cdot \mathbb{Z}.$$

Then for any independent pseudo-Anosov  $g, h \in \Gamma$ , it follows from Proposition A.4 that

$$(A.2) \quad \log[g^+, h^+, g^-, h^-] \in c \cdot \mathbb{Z}.$$

Denote by

$$(A.3) \quad \Lambda(\Gamma) := \overline{\{[g^+] : g \in \Gamma \text{ pseudo-Anosov}\}} \subset \mathcal{PMF}$$

the limit set of  $\Gamma$ . Since  $\Gamma < \text{Mod}(S)$  is non-elementary,  $\Lambda(\Gamma)$  is the unique  $\Gamma$ -minimal subset of  $\mathcal{PMF}$  and is perfect [MP89, Theorem 4.1, Proposition 5.2]. Moreover, the north-south dynamics of pseudo-Anosov mapping classes also implies that

$$(A.4) \quad \{([g^+], [g^-]) : g \in \Gamma \text{ pseudo-Anosov}\} \subset \Lambda(\Gamma) \times \Lambda(\Gamma)$$

is a dense subset (cf. proof of Proposition A.4). Hence, for  $x, y, z, w \in \Lambda(\Gamma)$  such that  $i(x, z)i(y, w)i(x, w)i(y, z) > 0$ , it follows from Equation (A.2) that

$$(A.5) \quad \log[x, y, z, w] \in c \cdot \mathbb{Z}.$$

We will deduce a contradiction from Equation (A.5).

Let us now turn to the train track theory. We refer the reader to [Thu22], [PH92], [Pap83] for a more comprehensive exposition on train tracks and their various properties. For a maximal and recurrent train track<sup>6</sup>  $\tau$  on  $S$ , let  $V_\tau$  be the real vector space of weights on the branches of  $\tau$  satisfying the switch condition, and let  $K(\tau) \subset V_\tau$  be the non-empty open convex cone consisting of weights which are positive on every branch of  $\tau$ . Then there exists a natural  $\mathbb{R}_{>0}$ -equivariant map  $\phi_\tau : K(\tau) \rightarrow \mathcal{MF}$  which is a

<sup>6</sup>We only need the following properties of maximal and recurrent train tracks, not their precise definitions. For this reason, we omit their definitions.

homeomorphism onto its image. Moreover, the image  $U(\tau) := \phi_\tau(K(\tau)) \subset \mathcal{MF}$  is a non-empty open cone in  $\mathcal{MF}$  whose elements are carried by  $\tau$  (cf. [Pap83, Section II.3]). The charts  $(U(\tau), \phi_\tau)$ 's form an atlas of  $\mathcal{MF}$ , giving rise to a piecewise-linear structure [Thu22, Proposition 9.5.8] (cf. [Pap08, Proposition 4.3]). Each  $(U(\tau), \phi_\tau)$  is called the train track chart associated with  $\tau$ .

In the rest of the proof,

we fix a pseudo-Anosov  $g \in \Gamma$ .

Then, after replacing  $g$  with its positive power if necessary, there exists a train track chart  $(U(\tau), \phi_\tau)$  containing  $g^+ \in \mathcal{MF}$  such that,  $g^2 U(\tau) \subset U(\tau)$  and the  $g^2$ -action on  $U(\tau)$  induces an action on  $V_\tau$  given by an invertible linear map  $A : V_\tau \rightarrow V_\tau$  with the leading eigenvalue  $\alpha := \lambda_g^2$  ([Pap83, Section IV], [PP87, Theorem 4.1]).

Since  $\Lambda(\Gamma)$  is perfect, there exists a pseudo-Anosov  $h \in \Gamma$  such that  $g$  and  $h$  are independent and  $h^+ \in U(\tau)$ ; in particular,  $i(h^+, g^\pm) > 0$ . We simply write  $z := h^+ \in \mathcal{MF}$ . Then for each  $n \in \mathbb{N}$ ,

$$[g^+, g^n z, g^-, g^{-n} z] = \frac{i(g^+, g^{-n} z) i(g^n z, g^-)}{i(g^+, g^-) i(g^n z, g^{-n} z)} = \frac{i(g^+, \lambda_g^{-n} g^{-n} z) i(\lambda_g^{-n} g^n z, g^-)}{i(g^+, g^-) i(\lambda_g^{-n} g^n z, \lambda_g^{-n} g^{-n} z)}.$$

By Lemma A.3(1), we have  $\lambda_g^{-n} g^{\pm n} z \rightarrow \frac{i(z, g^\mp)}{i(g^+, g^-)} \cdot g^\pm$ , and hence

$$[g^+, g^n z, g^-, g^{-n} z] \rightarrow 1 \quad \text{as } n \rightarrow +\infty.$$

On the other hand, since  $i(g^+, g^-) i(g^n z, g^{-n} z) i(g^+, g^{-n} z) i(g^n z, g^-) > 0$ , it follows from Equation (A.5) that

$$[g^+, g^n z, g^-, g^{-n} z] = 1 \quad \text{for all large } n \in \mathbb{N}.$$

In other words,

$$i(g^n z, g^{-n} z) = \frac{i(g^+, g^{-n} z) i(g^n z, g^-)}{i(g^+, g^-)} \quad \text{for all large } n \in \mathbb{N}.$$

Note that  $i(g^n z, g^{-n} z) = i(z, g^{2n} z)$ , that  $i(g^+, g^{-n} z) = i(g^n g^+, z) = \lambda_g^n i(g^+, z)$ , and that  $i(g^n z, g^-) = i(z, g^{-n} g^-) = \lambda_g^n i(z, g^-)$ . Hence we have

$$(A.6) \quad i(z, g^{2n} z) = \lambda_g^{2n} \cdot \frac{i(g^+, z) i(z, g^-)}{i(g^+, g^-)} \quad \text{for all large } n \in \mathbb{N}.$$

One may compare the identity in Equation (A.6) with the one for eigenvectors of linear maps. In this regard, the above identity can be interpreted as saying that  $z$  becomes an eigenvector after iteration of  $g^2$ . In linear algebra, this already implies that  $z$  was in fact an eigenvector, as can be proved using the Cayley–Hamilton theorem, and this yields a contradiction. This is in a similar spirit to the contradiction deduced in [Kim06], for rank one symmetric spaces. To transfer this intuition to our piecewise linear setting, we will take a tensor product of the vector spaces associated with each of

the linear pieces. This allows us to handle all pieces simultaneously, while still living in a linear algebraic setting.

Now noting that  $[z] = [h^+] \neq [g^+]$ , it follows from ([Pap86, Proposition 3], [PH92, Section 3.4]) that there exists a maximal and recurrent train track  $\sigma$  on  $S$  and an associated train track chart  $(U(\sigma), \phi_\sigma : K(\sigma) \subset V_\sigma \rightarrow U(\sigma))$  such that  $g^+ \in U(\sigma)$  and such that the function

$$i(z, \phi_\sigma(\cdot)) \quad \text{is linear on } K(\sigma).$$

Let  $\psi : V_\sigma \rightarrow \mathbb{R}$  be the linear form such that

$$\psi(\cdot) = i(z, \phi_\sigma(\cdot)) \quad \text{on } K(\sigma).$$

By [Pap08, Proposition 4.3], there exist an open cone neighborhood  $W \subset K(\tau)$  of  $\phi_\tau^{-1}(g^+)$  with  $\phi_\tau(W) \subset U(\sigma)$  and convex cones  $W_1, \dots, W_m \subset W$  with non-empty interiors such that  $W = \bigcup_{j=1}^m W_j$  and the transition map

$$\phi_\sigma^{-1} \circ \phi_\tau \quad \text{is linear on } W_j \text{ for each } 1 \leq j \leq m.$$

For each  $j$ , let  $\phi_j : V_\tau \rightarrow V_\sigma$  be the linear map such that

$$\phi_j = \phi_\sigma^{-1} \circ \phi_\tau \quad \text{on } W_j.$$

Define a linear form  $L_j : V_\tau \rightarrow \mathbb{R}$  by

$$L_j := \psi \circ \phi_j.$$

Now define the function  $f_\tau : K(\tau) \rightarrow \mathbb{R}$  by

$$f_\tau(\cdot) = i(z, \phi_\tau(\cdot)).$$

Then we have

$$f_\tau = L_j \quad \text{on } W_j.$$

We simply write

$$v := \phi_\tau^{-1}(z) \in K(\tau).$$

The following observation is crucial in deducing the desired contradiction. The name is explained by Figure 3.

**Lemma A.5** (Bottom-of-the-Iceberg Lemma). *For each  $1 \leq j \leq m$ , we have*

$$L_j(v) \leq 0.$$

*Proof.* Let  $u \in \text{int } W_j$  and for  $t \in [0, 1]$ , let  $u_t := (1-t)u + tv \in K(\tau)$ . Then by the convexity of  $f_\tau$  proven in [Mir04, Theorem A.1] (cf. [AH24, Theorem 4.1]),

$$f_\tau(u_t) \leq (1-t)f_\tau(u) + tf_\tau(v).$$

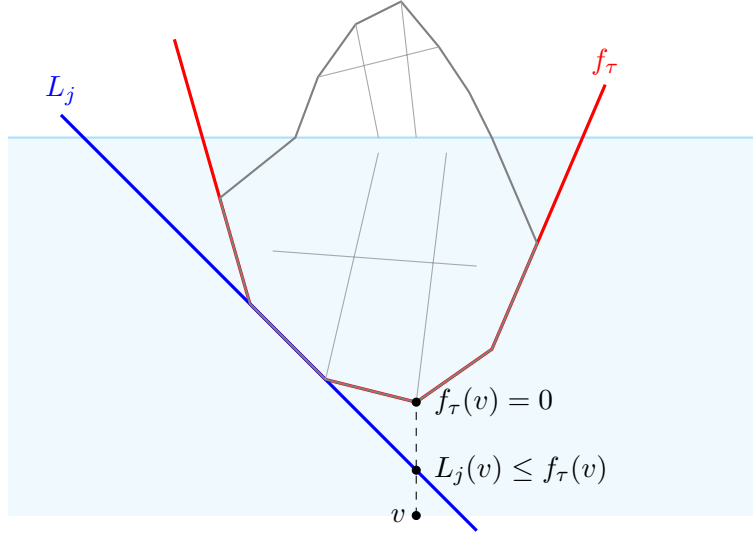
On the other hand, for small enough  $t > 0$ ,  $u_t \in W_j$  and hence

$$f_\tau(u_t) = L_j(u_t) = (1-t)L_j(u) + tL_j(v) = (1-t)f_\tau(u) + tL_j(v).$$

Combining these, we have

$$L_j(v) \leq f_\tau(v) = i(z, z) = 0.$$

□

FIGURE 3. Iceberg on  $f_\tau$ 

Now we set

$$C := \frac{i(g^+, z)i(z, g^-)}{i(g^+, g^-)} > 0.$$

Recalling that  $\alpha = \lambda_g^2$ , it follows from Equation (A.6) that

$$f_\tau(A^n v) = C \cdot \alpha^n \quad \text{for all large } n \in \mathbb{N}.$$

Since  $[g^{2n}z] \rightarrow [g^+]$  in  $\mathcal{PMF}$  as  $n \rightarrow +\infty$ , we have  $[A^n v] \rightarrow [\phi_\tau^{-1}(g^+)]$  as  $n \rightarrow +\infty$ , after projecting to the unit sphere in  $K(\tau) \subset V_\tau$ . Hence,

$$A^n v \in W \quad \text{for all large } n \in \mathbb{N}.$$

For each  $n \geq 0$ , set

$$P_n := \prod_{j=1}^m (L_j(A^n v) - C \cdot \alpha^n).$$

Then for all large  $n \in \mathbb{N}$ , the fact that  $A^n v \in W$  implies that  $L_j(A^n v) = f_\tau(A^n v) = C \cdot \alpha^n$  for some  $1 \leq j \leq m$ . This implies

$$(A.7) \quad P_n = 0 \quad \text{for all large } n \in \mathbb{N}.$$

Now for each  $1 \leq j \leq m$ , define the vector space  $V_j := V_\tau \times \mathbb{R}$  and consider the linear form  $\hat{L}_j : V_j \rightarrow \mathbb{R}$  defined by

$$\hat{L}_j(u, s) := L_j(u) - C \cdot s.$$

We also define  $A_j : V_j \rightarrow V_j$  by

$$A_j(u, s) := (Au, \alpha s),$$

which is an invertible linear map.

We then consider tensor products

$$\widehat{V} := \bigotimes_{j=1}^m V_j, \quad \widehat{L} := \bigotimes_{j=1}^m \widehat{L}_j, \quad \text{and} \quad \widehat{A} := \bigotimes_{j=1}^m A_j.$$

Then  $\widehat{L} : \widehat{V} \rightarrow \mathbb{R}$  is a linear form and  $\widehat{A} : \widehat{V} \rightarrow \widehat{V}$  is an invertible linear map.

Let

$$\chi_{\widehat{A}}(t) := t^d + c_{d-1}t^{d-1} + \cdots + c_1t + c_0$$

be the characteristic polynomial of  $\widehat{A}$ . Since  $\widehat{A}$  is invertible,  $c_0 \neq 0$ . By Cayley–Hamilton theorem,

$$\widehat{A}^d + c_{d-1}\widehat{A}^{d-1} + \cdots + c_1\widehat{A} + c_0\text{Id} = 0.$$

In other words, for each  $n \geq 0$ ,

$$\widehat{A}^{n+d} + c_{d-1}\widehat{A}^{n+d-1} + \cdots + c_1\widehat{A}^{n+1} + c_0\widehat{A}^n = 0.$$

In particular, setting

$$\widehat{v} := \bigotimes_{j=1}^m (v, 1) \in \widehat{V},$$

we have

$$\widehat{A}^{n+d}\widehat{v} + c_{d-1}\widehat{A}^{n+d-1}\widehat{v} + \cdots + c_1\widehat{A}^{n+1}\widehat{v} + c_0\widehat{A}^n\widehat{v} = 0,$$

and hence

$$\widehat{L}(\widehat{A}^{n+d}\widehat{v}) + c_{d-1}\widehat{L}(\widehat{A}^{n+d-1}\widehat{v}) + \cdots + c_1\widehat{L}(\widehat{A}^{n+1}\widehat{v}) + c_0\widehat{L}(\widehat{A}^n\widehat{v}) = 0.$$

Note that for each  $n \geq 0$ ,

$$\widehat{A}^n\widehat{v} = \bigotimes_{j=1}^m (A^n v, \alpha^n)$$

and hence

$$\widehat{L}(\widehat{A}^n\widehat{v}) = \prod_{j=1}^m (L_j(A^n v) - C \cdot \alpha^n) = P_n.$$

Therefore,

$$P_{n+d} + c_{d-1}P_{n+d-1} + \cdots + c_1P_{n+1} + c_0P_n = 0 \quad \text{for each } n \geq 0.$$

On the other hand, by Equation (A.7),  $P_n = 0$  for all large  $n \geq 0$ . Therefore, since  $c_0 \neq 0$ , backward induction gives  $P_n = 0$  for all  $n \geq 0$ . In particular,

$$P_0 = 0.$$

Since  $P_0 = \prod_{j=1}^m (L_j(v) - C)$ , this implies

$$L_j(v) - C = 0 \quad \text{for some } 1 \leq j \leq m.$$

However, by the Bottom-of-the-Iceberg Lemma (Lemma A.5),  $L_j(v) \leq 0$ . This contradicts  $C > 0$ , completing the proof of Theorem A.1.  $\square$

## APPENDIX B. TOPOLOGICAL MIXING OF TEICHMÜLLER GEODESIC FLOW

In this section, we prove Theorem 1.6, which we restate as Theorem B.1 below. Recall that for a non-elementary subgroup  $\Gamma < \text{Mod}(S)$ , we consider the subset

$$\Omega_\Gamma \subset \Gamma \backslash \mathcal{Q}^1\mathcal{T}$$

of quadratic differentials whose two associated projective measured foliations belong to the limit set  $\Lambda(\Gamma) \subset \mathcal{PMF}$  of  $\Gamma$ . Then  $\Omega_\Gamma$  is a non-wandering domain for the Teichmüller geodesic flow, and we indeed prove its topological mixing.

**Theorem B.1** (Topological mixing). *Let  $\Gamma < \text{Mod}(S)$  be a non-elementary subgroup. Then the Teichmüller geodesic flow on  $\Omega_\Gamma \subset \Gamma \backslash \mathcal{Q}^1\mathcal{T}$  is topologically mixing.*

It is originally due to Dal'bo [Dal00] that non-arithmeticity implies the topological mixing of geodesic flow, in negatively curved settings. Our deduction of Theorem B.1 from Theorem A.1 basically follows Dal'bo's ideas, but relies on our work on the dynamics of non-elementary subgroups acting on  $\mathcal{MF}$  in this paper.

**B.1. Bundle of quadratic differentials.** A (holomorphic) quadratic differential  $q$  on a (marked) Riemann surface  $X \in \mathcal{T}$  is a tensor locally given by  $\phi(z)dz^2$  where  $\phi$  is a holomorphic function with simple poles at the punctures. The vector space  $Q(X)$  of quadratic differentials on  $X$  is identified with the cotangent space of  $\mathcal{T}$  at  $X \in \mathcal{T}$ , and hence the bundle

$$\mathcal{QT} := \{(X, q) : X \in \mathcal{T} \text{ and } q \in Q(X)\}$$

of quadratic differentials is identified with the cotangent bundle of  $\mathcal{T}$ .

Since the Teichmüller metric is Finsler, the corresponding norm of  $q \in Q(X)$ ,  $X \in \mathcal{T}$ , is given by

$$\|q\| := \int_X |q| = \int_X |\phi(z)| |dz|^2,$$

which represents the area of  $q$ . Then the bundle

$$\mathcal{Q}^1\mathcal{T} := \{(X, q) \in \mathcal{QT} : \|q\| = 1\}$$

of unit-norm quadratic differentials is identified with the unit cotangent bundle of  $\mathcal{T}$ . The  $\text{Mod}(S)$ -action on  $\mathcal{T}$  naturally extends to actions on  $\mathcal{QT}$  and  $\mathcal{Q}^1\mathcal{T}$ . We refer the reader to [FM12] for more comprehensive exposition.

In this regard, the Teichmüller geodesic flow is given as a certain  $\mathbb{R}$ -action on  $\mathcal{QT}$ . For a more explicit description, we note that a non-zero quadratic differential  $(X, q) \in \mathcal{QT}$  determines real and imaginary measured foliations  $\text{Re}(q^{1/2})$  and  $\text{Im}(q^{1/2})$  on  $S$ , respectively, omitting the basepoint  $X$ :  $\text{Re}(q^{1/2})$  consists of a foliation obtained by taking the union of the zeros of  $q$  with the set of smooth paths in  $X$  whose tangent vectors at each

point evaluate to negative real numbers under  $q$ , and a transverse measure given by  $\mu(\alpha) := \int_\alpha |\operatorname{Re}(\sqrt{\phi(z)}dz)|$  where we write a local expression  $q = \phi(z)dz^2$ . The measured foliation  $\operatorname{Im}(q^{1/2})$  is defined by taking the paths in  $X$  whose tangent vectors evaluate to positive real numbers under  $q$  instead, and considering the transverse measure defined by integrating  $|\operatorname{Im}(\sqrt{\phi(z)}dz)|$ . These measured foliations  $\operatorname{Re}(q^{1/2})$  and  $\operatorname{Im}(q^{1/2})$  are also referred to as vertical and horizontal measured foliations of  $q$ , respectively. See ([FM12], [ABEM12]) for details.

The above correspondence defines a  $\operatorname{Mod}(S)$ -equivariant homeomorphism

$$\mathcal{QT} \setminus \{0\} \longrightarrow (\mathcal{MF} \times \mathcal{MF}) \setminus \tilde{\Delta}$$

$$q \longmapsto (\operatorname{Re}(q^{1/2}), \operatorname{Im}(q^{1/2}))$$

where  $\tilde{\Delta} := \{(x, y) \in \mathcal{MF} \times \mathcal{MF} : \exists z \in \mathcal{MF} \text{ s.t. } i(x, z) = i(y, z) = 0\}$  [GM91]. Noting that the Teichmüller geodesic flow commutes with the  $\operatorname{Mod}(S)$ -action on  $\mathcal{QT}$ , we denote the time- $t$  map of the Teichmüller geodesic flow by the right action  $a_t$ ,  $t \in \mathbb{R}$ . Then in terms of the above homeomorphism, we have

$$qa_t \longmapsto (e^t \operatorname{Re}(q^{1/2}), e^{-t} \operatorname{Im}(q^{1/2}))$$

for  $t \in \mathbb{R}$ .

The above discussion descends to  $\mathcal{Q}^1\mathcal{T}$  as follows: we have a  $\operatorname{Mod}(S)$ -equivariant homeomorphism

$$(B.1) \quad \mathcal{Q}^1\mathcal{T} \longrightarrow (\mathcal{MF} \times \mathcal{PMF}) \setminus \Delta$$

$$q \longmapsto (\operatorname{Re}(q^{1/2}), [\operatorname{Im}(q^{1/2})])$$

where  $\Delta := \{(x, [y]) \in \mathcal{MF} \times \mathcal{PMF} : (x, y) \in \tilde{\Delta}\}$ , and the Teichmüller geodesic flow can be described via

$$qa_t \longmapsto (e^t \operatorname{Re}(q^{1/2}), [\operatorname{Im}(q^{1/2})])$$

for  $t \in \mathbb{R}$ .

**B.2. Non-wandering domains.** Let  $\Gamma < \operatorname{Mod}(S)$  be a non-elementary subgroup. Since the  $\operatorname{Mod}(S)$ -action on  $\mathcal{Q}^1\mathcal{T}$  commutes with the Teichmüller geodesic flow, the Teichmüller geodesic flow descends to the bundle

$$\Gamma \backslash \mathcal{Q}^1\mathcal{T}$$

which is identified with the unit cotangent bundle over  $\Gamma \backslash \mathcal{T}$ . We also denote this induced Teichmüller geodesic flow on  $\Gamma \backslash \mathcal{Q}^1\mathcal{T}$  by  $\{a_t\}_{t \in \mathbb{R}}$ .

We set

$$\tilde{\Omega}_\Gamma := \left\{ q \in \mathcal{Q}^1\mathcal{T} : \left[ \operatorname{Re}(q^{1/2}) \right], \left[ \operatorname{Im}(q^{1/2}) \right] \in \Lambda(\Gamma) \right\} \quad \text{and} \quad \Omega_\Gamma := \Gamma \backslash \tilde{\Omega}_\Gamma.$$

Then it follows from the density of fixed points of pseudo-Anosov elements in  $\Gamma$  (Equation (A.4)) that  $\Omega_\Gamma$  is a non-wandering domain for the Teichmüller geodesic flow. Namely, for any non-empty open subset  $\mathcal{O} \subset \Omega_\Gamma$ , there exists a sequence  $\{t_n\}_{n \in \mathbb{N}} \subset \mathbb{R}$  with  $t_n \rightarrow +\infty$  such that

$$\mathcal{O} \cap \mathcal{O}a_{t_n} \neq \emptyset \quad \text{for all } n \in \mathbb{N}.$$

The rest of this section is devoted to the proof of *topological mixing* of the Teichmüller geodesic flow on  $\Omega_\Gamma$ , which was stated as Theorem 1.6 in the introduction. That is, for any non-empty open subsets  $\mathcal{O}_1, \mathcal{O}_2 \subset \Omega_\Gamma$ , we have

$$\mathcal{O}_1 \cap \mathcal{O}_2 a_t \neq \emptyset \quad \text{for all large } t.$$

**B.3. Proof of Theorem B.1.** Suppose to the contrary that there exist non-empty open subsets  $\mathcal{O}_1, \mathcal{O}_2 \subset \Omega_\Gamma$  and a diverging sequence  $\{t_n\}_{n \in \mathbb{N}} \subset \mathbb{R}_{>0}$  such that  $\mathcal{O}_1 \cap \mathcal{O}_2 a_{t_n} = \emptyset$  for all  $n \in \mathbb{N}$ . In other words,

$$\mathcal{O}_1 a_{-t_n} \cap \mathcal{O}_2 = \emptyset \quad \text{for all } n \in \mathbb{N}.$$

Let  $\tilde{\mathcal{O}}_1, \tilde{\mathcal{O}}_2 \subset \tilde{\Omega}_\Gamma$  be the  $\Gamma$ -invariant lifts of  $\mathcal{O}_1$  and  $\mathcal{O}_2$ , respectively. In the rest of the proof, we identify  $\tilde{\mathcal{O}}_1$  and  $\tilde{\mathcal{O}}_2$  with subsets of  $(\mathcal{MF} \times \mathcal{PMF}) \setminus \Delta$  as in Equation (B.1).

Using the density of fixed points of pseudo-Anosov elements of  $\Gamma$  in Equation (A.4) and the identification in Equation (B.1), we fix a pseudo-Anosov  $g \in \Gamma$  such that  $(g^+, [g^-]) \in \tilde{\mathcal{O}}_1$ , for some choice of the representative  $g^+ \in \mathcal{MF}$ . The corresponding quadratic differential generates the axis of  $g$  in  $\mathcal{T}$ .

We also fix an open neighborhood  $U \subset \Lambda(\Gamma)$  of  $[g^-]$  such that

$$\{g^+\} \times U \subset \tilde{\mathcal{O}}_1.$$

Then for any  $k, n \in \mathbb{N}$ ,

$$\left\{ \lambda_g^k e^{-t_n} g^+ \right\} \times g^k U = g^k (\{g^+\} \times U) a_{-t_n} \subset \tilde{\mathcal{O}}_1 a_{-t_n}.$$

Now for each  $n \in \mathbb{N}$ , let  $k_n \in \mathbb{N}$  be such that  $\lambda_g^{k_n} e^{-t_n} \in [1, \lambda_g)$ . After passing to a subsequence, we may assume that

$$(B.2) \quad \lambda_g^{k_n} e^{-t_n} \rightarrow \lambda \in [1, \lambda_g] \quad \text{as } n \rightarrow +\infty.$$

Then by Theorem 10.1,

$$(B.3) \quad \overline{\Gamma \cdot (\lambda g^+)} = \{x \in \mathcal{MF} : [x] \in \Lambda(\Gamma)\}$$

which uses the non-arithmeticity result (Theorem A.1). Note that while Theorem 10.1 is stated only for convex cocompact subgroups, it implies Equation B.3: since for any pseudo-Anosov  $h \in \Gamma$  independent of  $g$ , there exists a non-elementary convex cocompact subgroup of  $\Gamma$  containing powers of  $g$  and  $h$  [FM02, Theorem 1.4], and Equation (A.3) then gives the claim.

Now fix a point  $(x, [y]) \in \tilde{\mathcal{O}}_2$ . Since  $[x] \in \Lambda(\Gamma)$ , it follows from Equation (B.3) that there exists a sequence  $\{\gamma_j\}_{j \in \mathbb{N}} \subset \Gamma$  such that

$$(B.4) \quad \gamma_j(\lambda g^+) \rightarrow x \quad \text{as } j \rightarrow +\infty.$$

Fixing a metric on  $\mathcal{PMF}$ , which is metrizable (cf. [FLP79, Théorème V.4, Exposé 6]), denote by  $B_{1/j}([y]) \subset \mathcal{PMF}$  the open ball of radius  $1/j$  centered at  $[y]$ , for each  $j \in \mathbb{N}$ . Since  $\Lambda(\Gamma)$  is perfect [MP89, Proposition 5.2], for each  $j \in \mathbb{N}$  there exists  $[z_j] \in (\gamma_j^{-1} B_{1/j}([y]) \cap \Lambda(\Gamma)) \setminus \{[g^+]\}$ . Then there exists a diverging sequence  $\{n_j\}_{j \in \mathbb{N}} \subset \mathbb{N}$  so that  $[z_j] \in g^{k_{n_j}} U$  for all  $j \in \mathbb{N}$ . In other words,

$$(B.5) \quad [z_j] \in \gamma_j^{-1} B_{1/j}([y]) \cap g^{k_{n_j}} U \quad \text{for all } j \in \mathbb{N}.$$

Now consider the sequence of points

$$(B.6) \quad \left( \gamma_j \left( \lambda_g^{k_{n_j}} e^{-t_{n_j}} g^+ \right), \gamma_j [z_j] \right) \in \gamma_j g^{k_{n_j}} (\{g^+\} \times U) a_{-t_{n_j}} \subset \tilde{\mathcal{O}}_1 a_{-t_{n_j}},$$

for  $j \in \mathbb{N}$ . By Equation (B.5), we have

$$\gamma_j [z_j] \rightarrow [y] \quad \text{as } j \rightarrow +\infty.$$

Since the  $\text{Mod}(S)$ -action on  $\mathcal{MF}$  commutes with the  $\mathbb{R}_{>0}$ -action on  $\mathcal{MF}$  by scaling transverse measures, it follows from Equation (B.2) and Equation (B.4) that

$$\gamma_j \left( \lambda_g^{k_{n_j}} e^{-t_{n_j}} g^+ \right) = \frac{\lambda_g^{k_{n_j}} e^{-t_{n_j}}}{\lambda} \cdot \gamma_j(\lambda g^+) \rightarrow x \quad \text{as } j \rightarrow +\infty.$$

Therefore, the sequence of points in Equation (B.6) converges to  $(x, [y]) \in \tilde{\mathcal{O}}_2$  as  $j \rightarrow +\infty$ . Since  $\tilde{\mathcal{O}}_2$  is open, this implies that

$$\tilde{\mathcal{O}}_1 a_{-t_{n_j}} \cap \tilde{\mathcal{O}}_2 \neq \emptyset \quad \text{for all large } j \in \mathbb{N}.$$

This contradicts  $\mathcal{O}_1 a_{-t_n} \cap \mathcal{O}_2 = \emptyset$  for all  $n \in \mathbb{N}$ , and the topological mixing follows.  $\square$

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